STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

JOSEPH VON FISCHER

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: August 30, 2019

I. Introduction and Qualifications

Q. Please state your name, title, and business address.

A. My name is Joseph C. von Fischer. My title is Professor and Associate Chair of Biology, Colorado State University ("CSU"). My business address is 100 W. Lake Street, Fort Collins, CO 80523.

Q. On whose behalf are you submitting this testimony in this proceeding?

A. I am submitting this testimony on behalf of the Environmental Defense Fund ("EDF").

Q. Please provide a summary of your education and experience.

A. My primary research expertise is in understanding what processes on the land surface give the atmosphere its greenhouse gas composition. I earned my PhD in Ecology and Evolutionary Biology in 2002 through Cornell University's program in Biogeochemistry and Global Change. From 2002-2003, I conducted post-doctoral training in atmospheric composition at Princeton University. I have been on the faculty at CSU since 2003, where I am now a Professor and Associate Department Chair of Biology.

Over my career, I have won competitive research funding from diverse sources including National Science Foundation, National Aeronautics and Space Administration, United States Geological Survey, the Department of Energy, EDF, NYSEARCH, and Consolidated Edison Company of New York, Inc. ("Con Edison"). This funding has supported my research and allowed me to publish 37 scientific publications, which are routinely cited by other scientists.

Since 2012, I have been collaborating with EDF as the science lead on their "methane mapping" project, where high-sensitivity methane analyzers have been deployed on Google Street View cars. In this collaborative project, we have mapped 15 U.S. cities to date. Each city deployment has been associated with multiple meetings with the utility management team. These conversations have allowed me to deepen my understanding of how distribution systems are managed by utilities, especially with regards to leak surveys, survey technology and the incorporation of survey results into pipeline integrity management. Based on these deployments and the supporting scientific experiments associated with them, I have published 5 scientific papers on methane mapping to date and one industry paper describing how to deploy advanced leak detection to reduce emissions. I describe the results of these papers below.

Based on my training and research experiences, I have developed an expertise in understanding how natural gas leaks in distribution systems can be discovered by high-sensitivity methane analyzer systems, and how information about leak magnitude can be used to prioritize repair and replacement.

I have attached a copy of my curriculum vitae in Exhibit __ (JF-1).

Q. Have you previously filed testimony before regulatory or legislative bodies?

A. Yes, I have testified before the New York Public Service Commission ("Commission") in Case Nos. 16-G-0058, 16-G-0059 and 16-G-0061 and before the Illinois Commerce Commission in Case No. 16-0376. Please refer to Exhibit __ (JF-1) for a detailed listing of my testimonies.

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to provide recommendations to improve The Brooklyn Union Gas Company's d/b/a National Grid NY and KeySpan Gas East Corporation's d/b/a National Grid (collectively, "Companies" or "National Grid") proposed leak repair and pipe replacement activities, in order to advance New York climate policy to reduce natural gas leaks and methane emissions.

Q. Please provide a summary of your testimony and recommendations.

A. I recommend that National Grid deploy Advanced Leak Detection ("ALD") technologies to quantify their distribution system leakage, and to use information about leak size to prioritize pipeline replacement efforts and to repair the largest (i.e., fastest flowing) leaks as quickly as possible. Given that a small number of the largest "super emitter" leaks is typically responsible for the majority of total emissions, a program aimed at finding and rapidly repairing the largest leaks is a relatively easy way to reduce overall system emissions. Because quantifying and reducing emissions is a requirement of New York's recently enacted climate legislation, deployment of ALD technologies would provide National Grid a straightforward path to compliance.

I also note that the current incentive structure for reducing the Companies' leak backlog may be counter-productive. ALD technologies have been shown to help utilities find more leaks in their system than has been possible with traditional technology. Discovering more leaks would likely add leaks to the backlog. If the

incentive structure only rewards the Companies for declining leak backlog counts, the Companies are actually dis-incentivized to use a technology that discovers more leaks. For these reasons, I suggest that the incentive system be re-examined in light of the capabilities of this new technology.

Q. Are you providing any exhibits to your testimony?

A. Yes. I am attaching the following exhibits to my testimony:

- Exhibit __ (JF-1): Curriculum Vitae and List of Expert Testimony
- Exhibit __(JF-2): Pacific Gas and Electric Company Natural Gas Leakage Abatement Report, California Public Utilities Commission Rulemaking 15-01-008 (June 17, 2019)
- Exhibit__(JF-3): Picarro Emissions Quantification Results Final Report in Support of the Methane Leak Surveying Report for the Public Service Electric and Gas Company ("PSE&G") Gas System Modernization Program ("GSMP") II Program
- Exhibit__(JF-4): PSE&G Presentation "Replacement Main Prioritization: A Practical Application of Using Risk and Methane Emissions" (May 2, 2019)
- Exhibit __(JF-5): Kerans, Mike, Picarro Surveyor Leak Detection Study – Sacramento Side-by-Side Study (2012)
- Exhibit__(JF-6): Clark, Timothy, Picarro Surveyor Leak Detection Study – Diablo Side-by-Side Study (2012)

IV. Addressing Methane Leaks from Distribution Networks to Advance New York's Climate Policies

Q. You stated above that one of the purposes of your testimony is to provide recommendations in order to advance New York climate policy to reduce natural gas leaks and methane emissions. Please explain.

A. Natural gas leakage from distribution pipelines is widespread in U.S. cities, including many in New York State. As shown in the chart below, New York has a significant amount of leak prone pipe as compared to other states.

Top Ten States by Miles of Leak Prone Mains[1]

	Unprotected Bare Steel	Cast Iron	Ductile Iron	Copper	**Total Leak Prone Pipe**	Total miles
Grand Total	33,336	22,861	513	13	**56,722**	1,305,025
PA	5,932	2,525	170	2	**8,629**	48,335
NY	4,972	3,175	-	0	**8,147**	49,307
OH	6,197	197	1	1	**6,396**	58,759
NJ	550	3,911	24	1	**4,486**	35,007
TX	3,905	466	-	-	**4,371**	107,799
MA	1,146	2,925	1	0	**4,073**	21,714
CA	3,244	58	-	0	**3,302**	106,806
MI	352	2,389	-	-	**2,742**	59,731
WV	2,546	12	-	-	**2,557**	10,961
IL	28	1,152	205	-	**1,385**	62,168

This leakage is costly from the perspective of ratepayers and the environment. Methane, the principal component of natural gas, is a potent greenhouse gas that,

[1] Pipeline and Hazardous Materials Safety Administration. (2019). 2018 Gas Distribution Annual Report Data. Washington, D.C. Retrieved from https://www.phmsa.dot.gov/data-and-statistics/pipeline/gas-distribution-gas-gathering-gas-transmission-hazardous-liquids

pound for pound, traps 84 times more heat than carbon dioxide over the first 20 years it is released, and is therefore a significant contributor to climate change.[2] As acknowledged by the Companies, retirement of LPP is a "major component" toward reaching methane emission goals, as it "reduces gas losses and fugitive emissions of methane."[3]

Q. Has New York State recognized the importance of prioritizing leak repairs in distribution systems in addressing methane?

A. Yes. Governor Cuomo's May 2017 Methane Reduction Plan directs state agencies, including the Department of Public Service ("DPS"), to develop proposals and policies to inventory emissions and identify strategies for methane capture and elimination. The Plan specifically directs the DPS to "prioritize leak repairs in the distribution system."[4]

Q. Have other recent developments in New York State underscored the importance of addressing methane leaks from distribution networks?

A. Yes. The New York State legislature recently enacted the Climate Leadership and Community Protection Act ("Act" or "CLCPA"), which requires New York to

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[2] Myhre, G.; Shindell, D.; BrÂŐon, F.-M.; Collins, W.; Fuglestvedt, J.; Huang, J.; Koch, D.; Lamarque, J.-F.; Lee, D.; Mendoza, B.; Nakajima, T.; Robock, A.; G. Stephens, T. T.; Zhang, H. In *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; Stocker, T., Qin, D., Plattner, G., Tignor, M., Allen, S., Boschung, J., Nauels, A., Xia, Y., Bex, B., Midgley, B., Eds.; Cambridge University Press, 2013.

[3] *See, e.g.*, Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 29.

[4] Methane Reduction Plan at 6 (May 2017), https://www.dec.ny.gov/docs/administration_pdf/mrpfinal.pdf.

adopt measures to reduce statewide greenhouse gas ("GHG") emissions by 85 percent by 2050, with an additional goal of achieving net zero emissions across all sectors of the economy by that time.[5] The Act acknowledges the short-term climate impact of methane, recognizing that methane causes 84 times as much global warming as the same amount of carbon dioxide.[6] The Act also seeks to ensure that GHG emissions are "real, permanent, quantifiable, verifiable, and enforceable."[7]

Q. Have the Companies made any commitments to reduce methane emissions?

A. Yes, the Companies are establishing a target to reduce methane emissions from the gas distribution system 60 percent (from 1990 levels) by 2035.[8] The Companies acknowledge that retirement of LPP is a major component toward reaching this ambitious goal.[9]

Q. Please explain some of the challenges in addressing methane leakage from distribution pipelines.

A. I perceive two key challenges for addressing methane leakage: 1) there are more leaks in the natural gas system than utilities are aware of, and 2) the current regulatory structure hampers utility initiatives to improve leak detection effectiveness.

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5 https://legislation.nysenate.gov/pdf/bills/2019/S6599.

6 *Id.*, Section 75-0101(2).

7 *Id.*, Section 75-0109(3)(b).

8 Future of Heat Testimony at page 16.

9 Gas Infrastructure and Operations Panel at page 28.

Recent academic findings have demonstrated that observed methane emissions from cities are about twice that reported in the Environmental Protection Agency inventory.[10] In addition, as I describe in a 2018 scientific paper, my team observed that utility crews using traditional technologies were able to locate only 35% of the pipeline leaks that were found using advanced leak detection methods.[11] Interestingly, this result was not unique to our work; two studies by the utility PG&E found a similar fraction of "false negatives" where leaks exist but are not found.[12] Together, these findings suggest that there are many more leaks than those reported in utility inventories. Based on our own published work, and corroborated by the PG&E studies, my team and I estimate that, in areas where utilities are using traditional leak detection technologies, there are 2.4x as many leaks in the system as the utility is aware of.

And, as I explain in detail below, the motivation for utilities to find more leaks is hampered by the regulatory structures for utilities to find and abate leaks. Although there exist well-intended incentives for utilities to reduce the number of leaks in their backlog,[13] an unintended consequence of these policies is that they

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[10] G. Plant *et al.*, Large Fugitive Methane Emissions from Urban Centers Along the US East Coast, https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2019GL082635.

[11] Weller, Zachary *et al.*, Vehicle Based Methane Surveys for Finding Natural Gas Leaks and Estimating their Size: Validation and Uncertainty, Environmental Science and Technology (2018). If this detection rate is applied at the national scale, then the national inventory for the number of pipeline leaks in natural gas distribution infrastructure would increase by a factor of 2.4. *Id.* at 11925.

[12] Exhibit__(JF-5) and Exhibit__(JF-6).

[13] Gas Safety Panel at pages 53-54.

provide perverse incentives for utilities to not find and record new leaks in their inventories. This could hinder the State's ability to comply with the new climate legislation, which emphasizes the importance of ensuring a complete and accurate accounting of GHG emissions using the best available scientific, technological, and economic information.

Q. Please explain the role advanced leak detection and data analytics could play in addressing these challenges.

A. The idea of ALD is that high-sensitivity methane analyzers can be deployed in survey vehicles to identify where natural gas is leaking from distribution systems. The "advanced" part of this terminology refers in part to the methane sensors using a form of laser-absorption spectroscopy[14] that can more accurately and precisely determine the concentration of methane in the air. In contrast to "traditional technology," the sensors in ALD can quantify much smaller departures from the local background concentration and thus find more leaks. Moreover, ALD uses advanced data analytics that examine the overall neighborhood patterns in the methane concentration to discern when the concentrations are elevated. These sophisticated analyses can reveal subtler leak information than traditional technologies. Finally, ALD deployments can report not just the presence of a leak, but they can also estimate leak size. With leak size information, utilities can

[14] While my testimony refers to ALD, the Companies describe the technology as "CRS" or "CRDS," which is a type of laser absorption spectroscopy. Gas Safety Panel at page 45. While "CRDS" is one component of ALD, ALD also encompasses the advanced analytics or algorithms that utilize the data acquired from the ALD technology to estimate the methane flow rate attributed to a leak indication.

prioritize their repair and replacement efforts of non-hazardous leaks to fix the largest leaks fastest. This prioritization means that big leaks are not emitting methane for as long, thus reducing a utility's annual methane emissions. By simply re-organizing the timing of repair and replacement to fix the biggest leaks first, there is no additional cost.

As other utilities have demonstrated, advanced leak detection technology can be deployed in implementing "Super Emitter" programs, which aim to rapidly detect and repair large grade 3 leaks, leading to significant and rapid GHG emissions reductions.[15] Deploying these methods will also assist the Companies in ensuring a more accurate accounting of GHG emissions, consistent with the new climate legislation.[16]

Q. Could ALD also be used to prioritize pipeline replacement efforts?

A. Yes. Results from ALD yield information about leak location and magnitude. As we illustrated in the publication by Palacios and others for the utility Public Service Electric and Gas Company ("PSE&G"),[17] these leak size and density results can be aggregated by the utility to determine which stretches of pipe have the greatest

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[15] Exhibit __ (JF-2) at page 9.

[16] For example, both Companies provide a high level estimate of potential methane emissions reductions over the next several years based on each respective company's current facilities assuming LPP retirement pursuant to each company's proposed program. Gas Infrastructure and Operations Panel, Case 19-G-0309 at page 30; Gas Infrastructure and Operations Panel, Case 19-G-0310 at page 29.

[17] Palacios, V., S.R. George, J.C. von Fischer and K. Mohlin (2017) "Integrating leak quantification into Natural Gas Utility Operations" white paper in *Public Utilities Fortnightly* https://www.fortnightly.com/white-papers/integrating-leak-quantification-natural-gas-utility-operations.

emissions. After safety considerations have been made, leak magnitude can be used as a metric for prioritizing which pipelines are replaced first.

IV. CSU and EDF Collaboration to Survey Distribution System Leaks

Q. Please provide a brief overview of EDF and CSU's collaborative project to survey methane leaks from distribution systems, and engagement with gas utilities on the use of leak quantification to prioritize pipe replacement and leak repair efforts (the "Project").

A. For the last 7 years, I have collaborated with EDF and Google on a unique project aimed at gathering data on the location and relative size of methane leaks from natural gas distribution systems using three Google Street View ("GSV") cars that are specially equipped with methane sensors.

The goal of the Project is to document patterns in the location and magnitude of natural gas leakage from the distribution systems of different utilities, demonstrate the benefits of using leak quantification technology to prioritize utilities' pipe replacement and leak repair activities, and to create the impetus for, and facilitate, the integration of this technology into utilities' regular leak operations.

The GSV vehicles that collected data as part of this initiative used a Picarro methane concentration analyzer that has a closed-path design with a sampling frequency of one data point every half-second, resulting in samples every 13 feet when driving at 20 miles per hour. These vehicles are also equipped with a Hemispheres Global Positioning System ("GPS") Receiver to document patterns in

methane concentration in space. Drivers were asked to drive every public street in a study area at least twice to ensure accuracy. Re-driving helped confirm the persistence of elevated methane readings and characterize their emission rate more accurately.

Data collected in urban systems and during controlled release experiments informed our development of a data analysis algorithm to interpret data collected through this Project. We conducted a number of "controlled release" studies at CSU by releasing methane at known rates and then driving through the resulting plume with the GSV car. We analyzed this data for patterns in plume size and shape to develop an algorithm that predicts leak size from measured patterns in methane concentration. We also validated the leak rate estimates using comparisons with a number of more costly and labor-intensive approaches and found that the method we used had more than enough precision and accuracy to be used effectively in utility leak reduction efforts.

Q. In how many cities has EDF/CSU mapped methane leaks from natural gas distribution pipelines as part of the Project?

A. To date, EDF and CSU have published methane maps for leaks from natural gas distribution systems in 13 cities: Boston, MA; Burlington, VT; Chicago, IL; Dallas, TX; Elizabethtown, NJ; Indianapolis, IN; Jacksonville, FL; Los Angeles, CA; New York City (Staten Island), NY; Syracuse, NY; Birmingham, AL; Mesa, AZ; and

Pittsburgh, PA.[18] In addition, we have conducted regional mapping in collaboration with the New Jersey utility PSE&G[19] and with the New York utility Con Edison.[20]

Q. Please summarize key findings and implications for utilities' pipeline replacement and leak repair programs.

A. In general, we found more natural gas leaks (expressed as leak indications per mile of roadway) in systems where federal data indicates high fraction of older, leak-prone materials in the distribution systems. That is, a high abundance of leaks per mile correlates with a higher percentage of leak-prone mains on the relevant utility system.

Scientific studies from across the natural gas supply chain have found that the majority of methane emissions come from a few larger leaks, and data gathered through this Project is consistent with that broader trend.[21] A 2014 Stanford University study noted, based on many independent experiments, that a small number of large leaks are responsible for a large fraction of the leakage from natural gas systems.[22] Likewise, across the 6100 natural gas leak indications found in the

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18 https://www.edf.org/climate/methanemaps.

19 https://www.edf.org/climate/methanemaps/pseg-collaboration.

20 https://www.edf.org/climate/methanemaps/con-edison.

21 Adam R. Brandt, Garvin A. Heath, and Daniel Cooley, Methane Leaks from Natural Gas Systems Follow Extreme Distributions, *Environmental Science & Technology* 2016 *50* (22), 12512-12520DOI: 10.1021/acs.est.6b04303, https://pubs.acs.org/doi/abs/10.1021/acs.est.6b04303.

22 A. R. Brandt et al., "Methane Leaks from North American Natural Gas Systems", *Science*, Vol. 343, 14 February 2011, available at http://www.novim.org/images/pdf/ScienceMethane.02.14.14.pdf.

15 urban areas we have mapped to date, we found that the larges 16% of those leaks are responsible for 50% of the emissions. Thus, efforts aimed to find and fix those largest super-emitter leaks can have a disproportionate and cost-effective impact on the overall emissions.

Only by measuring leak flow rate (i.e. the volume of methane emissions released from a given leak per unit time) can utilities use this information, alongside safety information of course, to prioritize the largest leaks for repair or the leakiest pipe segments for replacement. I summarize our findings in the table below.

City	Miles surveyed	Leaks found	Leaks per mile	Total Emissions (L/min)	Emissions per mile
Birmingham, AL	345	168	0.49	592	1.7
Boston, MA	2,607	2,349	0.90	6,142	2.4
Burlington, VT	113	13	0.12	18	0.2
Chicago, IL	1,037	402	0.39	834	0.8
New York, NY*	200	380	1.90	1,029	5.1
Dallas, TX	705	414	0.59	1,226	1.7
Elizabethtown, NJ*	212	224	1.06	655	3.1
Indianapolis, IN	767	3	0.00	6	0.0
Jacksonville, FL	820	81	0.10	160	0.2
Los Angeles, CA	1,674	354	0.21	1,104	0.7

Mesa, AZ	164	1	0.01	1	0.0
Newark, NJ*	813	313	0.38	781	1.0
Pittsburgh, PA	321	235	0.73	553	1.7
Staten Island, NY	990	976	0.99	2,621	2.6
Syracuse, NY	493	212	0.43	539	1.1
SUM or MEAN	11,262	6,125	0.54	16,262	1.5

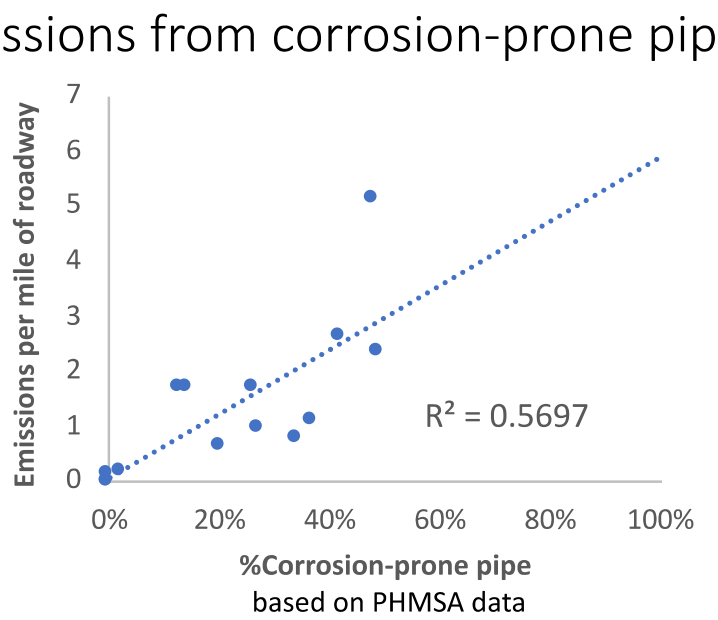
Our work shows that cities differ sharply in the number of leaks observed, depending, in part, on whether, and the extent to which, they had conducted accelerated pipeline replacement programs ("APRPs"). In general, cities that are known to have relatively newer, less leak prone distribution systems were found to have fewer leaks than cities with older, more leak prone systems. For instance, Burlington, VT and Indianapolis, IN registered 11 and 5 leaks, respectively, while we found hundreds or thousands of leaks in cities known to have older, more leak prone gas infrastructure such as Boston, MA, Staten Island, NY and Syracuse, NY.

For a more direct comparison, these leak counts should be normalized to the length of distribution lines or to the gas flow rate beneath the survey area, but appropriate data and/or infrastructure information were not in place to make this estimate. As a surrogate for this analysis, I calculated the frequency of leaks per unique km of roadway surveyed. This normalization did not alter our broad conclusions from an analysis of the raw data; there only were 6 leaks per 100 km

in Burlington and 0.4 per 100 km in Indianapolis, while Boston had 56 per 100 km, Staten Island had 63 per 100 km and Syracuse had 28 per 100 km.

In all cities, low leak flow rates were far more common (83% to 100% of cities' leaks) than medium (0% to 17%) or high (0% to 2%).

We estimated the average leak flow rates for each city as the sum of estimated leak flow rates per 100 km of roadway surveyed. When multiplied by the length of roadways in each city, we conservatively estimate that these leaks from Boston amount to at least 1300 tons methane yr^{-1}, 1000 tons methane yr^{-1} in Staten Island and 300 tons methane yr^{-1} in Syracuse.



The figure above shows the relationship between the percentage of corrosion-prone pipe and the estimated emissions per mile of corrosion-prone pipe. The corrosion-prone pipe data are drawn from the federal PHMSA database for the year when the surveys were conducted. Only 13 urban areas are shown because we

omitted results from New York, NY and Elizabethtown, NJ where the surveys were not a random sample of the area but instead a targeted survey of known areas with high abundance of leak-prone pipe.[23]

Many sectors of the natural gas supply system have a "fat tail problem" where large, infrequent leaks cause a significant proportion of the total system leakage.[24] The magnitude of the potential for leak reduction is stark: cities that were known to have relatively newer, less leak prone systems due to the implementation of APRP programs show an average 95% reduction in the normalized system leakage rate as compared to other cities.

Q. Have the results of this Project been published in peer reviewed scientific journals?

A. Yes, some of the results have been published, and more publications are underway. A description of the methodology was published in Environmental Science and Technology in 2017,[25] and a validation of false positives and leak size estimate was published in Environmental Science and Technology in 2018.[26] The approach for

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[23] This analysis has not been subject to peer review and was performed to illustrate the trends observed in the Project.

[24] A. R. Brandt et al., "Methane Leaks from North American Natural Gas Systems," *Science*, Vol. 343, 14 February 2011, available at http://www.novim.org/images/pdf/ScienceMethane.02.14.14.pdf.

[25] Joseph C. von Fischer *et al.*, Rapid Vehicle-Based Identification of Location and Magnitude of Urban Natural Gas Pipeline Leaks, Environmental Science & Technology (2017), https://pubs.acs.org/doi/abs/10.1021/acs.est.6b06095/.

[26] Zachary D. Weller *et al.*, Vehicle-Based Methane Surveys for Finding natural Gas Leaks and Estimating Their Size: Validation and Uncertainty, Environmental Science & Technology (2018), https://pubs.acs.org/doi/full/10.1021/acs.est.8b03135.

using leak size information for prioritizing pipeline repair and replacement was published in Public Utilities Fortnightly in 2017.[27] In 2018, we published a statistical study in the journal Environmetrics where we evaluate how greater survey effort finds more leaks.[28] A 2019 publication reports on updates to our data analysis algorithm and makes the algorithm publicly available.[29] Finally, in a second 2019 publication, we show how the GSV vehicle deployment could be used to estimate methane emissions from other sources, in this case fertilizer plants.[30]

I summarize each of these studies here:

- Von Fischer and others (2017): This is the first scientific paper to show that natural gas leak indications can be detected using ALD approaches. We outlined the methodology and described early findings in the first five cities we had mapped.

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[27] Virginia Palacios *et al.*, Integrating Leak Quantification into Natural Gas Utility Operations, Public Utilities Fortnightly (May 2017), https://www.fortnightly.com/white-papers/integrating-leak-quantification-natural-gas-utility-operations.

[28] Zachary D. Weller *et al.*, A calibration capture-recapture model for interfering natural gas leak population characteristics using data from Google Street View cars, Environmetrics (January 2018), https://doi.org/10.1002/env.2519/.

[29] Weller, Z.D., D. Yang* and J.C. von Fischer (2019) "An open source algorithm to detect natural gas leaks from mobile methane survey data." *PLoS ONE* 14(2): e0212287 https://doi.org/10.1371/journal.pone.0212287.

[30] Zhou, X., F.H. Passow, J. Rudek, J.C. von Fischer, S.P. Hamburg, and J.D. Albertson (2019) "Estimation of methane emissions from the ammonia fertilizer industry using a mobile sensing approach" *Elementa: Science of the Anthropocene.* 7(1) p19. https://www.elementascience.org/article/10.1525/elementa.358/.

- Weller and others (2018): we establish the efficacy of mobile methane surveys for managing local natural gas distribution systems by evaluating their ability to detect and locate natural gas leaks and quantify their emissions. In two cities, we found that three-quarters of leak indications from mobile surveys corresponded to natural gas leaks, but local distribution companies' field crews did not find most of these leaks, indicating that the national methane activity factor for leaks in local NG distribution pipelines is underestimated by a factor of 2.4. We found the median distance between mobile-estimated leak locations and actual leak locations was 19 meters. A comparison of emission quantification methods (mobile-based, surface enclosure, and tracer ratio) found that the mobile method overestimated leak magnitude for the smallest leaks but accurately estimated size for the largest leaks that are responsible for the majority of total emissions. Across leak sizes, mobile methods adequately rank relative emission rates for repair prioritization, and they are easily deployed and offer efficient spatial coverage.
- Palacios and others (2017): We describe how natural gas utilities can incorporate leak flow rate data into existing pipeline replacement and leak repair prioritization frameworks to more rapidly and efficiently reduce leakage on their system. Leak distributions typically demonstrate a "fat-tail," where a few, large leaks are responsible for the majority of lost gas volumes. Through ranking and ordering leak flow rate data, utilities can

identify a subset of the largest leaks to repair or the leakiest pipelines to replace, and capture more gas per dollar spent on leak repair or pipeline replacement. This benefits ratepayers, who pay for the cost of lost gas, and also carries broader environmental and societal benefits.

- Weller and others (2018): While new mobile monitoring technology has revolutionized our ability to measure pollutant levels over large regions, statistical methods for making inferences from data collected by these mobile systems are still being developed. In this paper, we introduce a new "capture–recapture" model to answer key questions from data collected by mobile monitoring systems. We apply our new method to characterize populations of natural gas leaks in urban areas using data collected by atmospheric methane analyzers placed on GSV cars. The new calibration capture–recapture ("CCR") model combines data from controlled methane release experiments and data collected from mobile air monitors to enable inference for several natural gas leak population characteristics, including the number of undetected leaks and the total methane output rate in a surveyed region. Our methodology is a novel application of capture–recapture modeling. The CCR model addresses challenges associated with using a capture–recapture model to analyze data collected by a mobile monitoring system such as a variable sampling effort. The CCR model provides a new framework for inferring the total number of leaks in natural gas distribution systems and offers critical insights for informing intelligent

infrastructure repair policy that is both cost effective and environmentally friendly.

- Zhou and others (2019): In this study, we estimate methane emissions from an important downstream consumer of natural gas, the ammonia fertilizer industry, which commonly uses natural gas as a feedstock and a fuel for the production of ammonia and other upgraded products. Using a GSV car equipped with a high-precision methane analyzer, we adopted a mobile sensing approach to measure methane mixing ratios along public roads that are downwind of the ammonia fertilizer plants. Useful data were collected from six plants, which represent >25% of the total number of U.S NG-based ammonia fertilizer plants, and use >20% of the total NG consumption by this industry. Based on the measured data, a source characterization model was applied to estimate the methane emission rates from the upwind plants. Assuming that the estimates are representative of emissions during normal operations of a plant, we calculated the NG loss rate (i.e. the ratio between NG emission rate and NG throughput). If the sampled plants are representative of the U.S. ammonia fertilizer industry, the industrial-averaged NG loss rate (± standard deviation) is estimated to be 0.34% (±0.20%), and the total methane emissions (± standard deviation) from this industry are estimated to be 29 (±18) Gigagram per year (Gg CH_4/yr) in 2015–2016. This is significantly higher than the reported methane emissions of 0.2 Gg CH_4/yr from the U.S. EPA's Facility Level Information

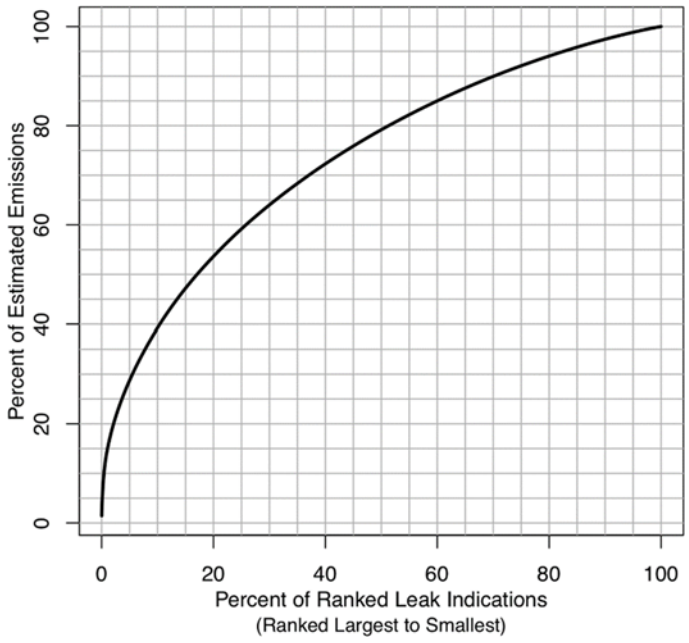
on Greenhouse Gas Tools (FLIGHT). This study begins to fill an important knowledge gap in quantifying methane emissions along the NG value chain, and demonstrates the capability of mobile sensing for characterizing emissions.

- Weller and others (2019) While survey platforms for ALD are commercially available, the associated data processing software largely constitute a black box due to their proprietary nature. In this paper we describe our updated, algorithm for identifying leak indications using data from mobile CH_4 surveys, and provide an under-the-hood look at the choices and challenges associated with data analysis. We also describe how our algorithm has evolved over time, and the data-driven insights that have prompted these changes. Applying our algorithm to data collected in 15 urban areas produced more than 6100 leak indications and estimates of the leaks' size. We use these results to characterize the distribution of leak sizes in local natural gas distribution systems. Mobile surveys are already an effective and necessary tool for managing natural gas distribution systems, but improvements in the technology and software will continue to increase its value.

Q. How can the use of advanced leak detection technology and leak quantification methodologies in leak prioritization ensure that ratepayer funding is deployed efficiently?

A. I have used a graphical tool called a "cumulative emissions curve" to illustrate the impact of eliminating the largest system leaks. A cumulative emission curve is shown below for the >6,100 leak indications that our Project has identified in the 15 urban areas where we have surveyed. To make this graph, we rank the leaks from largest to smallest, and then sum their emissions. The line on the graph thus illustrates how much of total emissions are caused by some percentage of the largest emissions. In the graph below, we can draw some direct conclusions about the role of the largest leaks in the system. For example:

- 20% of emissions come from the largest 2.5% of leaks
- 50% of emissions come from the largest 16% of leaks



Because a small fraction of the leaks are responsible for a large fraction of emissions, any effort that targets those large leaks for immediate repair will have a beneficially disproportionate effect on reducing overall system leakage.

V. Integration of Advanced Leak Detection into Utility Operations

Q. Have other utilities employed advanced leak detection, data analytics and quantification?

A. Yes. New Jersey's largest utility, PSE&G, received New Jersey Board of Public Utilities approval to implement a three year, $905 million pipe replacement program,[31] where PSE&G was required to consider data on the volume of methane emissions leaked from its pipes, in conjunction with other relevant factors, to identify those that are most in need of replacement.[32] Compared to a business as usual scenario, by using leak flow rates for prioritization, PSE&G achieved an 83% reduction of methane emissions early on by replacing one-third fewer miles of gas lines.[33]

As part of the second phase of its gas system modernization program in BPU Docket No. GR17070776, PSE&G committed to contract with a third party

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[31] *Decision and Order of the New Jersey Board of Public Utilities In The Matter Of Public Service Electric And Gas Company for Approval of a Gas System Modernization Program and Associated Cost Recovery Mechanism*, Docket No. GR15030272, November 16, 2015, retrieved from http://www.nj.gov/bpu/pdf/boardorders/2015/20151120/11-16-15-2F.pdf.

[32] Further information about this analysis can be accessed at https://www.edf.org/climate/methanemaps/pseg-collaboration.

[33] *Id.*

vendor to conduct a leak survey in 2018 on 280 miles of leak prone pipeline grids.[34] Leak survey data will be used to generate an "Estimated Flow Rate per Mile (Liter/min/mile)."[35] PSE&G will then develop a ranking threshold which will be used to prioritize grids for replacement in subsequent program years.[36]

Q. Has PSE&G shared the results from its Gas System Modernization Program?

A. Yes. The Methane Leak Surveying Report on PSE&G's Gas System Modernization Program II explains that "[w]ith methane maps and their aggregated emissions data…it is possible to make accurate, surgical construction decisions at the grid or individual pipeline section level as desired."[37] The report concludes:

> This variability shows the power of the methane mapping technique for providing additional granularity that can be used to maximize methane emissions reductions and/or maximize remediation of the maximum number of belowground leaks through changes to construction priorities based on these methane maps and associated data.[38]

A recent PSE&G presentation also concludes that its use of advanced leak detection technology is "better for the environment, [provides] less chance of non-

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[34] *In the Matter of the Petition of Public Service Electric and Gas Company for Approval of the Next Phase of the Gas System Modernization Program and Associated Cost Recovery Mechanism*, BPU Docket No. GR17070776, Stipulation of Settlement and Agreement at P 24 (April 18, 2018). The BPU approved this settlement in a June 1, 2018 order.

[35] *Id*.

[36] *Id.*

[37] Exhibit __(JF-3) at page 8.

[38] *Id.* at page 11.

hazardous leaks getting worse, and [results in] fewer potential customer calls/complaints."[39]

Q. Are you aware of other utilities' efforts to integrate advanced leak detection and data analytics into their operations?

A. Yes. Elizabethtown Gas Company ("Elizabethtown") recently became the first New Jersey utility to officially adopt and integrate advanced leak detection techniques into its operations as it implements its $300 million Infrastructure Investment Program ("IIP"). Pursuant to the settlement agreement in this case, Elizabethtown will integrate advanced leak detection technology information and methane emission flow rates in prioritizing its IIP projects.[40] The Peoples' Gas Light and Coke Company ("PGL") in Chicago, Illinois agreed to conduct a pilot program in which "leak flow rate data, collected by a contracted service provider or PGL using advanced leak detection and quantification technology, will be considered in prioritizing leak-prone pipe ("LPP") replacement under the [System Modernization Program]."[41] In addition, Peoples Gas of Pittsburgh committed to using advanced leak detection and data analytics to cut methane emissions from its

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[39] Exhibit __(JF-4) at slide 26.

[40] *In the Matter of the Petition of Elizabethtown Gas Company to Implement an Infrastructure Investment Program ("IIP") and Associated Recovery Mechanism Pursuant to N.J.S.A. 48:2-21 and N.J.A.C. 14:3-2A*, BPU Docket No. GR18101197, Stipulation of Settlement and Agreement at P 11 (May 30, 2019). The New Jersey Board of Public Utilities approved this settlement in a June 12, 2019 order.

[41] *Illinois Commerce Commission On its Own Motion v. The Peoples Gas Light and Coke Company*, ICC No. Docket 16-0376 at page 77 (January 10, 2018 Final Order).

distribution system by 50%.[42] After mapping and measuring the leaks from its underground pipes, Peoples will use the data to prioritize upgrades to achieve the greatest climate benefits.

Q. Have utilities in New York also explored the benefits of advanced leak detection and data analytics?

A. Yes, in 2016, Consolidated Edison Company of New York Inc. ("Con Edison"), EDF, and CSU conducted a pilot program to survey Con Edison's backlog of Type 3 leaks and characterize the leaks by size (i.e. flow rate). The leak flow rate data provided by EDF allowed Con Edison to repair the largest emitting non-hazardous leaks, representing an estimated reduction in nearly twice the amount of methane emissions compared to a business-as-usual scenario. EDF estimated that Con Edison reduced about 30% of the total emissions from surveyed areas. These results and an interactive map showing the leak locations are available on EDF's website.[43]

Q. Has Con Edison acknowledged the benefits of advanced leak detection?

A. Yes. The Company has stated that "[d]etection of gas leaks through state-of-the-art technology is critical to our comprehensive approach to risk management and

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[42] Environmental Defense Fund. January 8, 2019. Peoples Gas, EDF Unveil New Commitment to Help Protect the Climate by Cutting Methane Emissions From Pittsburgh Utility System in Half. Retrieved from: https://www.edf.org/media/peoples-gas-edf-unveil-new-commitment-help-protect-climate-cutting-methane-emissions.

[43] Environmental Defense Fund. Innovative collaboration fixes non-hazardous leaks faster. Retrieved from: https://www.edf.org/climate/methanemaps/con-edison Accessed on: May 20, 2019.

commitment to public safety."[44] Con Edison also notes that "through enhanced leak detection, we can identify, respond and remediate leaks more rapidly, reducing risk, keeping the public safe, and protecting the environment by reducing emissions of methane, a greenhouse gas."[45]

VI. The Companies' Leak Prone Pipe Repair and Replacement Efforts

Q. Please describe the pilot projects that EDF is conducting in collaboration with National Grid.

A. EDF and National Grid have collaborated on a range of pilot projects, in order to advance the use of new technological solutions, including advanced leak detection and quantification methods, to build on National Grid's efforts to enhance its system modernization and maintenance efforts. These projects are aimed at providing additional data and perspective around underground infrastructure condition, leakiness and program need, to help National Grid and the Commission balance the need for pipe replacement/repair efforts with the associated costs.

Q. Do the Companies propose to continue to explore the use of ALD technology to enhance leak program repair?

A. Yes, as explained by the Gas Safety Panel:

> During CY 2019, the Companies are continuing to assist Colorado State with its next phase of analysis of the 2018 CRD testing results. The Companies are providing access to and guidance on their gas system GIS mapping to enable the University to conduct geospatial analysis. The Companies will continue to

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[44] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company for Gas Service*, Case No. 19-G-0066, Direct Testimony of Gas Policy Panel at pages 15-16.

[45] *Id.* at page 16.

support these efforts and will consider the results and possible expanded application of the CRDS technology.[46]

Q. Do you have any recommendations regarding how the Companies should integrate advanced leak detection technology into their operations?

A. The Companies should conduct regular, comprehensive surveys of their infrastructure using ALD technology, especially in areas with high density of leak-prone pipe. It would serve the Companies well to set an accurate GHG baseline so that they can identify and quantify the largest sources of methane emissions in their service territories and receive regulatory credit for methane reductions from abating the largest leaks. The surveys should gather information on leak size so that the "super emitters" or leaks of exceptionally large size can be quickly found and repaired.

Q. Please explain what you mean by "super emitters."

A. Super emitters are the largest leaks on a utility's system, measuring 10 standard cubic feet per hour or greater. A "Super Emitter" Program identifies the highest emitting leaks for targeted replacement to satisfy emission reduction requirements.[47]

Q. Are you aware of any utilities that have implemented such a program?

A. Yes, Pacific Gas and Electric Company's ("PG&E") conducted its first "Super Emitter" leak survey in August 2018, where PG&E targeted the largest leaks and

[46] Gas Safety Panel, Case 19-G-0309 at pages 45-46.

[47] https://naturalgas.picarro.com/case_studies/super_emitters.

aimed to repair them promptly no matter the leak grade.[48] In 2018, PG&E found 220 "Super Emitter" leaks. The impact of the program was estimated at a reduction of 90 Mscf and PG&E expects emissions to continue to decline in the upcoming reporting years.[49]

Q. How would implementation of a Super Emitter program assist the Companies in complying with New York's new climate legislation?

A. In light of the CLCPA, finding and abating super-emitters is low hanging fruit that benefits ratepayers and advances the goals of the Act as well as safety. Because earlier reductions provide enhanced benefit in comparison to later in the terms of the Act, I recommend that the Companies pursue this type of program and receive regulatory benefit under the Act for doing so. If the Companies do not employ such a program now, as EDF witness Ferit Ucar observes, detection of large methane emissions at a later date "could cast doubt on earlier-demonstrated achievements of GHG reductions and hamper the Companies' ability to have future emissions trajectories deemed to be in compliance with the requirements of the Act."[50]

VII. Leak Backlog Incentive

Q. Please explain the current incentives the Companies receive for eliminating the highest volume Type 3 leaks.

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[48] Exhibit __ (JF-2) at page 3.

[49] *Id.* at page 9.

[50] Ucar Direct Testimony at page 17.

A. The current leak backlog performance metrics and incentives were established in the Companies' 2016 rate cases. The metrics set annual leak backlog targets for workable leaks (Types 1, 2, and 2A) and total leaks and imposed Negative Revenue Adjustments for failure to meet targets.[51] The Companies are incentivized to repair Type 3 leaks and are permitted to earn a positive revenue adjustment of one basis point for each 50 such leaks eliminated, up to a maximum of five basis points per year.[52]

Q. Please explain the Companies' proposal for the leak backlog metrics and incentives.

A. The Companies' proposed metrics and incentives are presented in the tables below:

KEDNY Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	3,100*	5 basis points
	CY 21	2,950 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	2,800 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	2,650 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 150 reduction from CY19 target

[51] Gas Safety Panel at pages 53-54.

[52] *Id.* at page 55. The Companies must achieve both annual backlog reduction targets to qualify for this incentive. *Id.*

KEDLI Leak Metric	Target		Negative Revenue Adjustment
Workable Leaks (Type 1, 2, 2A)	Each CY	≤ 25	10 basis points
Total Leaks (Type 1, 2, 2A, 3)	CY 20	7,750*	5 basis points
	CY 21	7,000 less Total Leak Reduction Adjustment Factor from CY20	
	CY22	6,250 less Total Leak Reduction Adjustment Factor from CY21	
	CY23	5,500 less Total Leak Reduction Adjustment Factor from CY22	

* CY20 target based on 750 reduction from CY19 target

KEDNY & KEDLI # High Emitting Type 3 Leaks Repaired	Positive Revenue Adjustment	Total Leak Target Reduction Subsequent Year
0- 49	None	0
50-99	1 basis point	50
100-149	2 basis points	100
150-199	3 basis points	150
200-249	4 basis points	200
>=250	5 basis points	250

Q. Do the current incentive mechanisms incentivize the Companies to deploy modern technology such as advanced leak technology that would find and quantify more leaks?

A. No. As I noted above, advanced leak detection technology and analytics are typically able to find many more existing and ongoing leaks than traditional technologies. At the same time, the current incentive structure rewards the utility primarily for reporting a lower number of leaks in their backlog at the end of the year. For these reasons, National Grid is currently disincentivized to use ALD, which could be expected to find more leaks. At the time of the Companies' 2016 rate case, ALD was not an established technology. Since that time, ALD has become a scientifically and commercially established technology, and it has been adopted by other utilities to find leaks and quantify their emissions.

Q. Do you have a recommendation to resolve this disconnect?

A. Yes. The current incentive structure should be revisited, as it ultimately serves to discourage the Companies from finding more leaks on its system and serves as an obstacle to achieving New York State climate and methane abatement goals and policies.

Q. Please further explain how your recommendation would work in practice.

A. I recommend that the Companies first complete a methane leak survey of its entire service territory using advanced leak detection technology. Using the information gathered from this initial survey, the Companies could then establish a system-wide baseline leak flow rate. Next, a volumetric leak reduction target could be

established within the current leak abatement incentive. In order to receive its annual maximum positive incentive, the Companies would be required to achieve a 50% reduction over three to five years which, according to our data, would require abatement of approximately the largest ca. 20% of leaks in its non-hazardous leak inventory. This objective could be met through a combination of pipeline replacement and Type 3 leak repairs, allowing the utility to optimize its approach to leak mitigation through pipeline replacements when necessary.

Q. Please explain why you are recommending a three to five year range for the Companies to achieve a 50% reduction in emissions reductions.

A. I am recommending a range of three to five years for two reasons: first, the current, estimated rate of advanced leak detection technology surveying is such that a system-wide survey could take several months to complete; and second, in recognition of the time that may be required for the Companies to integrate leak flow rate information into its geospatial information systems, and prioritization systems for leak repair and pipeline replacement. I believe it is possible for the Companies to achieve 50% emissions reductions within a three-year period, but also that some flexibility in the timeline is warranted to ensure that integration of the technology and information derived from its use is well thought out and positions the Companies to make efficient prioritization decisions over the long term.

Q. How does the three to five year range impact your recommendations regarding the Companies' incentive structures?

A. I recommend that the Companies use their proposed incentive structure based on the current leak backlog until a system-wide leak survey using advanced leak detection technology has been completed, and the Companies have developed plans for prioritizing leak repairs and pipeline replacements based on leak flow rate, after taking safety into consideration. After the Companies have made their prioritization plans available to the Commission, the new incentive structure based on percentage emission reductions should come into effect.

Q. Does this conclude your testimony?

A. Yes.

Exhibit __ (JF-1):

Curriculum Vitae and List of Expert Testimony of Joseph von Fischer

Joseph C. von Fischer

Curriculum Vitae

August 2019

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Google Scholar: https://scholar.google.com/citations?user=49ebnuwAAAAJ
Twitter: @Profvonfischer
LinkedIn: https://www.linkedin.com/in/joseph-von-fischer-1bb3218/

Research Interests

Greenhouse gases, climate change, ecosystem ecology, natural gas, urban ecosystems, microbial ecology, terrestrial biogeochemistry, anthropogenic methane sources, plant-soil-microbe interactions, stable isotopes, technology-society interactions, paleoclimate, grasslands, Arctic tundra.

Current Appointments

Professor, Department of Biology, Colorado State University, August 2018 – present.

Associate Chair, Department of Biology, Colorado State University. August 2014 – present.

Advising Faculty, Graduate Degree Program in Ecology, Colorado State University. 2003 – present.

Affiliate Faculty, Energy Institute, Colorado State University. 2017 – present.

Affiliate Scientist, Natural Resource Ecology Laboratory, Colorado State University, 2009 – present.

Education

Postdoctoral Fellow, Department of Geosciences, Princeton University, Princeton, NJ. 2001 – 2003. Funded by NOAA Climate and Global Change postdoctoral program. Advisor: Michael Bender.

Ph.D., Ecology and Evolutionary Biology, Cornell University, Ithaca, NY, 2002. "Gross Methane Transformations and Methanogenic Microsites along Natural Soil Moisture Gradients." Participant in the graduate program: Biogeochemistry and Environmental Change. Advisor: Lars Hedin.

B.A., Biology, Augustana College, Sioux Falls, SD. 1992.

Professional Experience

Associate Professor with tenure, Department of Biology, Colorado State University, August 2012 – 2018.

Merkator Fellow, Friedrich Schiller University, Jena, Germany. Host Kirsten Küssel. November 2013 – July 2014.

Visiting Scientist in Residence (sabbatical), Max Planck Institute for Biogeochemistry, Jena, Germany. Host Susan Trumbore. August 2103 – July 2014.

Assistant Professor, Department of Biology, Colorado State University, August 2007 – July 2012.

Research Scientist I, Natural Resource Ecology Laboratory, Colorado State University. 2003 – 2009.

Special Assistant Professor, non-tenure track, Department of Biology, Colorado State University, Fort Collins, CO. August 2003 – August 2007.

Visiting Researcher, Santa Fe Institute, Santa Fe, NM. May – July 2003.

Research Technician Michigan State University at the W.K. Kellogg Biological Station, Hickory Corners, MI. With Lars Hedin. 1992 – 1993.

Publications *(*graduate student author,* †undergraduate student author)

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35. Ostrom N.E., L.O. Hedin, J.C. von Fischer and G.P. Robertson. (2002) “Nitrogen transformations at a soil-stream interface: a stable isotope approach.” *Ecological Applications* 12(4) 1027-1043.

36. Cleveland C.C., A.R. Townsend, D.S. Schimel, H. Fisher, R.W. Howarth, L.O. Hedin, S.S. Perakis, E.F. Latty, J.C. von Fischer, A. Elseroad, M.F. Wasson (1999) “Global patterns of

terrestrial biological nitrogen (N_2) fixation in natural ecosystems." *Global Biogeochemical Cycles* 13:623-645.

37. Hedin L.O., J.C. von Fischer, N.E. Ostrom, B.P. Kennedy, M.G. Brown, and G.P. Robertson (1998) "Thermodynamic constraints on nitrogen transformations and other biogeochemical processes at soil-stream interfaces." *Ecology* 79(2):684-703.

38. von Fischer J.C. and L.L Tieszen (1995) "Carbon isotope characterization of four tropical forests in Luquillo, Puerto Rico." *Biotropica* 27(2):138-148.

Grants and Fellowships

Current awards

Co-PI "Development and Validation of Low-Cost Natural Gas Leak Detection Sensors and Analytics for Drone-Based and Handheld Deployments." $112,673 to CSU from NSF STTR with Manish Gupta of Nikira Labs. 2018-2019

PI, "Interpretation of Street-Level Methane Concentrations" $980,243 from Environmental Defense Fund with 4 CSU co-PI's. 2012-2019.

Past awards

PI, "Colorado State University Participation in Controlled Gas Release Testing" from NYSEARCH (the research arm of a group of natural gas utilities) with CSU's Jay Ham. $75,000. 2015-2017.

PI, "Methane uptake by grassland soils: Biogeochemistry, microbial ecology and integrative modeling." $1,200,000 from NSF DEB Ecosystems with 4 co-PI's. 2011-2016.

Partner Investigator (equivalent to co-PI), "Methane uptake of forest soils." AU$270,000 from the Australian Research Council Discovery grant program to S. Arndt (PI) and S. Livesley, 2011 – 2014.

Co-PI, "Site-Specific Nitrous Oxide Isotope Analyzer for Measuring Bioremediation," $108,340 subcontract for DOE Phase II SBIR grant with Los Gatos Research. 2013-2014.

Senior Personnel, Shortgrass Steppe Long Term Ecological Research Site. 2005 – 2014. $940,000 per year from NSF DEB, with 15 co-investigators.

Co-PI, "Site-Specific Nitrous Oxide Isotope Analyzer for Measuring Bioremediation," $40,000 subcontract for DOE Phase I SBIR grant with Los Gatos Research. 2012.

Partner-Investigator "International collaboration for Methane Biogeochemistry and Microbial Ecology" from University of Melbourne Research Office. $14,576 with S. Arndt and S. Livesley (Univ. Melbourne). 2009

Visiting Research Fellowship, University of Melbourne, November 2008.

Co-PI, LTER program "Catalyzing cross-site comparisons of microbial diversity and function: A proposed LTER Workshop" with PI Tom Schmidt (Michigan State Univ.) and 4 others. $9,000 from LTER Network Office. 2007.

Co-Leader, CSU Course Redesign Proposal for General Ecology (BIO320 and LAND220), with Indy Burke (CSU). $20,000. 2007.

PI, "Additions of Physical and Biogeochemical Tracers to Wetland Soils Can Reveal Mechanisms Underlying the Productivity-Methane Emission Relationship." with co-PI's

Dennis Ojima (Colorado State Univ.) and M. Todd Walter (Cornell Univ.). $500,000 from NSF DEB Ecosystems. 2005 – 2009, plus 2 REU supplements.

Faculty Fellow Award, National Academies Education Program in the Life Sciences, for the 2004 National Academies Summer Institute in Biology.

USGS Contract, EROS Data Center. Technical support for Carbon in US and Africa Program to investigate controls of carbon isotope composition of soil organic matter in the US Great Plains. $50,000. 2004 – 2005.

NOAA Post-Doctoral Fellowship, NOAA Climate and Global Change Program, for "How does the productivity of the biosphere respond to changes in climate? Global scale evidence from periods of rapid climate change as recorded in ice cores." $84,750. 2001 – 2002.

NASA Graduate Fellowship in Earth System Science $66,000 over 3 years. 1999 – 2001.

NSF RTG Graduate Fellowship, NSF Research Training Group in Biogeochemistry and Environmental Change, Cornell University. 1997 – 1999.

NSF RTG Graduate Fellowship, NSF Research Training Group in Ecology, Michigan State University at the W.K. Kellogg Biological Station. 1993.

Service & Administration

University: Notable invited, appointed & elected

Invited member, Building Committee for new Biology Department Building ($73M), 2014 – 2017.

Invited member, Provost's office Strategic Plan Area Review Committee, university-level funding recommendations for Faculty and Staff, AY 2014-15.

Invited Member, Executive Committee, CSU School of Global Environmental Sustainability, Fall 2012 to present.

Elected Member, Executive Committee, CSU Graduate Degree Program in Ecology, Fall 2011 – 2013.

Director, Colorado Laboratory for Environmental Mass Spectrometry, Colorado State University. 2004 – 2009.

University: other service

Scholar, School of Global Environmental Sustainability, since 2012.

Search committee Member

Vice President for Enrollment and Access, Spring 2017.

Faculty in Biogeochemistry, Dept. of Ecosystem Science & Sustainability, Fall 2018.

Chair, Search committee for faculty position, Science Education, Dept. of Biology, Fall 2014.

Chair, Search Committee for Biology Staff positions:

Undergraduate coordinator, Fall 2018

Academic Success Coordinator: Spring 2012, Fall 2012, Fall 2014, Spring 2017, Fall 2017 (2 positions).

Program Aide, Dept. of Biology, Spring 2015

Invited Member, Shortgrass Steppe Research and Interpretive Center Users Committee. 2011 – 2018.

Invited member, College of Natural Science Awards Committee, April 2013.

Invited Participant, CSU Institute for the Built Environment LENSES (Living Environments in Natural Social & Economic Systems) workshop for "green building" materials. May 19, 2010.

Undergraduate Affairs Committee, Department of Biology, Colorado State University,
Member: 2007 – 2009, 2011 – 2013
Chair: 2016 – present.

Co-organizer, NREL seminar series "Using stable isotopes to understand ecological processes and global change." Included 13 speakers, Fall 2009.

Science Steering Committee, Shortgrass Steppe Long-Term Ecological Research Site, Colorado State University, January 2009 – 2012.

Judge graduate and undergraduate presentations, annually since 2004:
Front Range Student Ecology Symposium,
CSU Celebrate Undergraduate Research and Creativity.

Co-Director and co-founder, Student Ecology Research Program. An enrichment program to improve the educational experience of undergraduate employee researchers through bi-weekly seminars and discussions. 2004 – 2010.

Member, Executive Steering Committee, Cornell Boyce Thompson Stable Isotope Lab, Cornell University 1998 – 2001.

University: Notable initiatives

Major renovations to first floor of Anatomy-Zoology building, making the space more attractive and useful for undergraduate students. Spring 2012. $85,000 from CSU College of Natural Sciences.

Outside Service

Co-organizer for the "CH4 Connections: The Methane Emissions Conference" hosted at CSU September 18-19, 2019 by the Gas Technology Institute and the CSU Energy Institute. Approximately 120 attendees from academia and industry. http://www.gastechnology.org/CH4/Pages/default.aspx

Co-organizer for the "CH4 Connections: The Methane Emissions Conference" hosted at CSU September 24-25, 2018 by the Gas Technology Institute and the CSU Energy Institute. Approximately 120 attendees from academia and industry. http://www.gastechnology.org/CH4/Pages/default.aspx

Co-organizer for the "CH4 Connections: The Methane Emissions Conference" hosted at CSU December 12-13, 2017 by the Gas Technology Institute and the CSU Energy Institute. Approximately 120 attendees from academia and industry. http://www.gastechnology.org/CH4/Pages/default.aspx

Invited Technical Advisory Committee member, Center for Methane Research at the Gas Technology Institute. 2017-present.

Invited expert review panelist, Google, April 2015.

Special Session Convener and chair, 14 presentations in the session "Processes and Mechanisms Controlling Soil Methane Oxidation" at the Annual Meeting of the American Geophysical Union, San Francisco, CA. with colleagues from the Univ. Melbourne, Australia: Benedikt Fest, Stefan Arndt and Philipp Nauer. December 2015.

Invited participant, Department of Energy ARPA-E (Advanced Research Projects Agency – Energy) Workshop on Natural Gas Methane. April 2013.

Invited Member, External Advisory Board for the NSF-funded Sevilleta Long Term Ecological Research project, University of New Mexico. 2013.

Associate Editor, Journal of Geophysical Research Biogeosciences, January 2010 – 2012.

Manuscript Reviews (incomplete list): *Annals of Botany, Biogeochemistry, Ecological Applications, Ecosystems, Global Change Biology, FEMS Microbiology Ecology, Geochimica et Cosmocimica Acta, Global Biogeochemical Cycles, ISME, Journal of Geophysical Research Biogeosciences, New Phytologist, Oecologia, PNAS, Soil Biology & Biochemistry, Teaching Issues & Experiments in Ecology.*

Review Panelist: NSF LTER Program, Washington D.C., April 2012

Review Panelist: DOE Terrestrial Ecology program, Washington D.C., January 2012.

Review Panelist: EPA STAR Graduate Fellowships, Washington D.C., March 2011.

Ad hoc Proposal Reviews: National Science Foundation, NASA, European Science Foundation, National Environment Research Council (UK).

Review Panelist, NASA Terrestrial Ecology Program, Washington, D.C., March 2009.

Invited Participant, National Ecological Observatory Network (NEON) Microbial Ecology planning meeting, Baton Rouge, LA. February 2008.

Invited Participant, National Center for Atmospheric Research, Junior Faculty Forum "Downscaling Climate Change: Extreme Events, Regional Impacts, and Ecosystems." Boulder, CO. 2005.

Invited Participant, North American Carbon Program special workshop on the global methane cycle, Durham, NH. 2002.

Member, Ecological Society of America (1998 – present).

Member, American Geophysical Union (1999 – present).

Member, European Geophysical Union (2013 – present).

Member, American Chemical Society (2017 – present).

Teaching

Instructor for “Ecosystems Ecology” (ECOL 610, 3cr.) Colorado State University. Fall semesters 2010 to present.

Instructor for "Ecology" (Life 320, 3cr.), Colorado State University. Fall semesters from 2003 to 2009.

Co-Instructor for "Global Change Ecology, Impacts and Mitigation" (BZ/NR 380, 3cr.), Colorado State University. Spring semesters from 2008 to present.

Co-Instructor for “Biology of Organisms” (LIFE103, 3cr.), Colorado State University, Spring semester from 2009 to present.

Instructor for Special Topics in Ecology (ECOL 592, 1cr.) “Microbial ecology and biogeochemistry.” Spring 2010.

Instructor for Special Topics in Ecology (EY592, 1cr.) “The struggle to unify human-scale and microbial ecology.” Spring 2004.

Co-founder and co-instructor “Soil Summer Institute” at Colorado State University. Two-week non-credit summer course on soil ecology and biogeochemistry. Annually since 2005. https://soilinstitute.nrel.colostate.edu

Teaching awards

Awarded "Excellence in Graduate Teaching and Mentoring" for teaching of ECOL 610 Ecosystem Ecology, CSU Dept. of Biology, Spring 2011.

Awarded "Distinguished Member" CSU National Society of Collegiate Scholars, 2009

Nominated for "Best Teacher Award" CSU Alumni Association, 2008.

Nominated "CSU Honors Professor" CSU Undergraduate Honors Program, 2005.

Course redesign: Development of active learning modules for CSU Ecology courses (BIO 320 and LAND/SOCR 220). With Indy Burke, trained teams of graduate students and faculty to design active learning modules for ecology. These 10 modules are now available and are being used in CSU Ecology courses. 2007-2008.

Guest lectures at Colorado State University: Numerous

Advising & Supervision

Post-doctoral associates

Akihiro Koyama, 2011 – 2015. Now Assistant Professor, Dept. of Forest Science, Michigan State University.

Nels Johnson, 2011 – 2015. Now Mathematical Statistician, U.S. Forest Service, PSW Research station.

Sam Chamberlain, 2016. Now post doctoral researcher, Univ. California Berkeley.

Zachary Weller, 2017 – 2018. Now Assistant Professor, Dept. of Statistics, Colorado State University.

Graduate Advisees

Stephanie Owens (M.S., Graduate Degree Program in Ecology), completed 2008

Craig Judd (M.S., Graduate Degree Program in Ecology), completed 2011

Sam Dunn (Ph.D., Graduate Degree Program in Ecology), completed 2016

Paul Brewer (Ph.D., Graduate Degree Program in Ecology), completed 2016

Charlotte Alster (Ph.D., Graduate Degree Program in Ecology), completed 2017

Holly Martin (Master of Natural Science Education), completed 2019

Emily Stuchiner (Ph.D., Graduate Degree Program in Ecology), current

Graduate Committee, Colorado State University

Completed

Douglas Van Hoewyk (M.S., Biology) completed 2006

Eliana Bontti (Ph.D., Graduate Degree Program in Ecology) completed 2008

Mark Gathany (Ph.D., Graduate Degree Program in Ecology) completed 2008

Jeff Dereume (M.S., Geosciences) completed 2010

Sarah Evans (Ph.D., Graduate Degree Program in Ecology) completed 2012

Parker Kraus (M.S., Atmospheric Sciences) completed 2012
Stormy Lindblom (Ph.D., Biology) completed 2012
Brooke Osborne (M.S., Graduate Degree Program in Ecology) completed 2012
Warren Newby (M.S., Geosciences), completed 2012
Anna Waido (Ph.D., Education), completed 2013
Caroline Melle (M.S., Graduate Degree Program in Ecology), completed 2013
Damien Borcovsky (M.S., Geosciences), completed 2013
Syed Shah (Ph.D., Graduate Degree Program in Ecology), completed 2013
Andrew Horton (M.S., Graduate Degree Program in Ecology) completed 2013
David Millar (Ph.D., Graduate Degree Program in Ecology), completed 2014
David Hoover (Ph.D., Graduate Degree Program in Ecology), completed 2014
Will Horner (M.S., Geoscience), completed 2015
Catherine Compton (M.S., Geosciences), completed 2015
Kevin Wilcox (Ph.D., Graduate Degree Program in Ecology), completed 2015
Cameron Toloee (M.S., Computer Science), completed 2015
Judy Daniels (Ph.D., Soil & Crop Sciences), completed 2016
Alyssa Cochran (Ph.D., Botany), completed 2016
Steven Mantell (M.S., Civil & Environmental Engineering), completed 2016
Johnson Charles Arulswamy (M.S., Computer Science), completed 2016
Dana Flett, (M.S., Graduate Degree Program in Ecology), completed 2018
Cheryl Bowker (Ph.D., Graduate Degree Program in Ecology), completed 2018
Lydia Baldwin (M.S., Graduate Degree Program in Ecology), completed 2018
Benjamin Luck (M.S., Engineering), completed 2018
Jason Reynolds (Ph.D., Botany), completed 2019
David Markman (Ph.D., Graduate Degree Program in Ecology), completed 2019

Current
Kathryn Schuller (M.S., Geoscience)
Nora Flynn (Ph.D., Soil & Crop Sciences)
Kim Hiyoto (PhD, Chemistry)
Kelechi Nwaokorie (M.S., Civil & Environmental Engineering)
Sajjad Karimi (Ph.D., Civil and Environmental Engineering)
Danielle Lin Hunter (Ph.D., Graduate Degree Program in Ecology)
Mallorie Hsieh (Master of Natural Science Education)
Sara Knighton (Master of Natural Science Education)

Joel Spansel (M.S. Geosciences)

Leena Vilonen (Ph.D., Graduate Degree Program in Ecology)

International Graduate Committees

Benedikt Fest, (Ph.D., Plant Biology) University of Melbourne, Australia. Completed 2013

Saadat Malghani, (Ph.D., Biogeochemistry), Frierich, Schiller University, Jena, Germany. Completed 2014

Tobias Jochum (Ph.D., Chemical Physics), Frierich Schiller University, Jena, Germany. Completed 2017.

Martin Nowak, (Ph.D., Biogeochemistry), Frierich Schiller University, Jena, Germany. Current.

Undergraduate Honors Theses

Hannah Shadis (advisor) 2005

Natalie Akagi (committee member) 2006

Lilli Welch (committee member) 2007

Dyllon Martini (committee member) 2008

Elin Franzen (committee member) 2008

Stephen Meyer (advisor) 2009

Elizabeth Moses (advisor) 2011

Olivia Shoup (committee member) 2012

Jamie Bunker (advisor) 2013

Jenna Tustin (advisor) 2013

Undergraduate Lab Researchers (n=31 since 2004)

Chris Anderson (2009), Liesel Brutlag† (2009), Jamie Bunker (2010-2011), Ariel Carmichael* (2011- 2012), Mariah Coler† (2011), Sara Cummings* (2005), Tanner Hill* (2005 - 2009), Lisa Fernholz* (2011- 2013), Nicholas Framsted (2015 - 2016), Claire Griebenow* (2011- 2013), David Hoesly (2005-2006), Rachel Jorgensen (2011- 2012), Jeff Kimbrel* (2004-2006), Paige Koutlas (2012 – 2014), Jason Lutz (2006), Elizabeth Mortensen* (2008-2010), Darcy MacPhee (2005), Claire Ojima† (2004), Deanna Olson (2012 - 2014), Shreya Pokharel (2011 - 2013), Ashley Quick Bear (2012 - 2013), Evan Rosenlieb* (2011 - 2013), Richard Siller* (2005), Jacqueline Sitko (2011- 2014), Matthew Stout (2011), Kelsey Thompson (2011 - 2014), Lucas Thompson (2012- 2014), Mike Weltzer* (2006), Sere Williams (2006), Breana Wolfe (2005), Duck Keun Yang (2015 – 2016)*.

*went on to graduate school (n=11), †went on to professional school (n=3)

Research technical staff

Anita Kear, 2004 – 2006

Evan Rosenlieb, 2011 – 2014

Jaqueline Sitko, 2013 – 2014

Claire Greibenow, 2013 – 2014

Adam Gaylord, 2014 – 2015

Duck Keun Yang, 2016 – present

Invited Scientific Presentations *(*international)*

2018 "New ways to find natural gas leaks in cities: physics of development and the human story of adoption." Department of Physics seminar, Colorado State University.

2018* "New technologies help reduce urban natural gas emissions." Keynote talk at the Global Methane Forum, April 16, 2018, Toronto, Canada.

2017 "Vehicle-based methane sensing for finding natural gas leaks and prioritizing repairs" Western Regional Gas Conference, San Diego, CA.

2015 "Rapid Identification of Location and Magnitude of Urban Natural Gas Leaks" 2015 International Emission Inventory Conference "Air Quality Challenges: Tackling the Changing Face of Emissions"

2015 "Methane uptake in Great Plains grassland soils: does microbial community composition matter for ecosystem function?" Graduate student invited seminar for University of Wyoming, Department of Botany.

2014* "Methane oxidation in upland soils: a model system for synthesis in biogeochemistry and microbial ecology." Seminar for Department of Soil Ecology, University Freiburg, Germany.

2014* "Isotopic studies of trace gas biogeochemistry" Seminar for Bio-Geo seminar, Institute for Geosciences, Friedrich Schiller University, Jena, Germany.

2013* "When does knowledge of microbial community composition improve biogeochemical models?" Seminar for Department of Geology, Stockholm University, Sweden.

2013* "Physical, biological and social controls of methane emissions: an overview of my interests & invitation to new collaborations." Colloquium seminar, Max Planck Institute for Biogeochemistry, Jena, Germany.

2013 "Mechanisms controlling terrestrial and aquatic methane release from the Arctic Coastal Tundra at Barrow, Alaska." Boston University Biogeochemistry Seminar Series, Boston, MA.

2012 "Interpretation of street-level methane concentrations." Robertson Scholar Program, Duke University.

2012 "When does knowledge of microbial community composition improve ecosystem models?" Biology Department seminar, University of Northern Colorado, Greely.

2012 "Boil yourself alive in a test tube: using the scientific method to make career choices." CSU Greek Month of the Scholar academic achievement event.

2012* "When does knowledge of microbial community composition improve biogeochemical models?" Seminar for Max Planck Institute for Biogeochemistry, Jena, Germany.

2011 "Boil yourself alive in a test tube: using the scientific method to make career choices." CSU College of Natural Sciences Mentoring Program.

2011 "Climate Change: The Physical Scientific Basis" for the Hubert H. Humphrey International Fellows program, hosted by the CSU Office of International Programs.

2011 "Plant allometry outperforms biogeochemical predictors of methane emission from the Arctic Coastal Tundra at Barrow, Alaska." Dept. of Botany seminar, University of Wyoming.

2010 "Soil microbes that consume greenhouse gases: do they use spoons or forks?" Biology Department Seminar, Augustana College, Sioux Falls, SD.

2010 "When does knowledge of microbial community composition improve biogeochemical models?" Seminar for Program in Biogeochemistry and Environmental Biocomplexity, Cornell University, Ithaca, NY.

2010 "Mechanisms controlling terrestrial and aquatic methane release from the Arctic Coastal Tundra at Barrow, Alaska." Dept. of Geography colloquium, University of California, Berkeley, CA.

2009 "Mechanisms controlling terrestrial and aquatic methane release from the Arctic Coastal Tundra at Barrow, Alaska." Institute for Arctic and Alpine Research (INSTAAR), University of Colorado, Boulder, CO.

2008* "Methane oxidation in upland soils: peeling away physics to see the ecology" University of Melbourne at Creswick, Creswick, NSW, Australia.

2008 "Methane oxidation in upland soils: putting the *bio* in biogeochemistry," Michigan State University, Kellogg Biological Station, Hickory Corners, MI.

2007 "Ecosystem-scale controls on methane emissions" NOAA Climate Monitoring & Diagnostics Laboratory, Boulder, CO.

2006 "Gas transport vs. methane production: What drives methane emission rates in *Typha* wetlands?" Special session on methane biogeochemistry. Annual Meeting of the American Geophysical Union.

2005 "Evidence and implications of habitat diversity for soil microbes from upland and wetland soils." Dept. of Geology, Baylor University, Waco, TX.

2005 "5 things that everyone should know about global climate change." Honors program, Colorado State University.

2005 "How much bio in biogeochemistry? Physical vs. biological controls of soil methane fluxes." Dept. of Earth System Sciences at the University of California, Irvine.

2004 "Linking microbial community properties to ecosystem function: methane consumption on the Shortgrass Steppe" Shortgrass Steppe LTER seminar series, Colorado State University.

2004 "Less than 1% of soil volume is anoxic, but it matters for upland ecosystem biogeochemistry." Dept. of Soil and Crop Sciences, Colorado State University.

2004 "Less than 1% of soil volume is anoxic, but it matters for upland ecosystem biogeochemistry." Institute for Arctic and Alpine Research (INSTAAR), University of Colorado, Boulder.

2004 "Soil heterogeneity: Evidence and implications of anoxic microzones in oxic soils." Dept. of Soil Science, University of Wisconsin, Madison.

2003 "Diverse perspectives on climate change: soil anoxic microsites and glacial changes in global photosynthesis." Santa Fe Institute, Santa Fe, NM.

2002 "Microsites, Ecosystems and Global Change" Natural Resources Ecology Laboratory, Colorado State University.

2002* "The triple isotopic composition of molecular oxygen reveals past rates of primary production." Biogeomon International Symposium on Ecosystem Behaviour, Reading, UK.

Conference presentations *(not including secondary authorships)*

2018 "A 15-city survey of urban natural gas leaks." Oral presentation at the Annual Meeting of the American Geophysical Union, Washington, D.C.

2017 "Vehicle-based Methane Mapping Helps Find Natural Gas Leaks and Prioritize Leak Repairs." Oral presentation at the Annual Meeting of the American Geophysical Union, New Orleans, LA. Thursday, Dec. 14, 2017

2016 "Mobile natural gas leak surveys indicate that two utilities have high false negative rates" Oral presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA. Monday, Dec. 12, 2016

2015 von Fischer, J.C. and J. Ham. "Rapid Identification of Location and Magnitude of Urban Natural Gas Leaks" EPA 2015 International Emission Inventory Conference "Air Quality Challenges: Tackling the Changing Face of Emissions" San Diego, California - April 13 - 16, 2015

2015 von Fischer, J.C., A. Koyama, N.G. Johnson and C.T. Webb "A hierarchical examination of methane uptake: field patterns, lab physiology, community composition and biogeography" Poster presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2013 von Fischer, Joseph Jessica Salo, Claire Griebenow, Linde Bischak, Daniel Cooley, Jay Ham, and Russ Schumacher. "Development of an algorithm to meaningfully interpret patterns in street-level methane concentrations" Solicited oral presentation at the annual meeting of the European Geophysical Union, Vienna Austria. April 2013

2012 Jorgensen, R.J., A. Koyama, J.C. von Fischer and M. Gupta. "Exploring potentials of a Los Gatos Research isotopic N2O analyzer capable of real time measurements." Poster presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2010 von Fischer, J.C., A. Angert, D. Augustine, C. Brown, F. Dijkstra, J. Derner, R. Hufbauer, N. Fierer, D. Milchunas, J. Moore, H. Steltzer, M. Wallenstein. "Timescales of biogeochemical and organismal responses to individual precipitation events." Organized oral session at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2009 von Fischer, J.C. "Criteria to demonstrate that microbial community composition is important for biogeochemical function." Organized oral session at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2009 von Fischer, J.C., C.R. Judd and C.T. Webb “The intersection of methanotroph community ecology and methane biogeochemistry.” Organized oral session at the Ecological Society of America, Albuquerque, NM.

2008 von Fischer, J.C. and R. Rhew “Methane emission rates from the Arctic coastal tundra at Barrw are log-normally distributed: Is this a tail that wags climate?” Oral presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2007 von Fischer, J.C., G. Ames, R. Rhew and W.C. Oechel "Methane emission rates from the Arctic coastal tundra at Barrow: temporal and spatial variability and response to an experimental carbon addition." Poster presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2006 von Fischer, J.C. and G. Butters “Physics or Biology? What controls variability in methane flux rates on the Shortgrass Steppe LTER site?” Poster presentation at the LTER All Scientists Meeting, Estes Park, CO.

2006 von Fischer, J.C. "Biology is more important than diffusion for regulating methane fluxes on the Shortgrass Steppe LTER." Oral presentation at the Annual Meeting of the Ecological Society of America.

2005 von Fischer, J.C., “Diffusion or biology? What drives variability in methane consumption of upland soils?” Poster presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2004 von Fischer, J.C. and L.L. Tieszen “Soil Organic Matter $\delta^{13}C$ across the Great Plains grasslands: both rainfall and temperature control C_3 vs. C_4 productivity. Oral presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2004 von Fischer, J.C. and R.J. Thelwell “Rates and importance of methane production in oxic soils.” Oral presentation at the Annual Meeting of the Ecological Society of America, Portland OR.

2004 von Fischer, J.C., M.L. Bender and T. Blunier “A record of global photosynthesis to 150ka derived from the triple isotopes of O_2 in polar ice cores.” Oral presentation at the Annual Meeting of the American Geophysical Union, Montreal, Quebec, Canada.

2002 von Fischer, J.C., M.L. Bender and T. Blunier “Triple isotope composition of O_2 and the record of global O_2 production.” Poster presentation at the Annual Meeting of the American Geophysical Union, San Francisco, CA.

2002 von Fischer, J.C., and L.O. Hedin “Do soil microsites drive methane flux?” Oral presentation at the Annual Meeting of the Ecological Society of America, Tucson, AZ.

Technical briefings

2015 von Fischer, J.C. “State of science on natural gas leak quantification.” California Public Utility Commission, deliberation of SB1371. San Francisco, CA. 27 October 2015. Led to adoption of mobile methane mapping as a “best practice” for leak detection. http://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M159/K670/159670815.PDF

Expert testimony

2016 von Fischer, J.C. "Direct testimony of Joe von Fischer on behalf of the Citizens Utility Board" State of Illinois – Illinois Commerce Commission vs. People's Light and Coke Company. Investigation into the cost, scope, schedule and other issues related to the Peoples Gas Light and Coke Company's natural gas system modernization program and the establishment of Program policies and practices pursuant to Sections 8-501; Case No. 16-0376. Written testimony submitted 11 October, 2016. Oral cross examination November 2016.

2016 von Fisher, J.C. "Direct Testimony of Joe von Fischer on behalf of Environmental Defense Fund" State of New York – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Gas Service; Case No. 16-G-0061. Direct testimony submitted May 27, 2016. Rebuttal testimony submitted June 17, 2016.

2016 von Fisher, J.C. "Direct Testimony of Joseph von Fischer" State of New York – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of the Brooklyn Union Gas Company for Gas Service; Case No. 16-G-0059. Direct testimony submitted May 20, 2016.

2016 von Fisher, J.C. "Direct Testimony of Joseph von Fischer" State of New York – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation for Gas Service; Case No. 16-G-0058. Direct testimony submitted May 20, 2016.

Outreach, public presentations & webinars

2019 "Prioritizing repairs and finding more natural gas leaks using advanced leak detection technologies." Downstream Natural Gas working group meeting at Xcel Colorado headquarters, Denver, CO. Host: Brian Jones, MJ Bradley & Associates.

2019 "Technology innovations bring win-win opportunities for business and climate" Invited presentation to the Energy Fellows Institute of the Colorado Cleantech Industries Association, Fort Collins, CO. May 17, 2019. Host: Mary Austin.

2018 Briefing webinar for the investor advocacy group *As You Sow*. Webinar was associated with report "The End of the Line" on natural gas leakage from local distribution utilities that I helped to review. 12 July 2018. https://www.asyousow.org/reports/2018/7/10/end-of-the-line

2018 "Innovations in Oil & Gas: Methane Leak Detection" Invited talk to Colorado Cleantech Industries Association "Energy Connections" monthly breakfast meeting. Denver, CO July 17, 2018

2018 "Technology innovations bring win-win opportunities for business and climate" Invited presentation to the Energy Fellows Institute of the Colorado Cleantech Industries Association, Fort Collins, CO. May 18, 2018. Host Mary Austin.

2018 "Methane and Cities" Metropolitan Mayors Caucus (Greater Chicago Metro Area), Environmental Committee meeting on Smart Cities. Chicago City Hall. 20 March 2018.

2017 "Technology innovations bring win-win opportunities for business and climate" Invited presentation to the Energy Fellows Institute of the Colorado Cleantech Industries Association, Fort Collins, CO. May 12, 2017. Host Mary Austin.

2017 "Greenhouse gases and climate change" Invited presentation to Biology classes at Brookings High School, Brookings, SD. 5 May 2017.

2017 "Greenhouse gases and climate change" Invited presentation to AP Science classes at Fort Collins High School. 21 April 2017. Host Holly Martin.

2017 "Hunting down natural gas leaks with Google Street View cars" Invited TEDxCSU talk at Colorado State University. 4 March 2017. https://www.youtube.com/watch?v=wCQtWhFf5I0

2016 Invited panelist at the CSU Energy Transition Symposium, Colorado State University. 29 September 2016.

2013 Training workshop on Methane Leakage from the Natural Gas Supply Chain for the Duke University Robertson Scholar Program. Colorado State University, January 16-20. Co-hosted with Steve Hamburg, Chief Scientist, Environmental Defense Fund.

2011 Updated the Wikipedia entry on *Atmospheric Methane* and created a new entry for *Wetland Methane Emissions* with CSU undergraduate student Olivia Shoup, as part of her CO301 class on Technical Writing. Edits are as user "OSJVFCO301" on 27 February 2011.
http://en.wikipedia.org/wiki/Atmospheric_methane
http://en.wikipedia.org/wiki/Wetland_Methane_Emissions

Selected popular press and other coverage

2014-18 Extensive popular press coverage for work with the Environmental Defense Fund to map natural gas leaks using Google Street View cars. Notable pieces include:

Featured in USA Today story "A gas leak every mile: Google Street View cars find problems that utilities don't."
https://www.usatoday.com/story/news/nation/2018/11/02/google-cars-sniff-natural-gas-leaks-old-utility-pipelines-explosions/1683173002/

Live radio appearance on Fort Collins, CO radio station KCSU Rocky Mountain News radio hour. 13 February 2018. http://kcsufm.com/2018/02/rocky-mountain-review-february-13-2018/

Featured in news story for US State Department's Voice of America. 3 August 2017
Text: https://www.voanews.com/a/mapping-methane-leaks/3970924.html
Video: https://www.voanews.com/embed/player/0/3970782.html

Featured in video news story for China Global Television Network, 12 June 2017
https://www.youtube.com/watch?v=bQIIIZs8Qh0&feature=youtu.be

Live radio interview on San Francisco, CA station KCBS. 4 minutes. 24 March 2017.

Live video interview on Canada's Business News Network. April 5, 2017.
http://www.bnn.ca/video/sniff-sniff-google-cars-map-methane-leaks~1093803

Featured in *Washington Post* article "These Google Street View cars just got a new job – spotting harmful methane leaks in big cities" March 22, 2017.
https://www.washingtonpost.com/news/energy-environment/wp/2017/03/22/these-google-street-view-cars-just-got-a-new-job-spotting-harmful-methane-leaks-in-big-cities/?utm_term=.70130c18ff01

Featured in a front-page story "Mapping the city's natural gas leaks — via lasers." in the *Pittsburgh Post Gazette* 14 November 2016. http://powersource.post-

gazette.com/powersource/policy-powersource/2016/11/14/Mapping-the-city-s-natural-gas-leaks-via-lasers/stories/201611130134

Quoted in "There's a new way to replace old gas mains—but not in Chicago" Crain's Chicago Business October 22, 2016 http://www.chicagobusiness.com/article/20161022/ISSUE01/310229991/theres-a-new-way-to-replace-old-gas-mains-but-not-in-chicago

2013 Quoted in article about mapping methane leakage from cities "Uprising" in *Matter*, February 2013. https://www.readmatter.com/uprising-preview/

2012 Quoted in article about the role of methane in rapid global warming. "Global Warming: Faster than Expected?" *Scientific American*, November 2012.

2010 Presentation at 2009 Annual Meeting of the Ecological Society of America received expanded coverage in a special report:
Yuko Hasegawa, John E. Hobbie (2010) Organized Oral Session 27. Application of Molecular Knowledge of Microbes to Studies of Ecological Processes: Why the Integration is so Challenging. *Bulletin of the Ecological Society of America*: Vol. 91, No. 1, pp. 68-79. doi:10.1890/0012-9623-91.1.68

2010 Quoted in article about new instruments that allow scientists to more easily measure greenhouse gas emissions. "Greenhouse Gases: Who's Cheating?" in *Bloomberg Business Wee*k, 11January 2010, p18-19. http://www.businessweek.com/magazine/content/10_02/b4162018066076.htm?chan=magazine+channel_new+business

2009 Research photographs and discussions with artist John Kelly included in his book Flood Cycle: Notes from a Changing Planet. Signal Books, Oxford, U.K. 2009

2009 Figures from von Fischer et al. (2009) featured in "Image Highlights" section on the website of *Journal of Geophysical Research Biogeosciences*.

2009 Mentioned in article "Ice Capades" *Outside Magazine*, January 2009 issue. http://outside.away.com/outside/destinations/200901/arctic-ocean-global-warming-4.html

2008 Interview with Monkia Seynsche on Deutschlandradio (German Public Radio) February 13 discussing research on methane emissions from the Arctic. http://www.dradio.de/dlf/sendungen/forschak/738992/

Exhibit __ (JF-2):

Pacific Gas and Electric Company Natural Gas Leakage Abatement Report, California Public Utilities Commission Rulemaking 15-01-008 (June 17, 2019)

Pacific Gas and Electric Company

Natural Gas Leakage Abatement Report

In partial fulfillment of

Rulemaking (R.) 15-01-008 to Adopt Rules and Procedures Governing Commission Regulated Natural Gas Pipelines and Facilities to Reduce Natural Gas Leaks Consistent with Senate Bill 1371, Leno.

And in Response to Data Request
Pacific Gas and Electric Company R.15-01-008
2019 Annual Report

By:

Date: 6/17/19

Introduction

Pacific Gas and Electric Company (PG&E) is providing the following data[1] and explanations to comply with Senate Bill (SB) 1371 (Leno, 2014), Section 2, Article 3, and to respond to the California Public Utilities Commission's (CPUC) data request in Order Instituting Rulemaking (OIR) 15-01-008.[2]

Pursuant to SB 1371, Leno – Natural gas: leakage abatement, the CPUC requests that the following information be transmitted to the CPUC and the California Air Resources Board (CARB):

(1) A summary of changes to utility leak and emission management practices from January 1st, 2018 to December 31st, 2018. The report must include a detailed summary of changes, including the reasoning behind each change and an explanation of how each change will reduce methane leaks and emissions.

Response:

In 2018, PG&E made several programmatic changes to its leak and emission management practices in an effort to reduce methane emissions. Most notably, PG&E increased the distribution pipeline survey frequency from a four-year survey cycle to a three-year survey cycle, consistent with Best Practice 15 (BP-15). In order to increase survey frequency, PG&E employed greater numbers of leak survey personnel and continued to invest in the vehicle-based leak detection system, Picarro. This change in survey frequency is helping reduce methane emissions through detection and subsequent repair of leaks on a more expedited timeline.

Driven by Best Practice 16 (BP-16), PG&E also implemented its first year of the annual Special Leak Surveys. To meet the scope of BP-16, PG&E conducted a special leak survey on selected vintages of distribution pipes. The material focus of the special leak survey was pre-1940 steel and pre-1975 Aldyl-A vintages, both shown to have higher leak rates. The survey targeted pipe segments of these vintages with leak histories and resulted in the identification and repair of more leaks and thus reduction

[1] As described in Data Request Pacific Gas and Electric Company R15-01-008 2019 Annual Report.

[2] Data request provided by the CPUC Safety and Enforcement Division in a letter dated April 5, 2019.

of methane emissions. PG&E is assessing the results of the special leak survey targeting the select vintage pipes and plans to continue the survey in 2019.

Additionally, PG&E conducted its first "Super-Emitter" leak survey starting in August 2018. The "Super-Emitter" program, approved as a part of PG&E's Compliance plan for Best Practice 21 (BP-21), is an application of recently-developed methods that can determine the approximate flow rate of a leak accurately enough to categorize the leak as larger than a threshold value. Through this annual dedicated survey of its full territory, PG&E targets the largest leaks and aims to repair them promptly no matter the leak grade. In 2018, PG&E found 220 "Super-Emitter" leaks and repaired 128 of them.[3] PG&E plans to repair the remainder of these "Super-Emitter" leaks in 2019 and plans to continue conducting the "Super-Emitter" leak survey and repair program in 2019.

In 2018, PG&E also focused on repairing aging (i.e., discovered prior to 2018) grade 3 leaks. PG&E repaired approximately 2,000 pre-2018 belowground distribution leaks and approximately 4,100 pre-2018 aboveground distribution leaks during 2018. This is an increase as compared to the approximately 370 pre-2017 belowground and 3,700 pre-2017 aboveground distribution leaks PG&E repaired in 2017. Total aboveground grade 3 leak repairs increased from approximately 16,000 in 2017 to approximately 23,000 in 2018. Total belowground grade 3 leak repairs increased from approximately 400 in 2017 to approximately 2,500 in 2018. Note that belowground leaks on services and mains and aboveground leaks on mains are reflected in Appendix 4 while aboveground leaks on services are reflected in Appendix 6 since they pertain to the Meter Set Assemblies. The majority of these grade 3 leak repairs are presented in Appendix 6. Due to the population-based emission factor methodology currently employed in the Appendix 6 reporting template, the decrease in emissions from PG&E's efforts to repair grade 3 leaks is not reflected in the reporting submittal for aboveground service leaks.

PG&E also made improvements to its transmission leak detection practices in 2018. Specifically, PG&E conducted the Class 3 pipelines survey using the aerial transmission leak detection technology and

[3] The planned 2018 "Super-Emitter" survey was completed in early 2019.

implemented increased use of more sensitive leak detection equipment (Detecto Pak-Infrared) by the leak survey team.

At its transmission compressor stations and underground storage facilities, PG&E conducted comprehensive component-by-component leak surveys for each quarter in 2018 in compliance with the California Air Resources Board Greenhouse Gas Emission Standard for Crude Oil and Natural Gas Facilities (herein referred to as the CARB Oil and Gas Rule or COGR), effective January 1, 2018.[4] The surveys resulted in quick identification and repair of more leaks and thus reduction of methane emissions.

In 2018, PG&E expanded its use of cross compression to reduce blowdown emissions from local transmission assets. Cross compression efforts were previously targeted at transmission backbone assets, but PG&E is actively spreading awareness of this practice along with the benefits of drafting and bundling of work in order to reduce emissions from all sectors of PG&E's gas transmission division. These practices led to less gas released from blowdowns in 2018 for both local transmission and transmission backbone in spite of increased maintenance activity. PG&E also worked to further define roles and responsibilities for the development of standard procedures to identify and execute methane mitigation measures on transmission maintenance projects (drafting, cross-compression, bundling, and flaring) in application of Best Practice 5 and Best Practice 7. These procedures will support the identification and evaluation of blowdown reduction opportunities and their implementation across the organization.

Finally, PG&E has continued investing in research to improve the leak detection process in application of Best Practice 17 and Best Practice 19. In 2018, PG&E conducted pilot testing of UV-based leak detection drones at compressor stations and transmission regulation stations. In addition to the drone survey research in 2018, PG&E used Picarro data to assess emissions from distribution regulation stations in an effort to refine the station-based emission factor when reporting emissions. Future research is needed, particularly regarding how to identify leaks not visible from the Picarro vehicle; however, the results thus far suggest that the distribution

[4] The CARB Oil and Gas Rule is promulgated under 17 CCR, Division 3, Chapter 1, Subchapter 10, Article 4, Subarticle 13, Section 95668(h)(2).

regulation stations emit less than the currently utilized emission factor estimates. The development of site-specific methane emission assessment is critical for directing abatement efforts on these assets that represent about 26% of PG&E system total emissions as calculated in the present report.

Changes Between Past Year and 2018 Reporting

In 2018, PG&E made several changes which affected reported emissions. Overall, total reported emissions in 2018 are about 9% lower than in 2017 and PG&E did make substantial emission reductions in several specific categories. A discussion of changes to the 2018 reporting is included below. These changes can be attributed to three causes:

- Advances in abatement efforts
- Changes in annual activity
- Improvements in reporting practices

Advances in Abatement Efforts

- Transmission Pipeline Leaks: While transmission pipeline leak emissions are based on total transmission pipeline mileage and did not result in a substantial year-over-year emission change, PG&E did observe an approximately 16% increase in the number of leaks found in 2018 as compared to 2017. PG&E estimates this increase in leaks found was due to the more widespread use of sensitive leak detection equipment by the leak survey team and the increase in aerial leak detection flights conducted in 2018 as compared to 2017.

- Transmission Pipeline Component Leaks: Transmission pipeline component leaks are a subset of total transmission pipeline leaks, and the increase in transmission pipeline leaks found in 2018 versus 2017 is consistent with the increase in transmission pipeline component leaks. In particular, the increase in the proportion of aboveground leaks is consistent with the increase in component leaks as components are more typically located aboveground whereas leaks on the body of the pipeline are typically located underground.

- Transmission Pipeline Blowdowns: PG&E's transmission pipeline and transmission M&R station blowdowns resulted in a much smaller gas release in 2018 as compared to 2017, due to increased abatement efforts as well as changes in the types of scheduled project work. In 2018, PG&E expanded its

use of cross compression to reduce blowdown emissions from local transmission assets. In 2018, PG&E diverted approximately 80% of all transmission blowdowns due to drafting and cross compression efforts as seen in Table 1 below. This demonstrates the advancement in abatement as compared to the 72% diverted in 2017.

Table 1. 2018 Transmission Pipeline and M&R Station Abatement Activities

Pipeline Activity Type	Total Gas Volume (Mscf)
Drafting	119,837
Cross Compression	504,320
Blowdown	159,215
Total Gas Baseline	*783,372*
Percent Gas Diverted to Drafting or Cross Compression	*80%*

- Transmission M&R Station Leaks and Emissions: The 2018 count of farm tap regulator sets decreased by approximately 10%, including farm tap regulator sets that were designated as active the entire year as well as farm tap regulator sets that were active for only part of the year (i.e., farm tap regulator sets that were brought online or taken offline in 2018). This approach is consistent with the 2017 reporting method. The reduction in farm taps in 2018 can be attributed to PG&E's 2KA Program which targets obsolete components within PG&E's transmission system with the goal being to rebuild or remove farm taps containing such components. PG&E's reduction in farm taps is an abatement activity to remove existing assets which results in a reduction of methane emissions. Changes to other station counts (i.e., inter and intra connects) were minimal compared to 2017 reported values.

  Additionally, in 2018 PG&E removed eight high bleed pneumatic devices from transmission M&R stations which resulted in an emissions reduction. However, due to the station population-based emission factor used in the transmission M&R station leaks reporting template (Appendix 1), the emission reductions from this change are not reflected in this report.

- Transmission Compressor Station and Underground Storage Facility Component Leaks: PG&E conducted quarterly comprehensive leak surveys pursuant to the requirements of the COGR, which became effective January 1, 2018, whereas, in 2017, only a single comprehensive leak survey was available.

The additional surveys contributed to a 116% increase in identified leaks at transmission compressor stations and a 28% increase in identified leaks at underground storage facilities. While the 2018 surveys yielded more identified leaks as compared to 2017, the average number of days leaking decreased by approximately half due to the increased survey frequency. Fewer days leaking corresponds to a lesser amount of methane released. Additionally, in 2018, PG&E differentiated compressor and non-compressor component leaks at transmission compressor stations to use the appropriate emission factors as listed in Appendix 9.[5] Emission factors for transmission underground storage component leaks were also revised in accordance with Appendix 9. These changes resulted in a greater number of reported leaks for both transmission compressor station and underground storage facilities, a larger emission estimate for transmission compressor stations and a smaller emission estimate for transmission underground storage facilities than in 2017.

- Transmission Compressor Station and Underground Storage Facility Component Vented Emissions: In 2018, PG&E converted many pneumatic devices to those which utilize air instead of gas, replaced high bleed devices with intermittent bleed devices, or otherwise removed them as demonstrated in Table 2, resulting in a slight decrease in devices and associated methane emissions for various facilities. Though the devices noted in Table 2 were removed or replaced prior to the end of 2018 and thus did not operate for the entire year, the devices are included in the total count of venting devices in Appendix 8.

[5] For direct comparison to 2017 methane emissions only, emissions from the 2018 leaks were estimated using only the non-compressor component emission factors, consistent with the previous year Appendix 9 methodology. This analysis demonstrated a net decrease in methane emissions indicating that the net increase in methane emissions visible in Appendix 8 for 2018 is due to the change in emission factor which overrides the benefit achieved by the significant decrease in number of days leaking.

Table 2. Summary of Decrease in Pneumatic Devices for Transmission Compressor Stations and Underground Storage Facilities during 2018

Facility	Count of Removed/Replaced Pneumatic Devices	Timing of Decrease
Pleasant Creek Underground Storage Facility	44 – high bleed	March 2018
McDonald Island Underground Storage	3 – high bleed	November 2018
Delevan Compressor Station	3 - high bleed 13 - intermittent bleed	October 2018
Kettleman Compressor Station	1 – high bleed 1 – intermittent bleed	November 2018
Gerber Compressor Station	2 - high bleed 11 - intermittent bleed	December 2018

- Distribution Main & Service Pipelines, Pipeline Leaks: In 2018, PG&E moved from a four-year leak survey cycle to a three-year leak survey cycle which decreased the estimated number of unknown leaks in application of BP-15. As predicted by the methane emission calculation models, PG&E also observed an increase in the number of found leaks. This increase will level off when the transition to a three-year leak survey cycle is completed because the larger survey area associated with higher survey frequency will be fully compensated by the shorter time between surveys[6] providing the full benefit of reducing the number of unknown leaks. Calculations were performed in accordance with the equations established at and presented to the CPUC workshop on January 17, 2019. Note that PG&E utilized the 2018 annual leak rate to estimate unknown leaks instead of the average 2016-2018 leak rate. PG&E made this change as the 2017 annual leak rate does not appropriately represent PG&E's transition to a three-year leak survey cycle and the average leak rate across 2016, 2017, and 2018 does not show the benefit of the new abatement measures PG&E has implemented to move to a three-year survey cycle in 2018.

  While PG&E saw an increase in the number of distribution leaks in 2018, PG&E crews repaired approximately 50% more leaks in 2018 than in 2017.

  In addition to the change in the number of leaks found and repaired in 2018, PG&E also conducted a special leak survey on vintage materials in accordance with BP-16 (discussed in the response to Question 1 above). The impact of the program is an estimated reduction of over 7,800 Mscf. This impact should

[6] PG&E 2018 Leak Abatement Compliance Plan pp.Atch1-139 to Atch1-149.

grow next year since less time will be left for new leaks to appear on these materials.

Additionally, PG&E conducted its Super Emitter program in accordance with BP-21. This program aims to rapidly detect and repair leaks larger than 10 standard cubic feet per hour (scfh). It uses the mobile leak detection system developed by the company Picarro and an algorithm estimates the size of leaks. The mobile system was driven across the full territory and led to the identification of 220 larger leaks (or super emitters) including 86 in the areas covered by the compliance surveys and 134 in the areas not covered by compliance surveys. Three grade 3 leaks from previous year surveys were also detected. Following the methodology presented in PG&E 2018 Leak Abatement Compliance Plan[7] and during the January 17, 2019 workshop, emissions were calculated by assigning a different emission factor to super emitters and other leaks. The impact of the program was estimated at a reduction of 90 Mscf. Similar to the vintage material special leak survey mentioned above, this impact is expected to grow in 2019 since less time will be left for large leaks to appear.

Overall, the combination of these various leak survey changes and abatement efforts resulted in a modest decrease of approximately 8,600 Mscf as compared to 2017. The decrease in emissions is expected to grow in the upcoming reporting years.

- Customer Meters Aboveground MSA Leaks: In 2018, and consistent with its approach in 2017, PG&E included all leaks which occurred on meter set assemblies aboveground level in Appendix 6 to be consistent with other reporting utilities. Figure 1 below has been provided to illustrate how PG&E grades aboveground leaks associated with meter sets. As seen in Figure 1, leaks that occur below the gas service shut-off valve are graded, whereas leaks that occur above the gas service shut-off valve are non-graded.

[7] PG&E 2018 Leak Abatement Compliance Plan pp. Atch1-187 to Atch1-200.

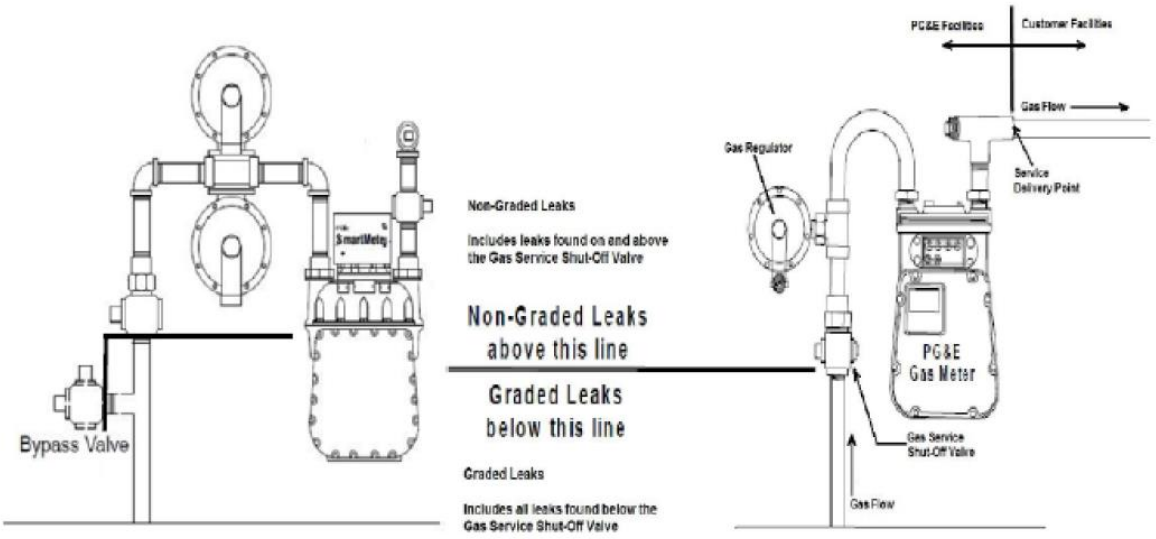


Figure 1: Graded and Non-Graded Leak Locations

In 2018, PG&E had an approximately 46% increase in aboveground gradable and non-gradable leaks, which was expected due to the change to a three-year distribution pipeline leak survey and is consistent with the increased frequency of distribution pipeline survey discussed in the paragraph above. This increase was not fully represented in the methane emissions since a population-based emission factor is used for Meter Set Assemblies (Appendix 6) which includes aboveground gradable service leaks and aboveground nongradable leaks. Moving forward, the use of a leak-based emission factor (instead of a population-based emission factor) will allow operators to represent repaired leaks from this reporting category appropriately.

- Underground Storage Facilities Storage Leaks and Emissions: Consistent with 2017, surface casing test data were available for McDonald Island and Los Medanos. The 2018 test data resulted in much lower emissions from a similar source count. Using the same method as 2017 to estimate the surface casing emission factor for Pleasant Creek (extrapolation from the other sites' test data) resulted in a negative value, thus the average emissions per well were utilized as the emission factor instead. Overall surface casing emissions decreased by 80%.

Changes in Annual Activity

In 2018, PG&E conducted different types of work compared to previous years which resulted in emissions changes. Such changes include the following:

- Transmission Pipeline Damages: In 2018, PG&E experienced fewer damages than in prior years. PG&E recognizes that changes in damages result in significant part from factors out of PG&E's control, which is why the reduction of 14,000 Mscf compared to 2017 is listed here as a change in activity. Nevertheless, PG&E has made important efforts in reducing pipeline dig-ins and continues to strengthen this program as part of Best Practice 24.

- Transmission Pipeline Odorizers: PG&E experienced an increase in odorizer emissions in 2018 as compared to 2017 due to the addition of 30 odorizers in its Meridian District and 8 odorizers in its Rio Vista District that were part of the Odorization Project conducted in 2018, in which PG&E required California natural gas producers to install odorization facilities. The Odorization Project was implemented as a safety measure to install odorizers at all gas delivery points into the PG&E system. This safety measure was implemented to ensure all gas is odorized during all various possible flow scenarios.

- Transmission Compressor Station and Underground Storage Facility Compressor Emissions: For transmission compressor stations, overall compressor pressurized operating hours increased compared to 2017 due to a higher natural gas throughput to meet higher residential and industrial customer demand. PG&E experienced an approximately 16% increase in demand through its Baja (southern) route and an approximately 2% increase through its Redwood (northern) route. This trend is consistent with the compressor operating data where compressor stations along the Baja route experienced a greater overall increase in compressor pressurized operating hours as compared to the Redwood route.

- Transmission Compressor Station and Underground Storage Facility Blowdowns: In general, blowdown emissions increase or decrease as the compressor operating hours increase or decrease since blowdowns are conducted to perform maintenance; however, blowdowns can also be project related. Overall the compressor stations had higher compressor operating hours in 2018 than in 2017, leading to increased blowdown volumes. Hinkley,

Kettleman, and Topock Compressor Stations also experienced larger volume clearances for leak repair activities in 2018 as compared to 2017. At Burney Compressor Station, the commissioning activities of the new compressor unit contributed to an increase in both unit and station blowdowns. Similarly, Gerber Compressor Station experienced increased blowdown activity due to the commissioning of the new station and unit controls system. In contrast, the underground storage facilities saw decreased operating hours in 2018 as compared to 2017, with a corresponding decrease in blowdown emissions.

- Distribution M&R Station Leaks and Emissions: In 2018, the total number of distribution M&R stations decreased; however, total emissions from stations increased due to changes in the individual categories of distribution M&R stations, some of which have higher emission factors than others.

Improvements in Reporting Practices

In 2018, PG&E expanded data collection efforts and refined existing data sets to provide more robust and detailed information for the 2018 reporting year. Changes include the following:

- Transmission M&R Station Blowdowns: In 2018, PG&E utilized transmission blowdown data collected for 40 Code of Federal Regulations Part 98, Subpart W greenhouse gas (GHG) reporting for both transmission pipeline blowdown data and transmission M&R station blowdown data. While PG&E had used this data source in prior years to report transmission pipeline blowdowns, improvement in the level of detail included in the reported data allowed PG&E to extract which blowdowns occurred at transmission M&R stations in 2018. PG&E considers this an improvement in reporting as the previous method for quantifying transmission M&R station blowdowns involved the following broad assumptions: (1) each station which underwent a preventative or corrective inspection during the year was completely blown down; and (2) the volume of blowdown was based on a literature value emission factor. Note that while the GHG reporting data does not include blowdowns for chambers that are less than 50 standard cubic feet (scf), PG&E estimates the total volume possibly not included is less than 14 Mscf.[8] PG&E

[8] In 2018, there were 274 preventative and corrective inspections (known as N1 and N3 inspections) on transmission M&R stations. Many of the blowdowns that occurred during these inspections are included in the GHG reporting data; however, the worst case scenario assumption would be that if all 274 inspections resulted in blowdowns below 50 scf in chamber volume this would equate to approximately 13.7 Mscf of unaccounted emissions.

believes that the quantification of blowdown emissions using this new method, despite some exclusion of small blowdown events, is more accurate than using a literature emission factor as was the case with the previously employed method.

While PG&E has made efforts to increase abatement practices, such as cross compression, at transmission M&R stations in 2018, the change in methodology from reporting year 2017 to reporting year 2018 does not allow direct year-over-year comparison. The new emission estimation methodology will allow PG&E to track blowdown abatement efforts at transmission M&R stations in future reporting years.

- Transmission Pipeline Component Emissions: In 2018, PG&E made improvements to its transmission pipeline asset reporting system by augmenting its Gas Transmission Geographic Information System (GT-GIS) reporting with assets captured in another system of record, SAP. This change in reporting led to an increase in reported transmission vented components, in particular, an increase in pressure relief devices. Many of the devices reported for 2018 that were not reported in previous years are located near gas receipt points at the edge of PG&E's transmission system, positioned close to the asset change of ownership location. These devices were not captured in the reporting system used in prior years (the GT-GIS system) due to technical mapping constraints; however, PG&E is actively working to streamline the use of both GT-GIS and SAP to capture all applicable assets. The greatest impact from this change is visible in the pressure relief valve count which increased from 14 to 81 devices. The pneumatic count also increased from 19 to 33 devices as compared to 2017. In addition, in 2018 PG&E applied the new pressure relief valve emission factors included in Appendix 9.

- Transmission Compressor Station and Underground Storage Facility Component Vented Emissions: In 2018, PG&E collected more detailed data from individual facilities on all venting components as part of an inventory for the COGR resulting in a higher device count. The overall device count at transmission compressor stations increased by 12% in 2018 while the overall device count at Underground Storage Facilities increased by 3% despite the reduction in pneumatic devices noted in Table 2 above. The largest increases in pneumatic counts occurred at Hinkley and Topock Compressor Stations due to the change in pneumatic device classification for compressor unloaders. Annual emissions in 2018 were refined using actual operating days of devices

where available instead of assuming 365 operating days for each device. Finally, in 2018, PG&E applied the new pressure relief valve emission factors included in Appendix 9. For transmission compressor stations, the new pressure relief valve emission factor required the specification of compressor or non-compressor component.[9] These changes resulted in a greater number of reported vented devices for both transmission compressor stations and underground storage facilities, a larger emission estimate for transmission compressor stations and a smaller emission estimate for transmission underground storage facilities than in 2017.

- Distribution Main & Service Pipeline Damages and Customer MSA Damages: PG&E experienced an approximately 9% increase in the number of distribution pipeline damages in 2018 due to increased construction activity in its territory. At the same time, PG&E's dig-in ratio has decreased from 1.89 dig-ins for every 1,000 USA tickets in 2017 to 1.61 dig-ins for every 1,000 USA tickets in 2018. PG&E also repaired 13% more distribution pipeline damages in less than one day in 2018 as compared to 2017 and emissions decreased by 71%. This reduction in emissions is partially due to a refined calculation methodology for distribution pipeline damages as well as PG&E's increased repair efforts. In previous reporting years, if a distribution damage leak was open for less than one day, PG&E estimated emissions using a mathematical formula which models gas flowing through an orifice with the time input factor represented by the time PG&E took to repair the leak. In 2018, PG&E revised this methodology by accounting for the time before the gas was shut-off as this time actually represents the duration of gas release. The average gas shut-off time for distribution damages observed on all emergency calls was used since the current database structure of emergency events and leak detections/ repairs does not provide a way to identify the gas shut-off time for each leak individually. As part of PG&E's commitment to improving safety, PG&E is focused on decreasing the time taken to shutoff gas after damage occurs and expects to see a decrease in this time in future reporting years.[10]

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[9] For direct comparison to 2017 methane emissions only, emissions from the 2018 components were estimated using only the non-compressor component emission factors, consistent with the previous year Appendix 9 methodology. This analysis demonstrated comparable methane emissions indicating that the net increase in methane emissions visible in Appendix 8 for 2018 is due to the change in emission factor which overrides the benefit achieved by the reduced high bleed device count and the refined operating day input.

[10] Per PG&E 2019 Gas Safety Plan published March 15, 2019, page 33, available here: https://www.pge.com/pge_global/common/pdfs/safety/gas-safety/safety-initiatives/pipeline-safety/2019GasSafetyReport.pdf.

Additionally, 2018 was the first full calendar year in which PG&E implemented a new system (the Event Management [EM] Tool) to track damages that occurred specifically on meter sets (as seen in Appendix 6), which are a subset of the total distribution pipeline damages listed in Appendix 4. This change addresses the comment on page 35 of the California Air Resources Board and California Public Utilities Commission Joint Staff Report-Analysis of the Utilities' June 15, 2018, Natural Gas Leak and Emission Reports published on December 21, 2018. In order to account for the fact that MSA damages are a subset of total distribution damages, PG&E has subtracted the total count of damage leaks and total emissions from damages quantified in Appendix 6 from the count and emissions quantified for damages in Appendix 4. PG&E is working towards refining its current reporting systems so that this overlap between Appendix 4 and Appendix 6 damages can be reconciled on a line-by-line basis using unique leak identifiers in the future. However, the data needed for this reconciliation process was not available for 2018.

- Underground Storage Facilities – Storage Leaks and Emissions: In 2017, PG&E included an estimate for wellhead leakage based on daily wellhead leak survey results and an emission factor for leaking wellheads developed through emission measurements taken at the Los Medanos Storage Facility using the Picarro system. To further refine the emissions estimate for 2018, PG&E identified the component types associated with the individual leaks in the daily wellhead leak survey results and applied the component-specific emission factor as listed in Appendix 9. Individual leak identification and repair dates were also incorporated in the calculation for the number of days leaking.

  Additionally, in 2018 PG&E included reporting of downhole safety valve (DHSV) testing emissions at the facilities where the testing occurs: McDonald Island and Los Medanos storage facilities. DHSVs are designed to isolate wellbore pressure and fluids in the event of an emergency or an extreme failure of surface equipment. While PG&E first identified the testing of DHSVs as potential methane emission sources during the 2018 reporting effort, PG&E confirmed that the testing did occur in previous years. Methane emissions from the three previous years of testing are summarized in Table 3.

Table 3. Methane Emissions from Historical DHSV Testing

Year	McDonald Island (Mscf)	Los Medanos (Mscf)	Total (Mscf)
2015	189.5	43.9	233.4
2016	156.3	59.5	215.7
2017	171.9	44.5	216.4

These changes contributed to an overall emissions decrease of 11% in 2018 as compared to 2017.

Conclusion

PG&E's advancements in abatement efforts, changes in annual activity, and improvements to reporting practices all contributed to a change in reported emissions from 2017. Table 4 below summarizes the effects which the 2018 advancements in abatement efforts, changes in annual activity, and changes to reporting practices had on year-over-year emission estimates.

Table 4. Effect of 2018 Changes on Total Reported Emissions

2018 Reporting Change	Change in Emissions Compared to Reporting Year 2017 (Mscf)	Percent Change in Emissions Compared to Reporting Year 2017
Advances in Abatement Efforts:	(181,882)	
Transmission Pipeline Leaks	(2)	(0.1%)
Transmission Pipeline Blowdowns	(133,307)	(50%)
Transmission M&R Station Leaks & Emissions	(21,695)	(3.6%)
Transmission Compressor Station Component Vented Emissions*	(219)	(2.2%)
Transmission Compressor Station and Underground Storage Facility Component Leaks (combined)*	(11,948)	(37%)
Distribution Main & Service Pipeline Leaks	(8,619)	(1.7%)
Customer Meter Leaks	(3,077)	(0.5%)
Underground Storage Facility Leaks and Emissions*	(1,180)	(23%)
Underground Storage Facility Component Vented Emissions*	(1,835)	(2.1%)
Change in Annual Activity:	(19,700)	
Transmission Pipeline Damages	(14,413)	(98.2%)
Transmission Pipeline Odorizers	48	32.8%
Transmission Compressor Station Compressor Emissions	10,160	57.9%
Transmission Compressor Station Blowdowns	17,603	54.2%
Distribution M&R Station Leaks & Emissions	25,249	3.5%
Underground Storage Facility Compressor Emissions	(726)	(8.1%)
Underground Storage Facility Blowdowns	(4,043)	(28%)
Unusually Large Leaks	(53,500)	(100%)
Other **	(78)	--
Improvements in Reporting Practices:	(89,964)	
Transmission Pipeline Component Emissions	23,471	438%
Transmission M&R Station Blowdowns	(59,132)	(70%)
Transmission Compressor Station and Underground Storage Facility Component Leaks (combined)*	1,668	5.2%
Transmission Compressor Stations Component Vented Emissions*	8,898	87%
Distribution Main & Service Pipeline Damages	(68,151)	(71%)
Customer Meter Set Assembly Damages	6,375	100%
Underground Storage Facility Leaks & Emissions*	628	(12%)
Underground Storage Facility Component Vented Emissions*	(3,721)	(4.3%)
Total Change:	*(291,545)*	*(9%)*

* Emissions from source categories with changes in multiple categories (i.e., "Advances in Abatement Efforts," "Change in Annual Activity," "Improvements in Reporting Practices") are listed under the respective categories with the change in emissions apportioned appropriately per category and estimated based on the year-over-year changes described in the report above.
** Emissions from categories listed in Appendix 8 which are not explicitly discussed above are included in "Change in Annual Activity." Such categories include transmission compressor station storage tank leaks and emissions, distribution blowdowns, distribution M&R station blowdowns, customer meter vented emissions, and underground storage dehydrator vent emissions.

In summary, PG&E's 2018 reported emissions are approximately 9% lower than in 2017. Many of the emission reductions can be attributed to reductions in blowdown emissions, reductions from damage emissions and faster repair timelines, as well as a reduction in emissions from component leaks at storage facilities.

(2) A list of new graded and ungraded gas leaks discovered, tracked by geographic location in a Geographic Information System (GIS) or best equivalent, by grade, component or equipment, pipe size, schedule and material, pressure, age, date discovered and annual volume of gas leaked for each, by month, from January 1st, 2018 through December 31st, 2018.

Response:

See Appendices

(3) A list of graded and ungraded gas leaks repaired, tracked by geographic location in a Geographic Information System (GIS) or best equivalent, by month, from January 1st, 2018 through December 31st, 2018. Include the grade, component or equipment, pipe size, schedule and material, pressure, age, date discovered, date of repair, annual volume of gas leaked for each and the number of days from the time the leak was discovered until the date of repair.

Response:

See Appendices

(4) A list of ALL open graded and ungraded leaks, regardless of when they were found, tracked by geographic location in a Geographic Information System (GIS) or best equivalent that are being monitored, or are scheduled to be repaired, by month, from January 1st, 2017 through December 31st, 2017. Include the grade, component or equipment, pipe size, schedule and material, pressure, age, date discovered, scheduled date of repair, and annual volume of gas leaked for each.

Response:

See Appendices

(5) System-wide gas leak and emission rate data, along with any data and computer models used in making that calculation, for the 12 months ending December 31st, 2018.

Response:

See Appendices

(6) Calculable or estimated emissions and non-graded gas leaks, as defined in Data Request Pacific Gas and Electric Company R.15-01-008 2019 Annual Report for the 12 months ending December 31st, 2018.

Response:

See Appendices

Exhibit __ (JF-3):

Picarro Emissions Quantification Results Final Report in Support of the Methane Leak Surveying Report for the Public Service Electric and Gas Company ("PSE&G") Gas System Modernization Program ("GSMP") II Program

Picarro Emissions Quantification Results Final Report

in Support of the Methane Leak Surveying Report for the PSE&G GSMP II Program

Prepared by Picarro, Inc.

December 14, 2018

Introduction

Picarro has completed mobile methane emissions measurements for use in the next phase of PSE&G's Gas System Modernization Program (GSMP II). Methane data was gathered along approximately 280 miles of Utilization Pressure Cast Iron ("UPCI") gas mains contained in 44 map grids. The replacement of mains within GSMP II will follow the prioritization based on the grid-based Leak Hazard Indices developed by PSE&G, and the Picarro methane emissions results will be used as a sub-prioritization metric within that framework. Including methane emission rate (volumetric flow rate) as part of the replacement prioritization process may result in the reduction of natural gas emissions and reduce the environmental impacts of such emissions. This document describes the measurement campaign results, data collection methodology, protocol and validation as well as details about Picarro's hardware, software and data analytics platform used to gather and process the data.

Picarro System Hardware

Picarro's mobile natural gas leak detection system is driven through a natural gas distribution infrastructure gathering methane, wind, atmospheric and GPS data which is later processed by Picarro's algorithms to detect and localize leaks and calculate methane emission rates. The Picarro hardware consists of the following elements (shown in figure 1 below) forming a completely integrated solution mounted in a vehicle:

- A parts-per-billion sensitivity gas analyzer based on Cavity Ring Down Spectroscopy (CRDS) measuring atmospheric gas composition and other tracers such as ethane
- An anemometer mounted on a mast for detecting wind speed, direction and wind variability
- Two antennas on the vehicle roof, one for the 4G wireless connectivity and one for sub-meter GPS vehicle positioning
- A 4G wireless router enabling the internet connection and data transmission to and from the Picarro Cloud and WiFi connection to the in-vehicle tablet

- A tablet computer which allows the operation and visualization of the system and data
- A supporting equipment module containing pumps, a backup battery, GPS receiver and various power supplies and gas handling equipment
- A gas inlet system mounted on the front of the vehicle

Air is continuously collected on the front of the vehicle routed to the gas analyzer via tubing. The entire system and accessories are directly connected to the vehicle battery.



Figure1. Picarro System Hardware.

Picarro Software & Data Analytics

The Picarro system identifies the characteristic signature of natural gas leaks by analyzing the methane plumes as they propagate in the atmosphere and intersect with the path of the vehicle. The system also measures atmospheric and meteorological conditions and uses algorithms to identify the origin and degree of hazard of the natural gas leak indication while virtually eliminating indications triggered on other non-natural gas sources of methane.

The most powerful feature of the Picarro system is its ability to combine information from multiple measurement sessions over a region, taking advantage of varying atmospheric conditions (wind direction, wind speed, atmospheric stability), to produce aggregated survey results over a certain period of time. This unique capability increases territory coverage with successive passes by the vehicle and allows statistics to be built up on location and risk for every leak indication. Reports and other data outputs can be generated from this processed data specific to the

intended use case – leak survey, forecasting, targeted emissions reduction, risk management, etc.

Picarro's Emissions Quantification Analytics is one of the analytical models that generates outputs and reports that can be applied to data taken by the Picarro vehicle. After multiple passes are driven in an area of interest where the vehicle path intersects methane plumes typically multiple times, the analytics process the data using four basic steps:

1. Calculate the emission rate from individual methane plume detections. Here, the methane concentration is represented as a function of distance along the vehicle's path and that "line integral" is evaluated, as described below, to calculate the flow (emission) rate.
2. Geographically associate (cluster) these detections to identify emission source locations.
3. Calculate "average" emission rate of each cluster using individual detections using a Bayesian framework.
4. Aggregate sources (clusters) over areas (grids) or pipe segments and sum emissions from individual sources to determine total emission rate and uncertainty.

The methane flow rate Q is derived from the volumetric flux equation which uses a "Mobile Flux Plane" measurement as input:

$$Q = u \iint [C(y,z) - Co]\, dy\ \ dz$$

where *C(y,z)* is the concentration at each measurement point of the cross-sectional area of the plume (the vehicle samples the concentration along a line through this plume in the *y* direction and the plume is assumed to be homogenous in concentration across this surface), C_0 is the background methane concentration, u is the mean wind speed (the wind is measured by the anemometer on the vehicle and is assumed to be roughly vertically constant over the size of the plume; the height of the plume is inferred from its measured width in the *y* direction). In standard engineering terminology the flux plane method is analogous to a control volume approach for quantifying gas flow rates. The vehicle drives downwind of the leak and captures methane emissions over a control surface along the vehicle's path. The inflow condition for the control volume is determined from highly sensitive measurements of the background methane concentration. The Picarro methane emission rate measurement system is consistent with the provisional EPA test method OTM 33 for gas leak detection and emissions quantification (EPA, 2014). For plume intersections where the angle of the wind is too shallow (i.e. the wind is along the direction of vehicle travel and the plume is propagating parallel or nearly parallel to the vehicle path) the wind direction data allows these plumes to be excluded from the analysis since their line integrals are not meaningful in this instance).

The power of the flux plane method for natural gas leak rate quantification is the simplicity of the approach. A prediction of emission rate is made directly by multiplying the measured crosswind concentration profile by the measured wind speed. Accurate emissions estimates are achieved through a combination of enhanced spatial resolution of the concentration profile, accurate measurements and models of the instantaneous vertical wind speed gradient, and averaging of multiple plume transects downwind of the leak. The fast response time (4 Hz) of the Picarro methane gas analyzer provides high spatial resolution in the crosswind direction. This produces a high-resolution concentration map without loss of spatial information content.

Comparison to Traditional Survey Equipment and Methods

The Picarro system takes methane data at a speed and scale not possible with traditional instrumentation, eliminating human bias and operator error associated with these legacy methods. It has been shown in over 60 field studies to consistently identify an average of *three times* as many gradeable leaks (and a three times more hazardous leaks) as compared to traditional survey equipment and methods. In comparison to traditional leak survey equipment, the Picarro hardware is 1000 times more sensitive, with the ability to detect methane and ethane at better than one part-per-*billion* (traditional systems have only 1-part-per-*million* methane sensitivity and do not use ethane to remove false positive leak indications from biogenic methane sources (sewer, etc.) as the Picarro system does. The system can take data at vehicle speeds over 40mph and in rain and snow conditions. The system's reliance on the wind enables it to sense leaks without driving directly over the gas main, and the analytics can rank methane plumes according to their potential hazard, emission rate and likelihood of emanating from an aboveground or belowground leak.

Comparing the Picarro system and analytics to other mobile methane detection technologies (including that which was used during the methane mapping done for GSMP I), there are some key technological advantages of the Picarro system which result in even higher-quality methane quantification results than were achieved for GSMP I.

The key advantage of the Picarro system and methodology is its use of wind information to both localize emission points and calculate emission rates. The use of wind information to calculate emission rates is critical to obtaining accurate results. The Picarro system also has a high collection rate and gas sampling rate so that gas plumes are measured with very high spatial resolution, resulting in high precision emissions quantification. Picarro's six-pass, two-night protocol results in high leak detection rates and high-precision emissions measurements. Picarro's analytics further improve these results by identifying and rejecting false-positive indications. Picarro's analytics can also statistically differentiate between aboveground and belowground leaks and therefore preferentially aggregates plumes judged to be

coming from belowground leaks for emissions quantification (i.e. it will preferentially exclude aboveground leak plumes from the emissions calculations).

Data Collection Methodology & Protocol

Picarro's Advanced Leak Detection technique utilizes the wind to bring methane plumes to Picarro's vehicle-based methane and atmospheric sensing platform. Picarro's data collection methodology is based on the ability of Picarro system to detect methane emissions below as well as at some distance away from the vehicle when the methane emission point is upwind of the vehicle. The reach of Picarro's Field of View coverage area is calculated at each point along the vehicle path to provide a documented record of survey coverage. This concept is shown in figure 2 below.

There are qualities of methane plumes that the system measures (size, emissions, concentration, ethane content, etc.) that allow analytics to predict the location and relative risk of the leak indication (i.e. if it is likely originating from a hazardous leak or not). Leak Indication Search Area (LISA) markers are computed for each potential leak to aid crews in pinpointing leaks for repair.

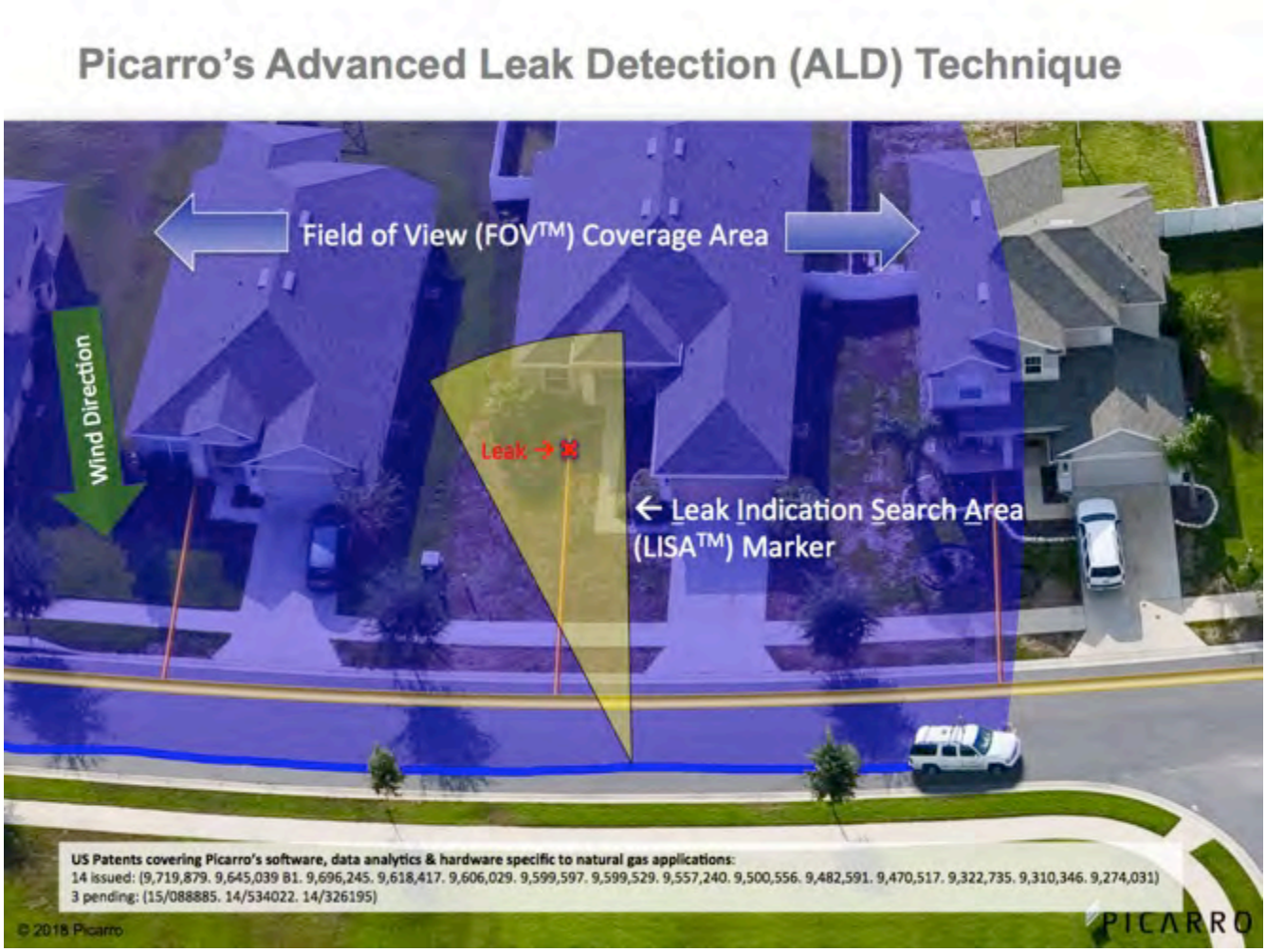


Figure 2. Picarro ALD technique concept.

Picarro collected data on PSE&G UPCI mains sections using a standard "three-drives" protocol that prescribes that each street along which mains were located be driven twice (one pass on each side of the street) and that this be repeated three times on at least two different nights (so that either two or four passes were completed on one night) between sunset and sunrise. This results in six passes per street along the defined sections of main. Data is taken at night to maximize plume detectability and minimize measurement noise due to higher atmospheric turbulence that is present during the day. Nighttime survey also avoids traffic that disturbs plumes.

The reasons for multiple passes over multiple nights is to collect data in a variety of wind conditions (multiple wind directions) to achieve complete Field of View coverage of the mains. Multiple passes also ensures the leak detection rate is >95% since the single-pass detection for a given leak is generally only 25-35% and the detection probability scales as the probability of independent events, reaching >95% for these belowground leaks that are generally near the vehicle. An example of one night's driving on a grid is shown below in figure 3.

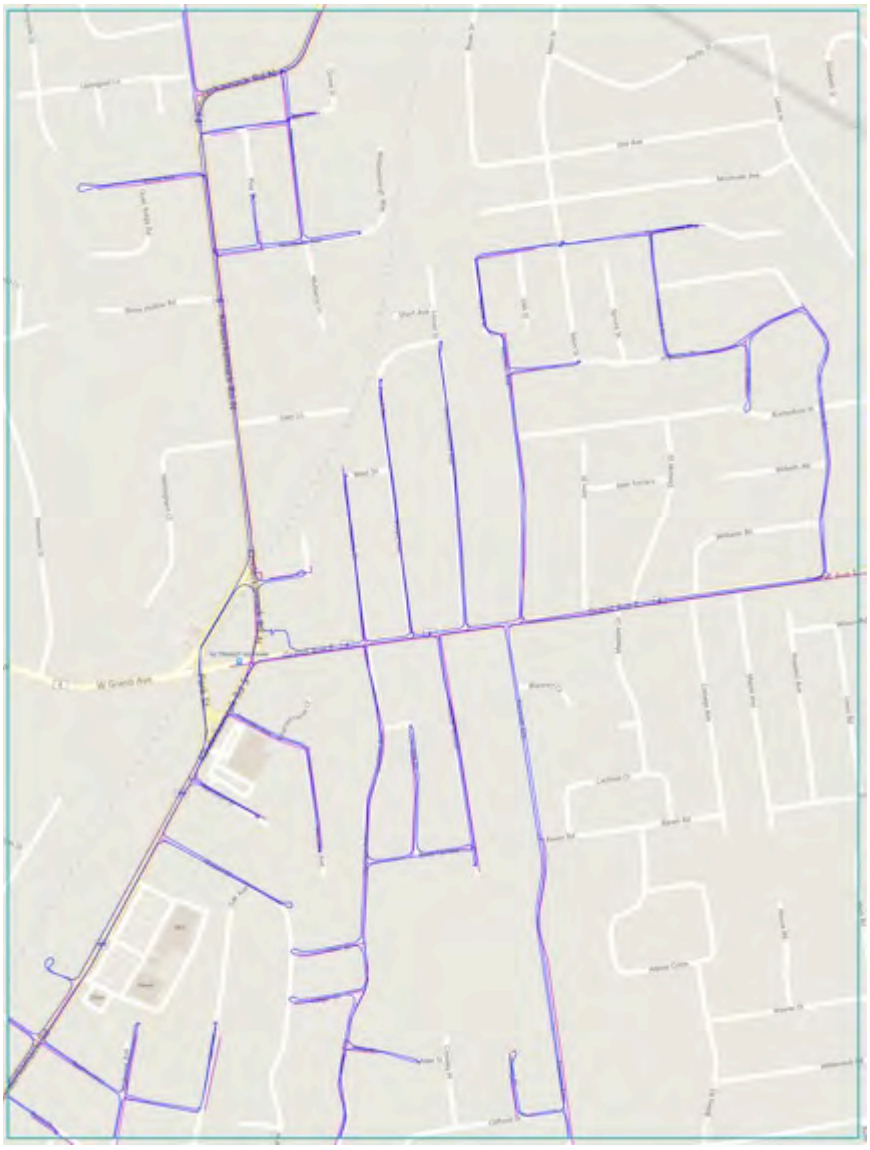


Figure 3. Driving example during one night (two passes per street). Vehicle breadcrumb (blue), UPCI mains (magenta).

Picarro provided a driver and vehicle outfitted with the Picarro equipment to accomplish this task, during the period from 10/23/18 through 11/27/18. PSE&G provided to Picarro shape files (for importation into Picarro's analytics platform) defining: 1) the 44 grid boundaries and 2) the sections of mains to be measured. Using the GIS data, the driver was able to visually identify specific streets to drive and capture data on and those to avoid. During post processing, only emissions that were measured along the sections of mains defined in the GIS were reported – any other data not associated with these mains was excluded and not processed. The system also automatically suspends data collection when the vehicle traverses outside a grid boundary.

PSE&G Results & Discussion

The complete set of tabulated numerical results is in Appendix I, and geographical methane maps for each grid along with summary statistics for each grid are presented in Appendix II).

In figure 4 below, various metrics are plotted against the total emissions per grid. For each grid, the number of measured emissions clusters and the estimated number of belowground leaks (as determined by Picarro's analytics) are plotted versus total grid emissions. Similarly, the emission rate per mile for each grid is plotted versus total grid emissions.

There is generally – but not always – a correlation between these metrics and total grid emissions. The fact that there are large departures from a perfect correlation shows that it would not be possible to accurately predict total grid emissions (nor leaks per mile) based on a "representative" per-mile emission rate (which might be inferred from pipe age, type, etc.) using an inventory approach. In other words, the emission rates and locations must be measured and mapped. With methane maps and their aggregated emissions data, however, it is possible to make accurate, surgical construction decisions at the grid or individual pipeline section level as desired.

It interesting to note that in the comparison of the number of emission clusters below (red squares) to the total grid emissions, the trend is linear up to about 20 liters/minute and then becomes nearly flat. The interpretation of this result – which has been observed in the study[1] of natural gas distribution system emissions previously – is that, a very few number of so-called "super emitters" (i.e. the largest emitting sources) are responsible for a significant fraction of the overall emissions. Here we see examples of where the number of clusters hardly changes, but the overall grid emissions more than doubles.

[1] Lamb BK, Edburg SL, Ferrara TW, Howard T, Harrison MR, Kolb CE, Townsend-Small A, Dyck W, Possolo A, Whetstone JR, 2015, Direct Measurements Show Decreasing Methane Emissions from Natural Gas Local Distribution Systems in the United States, *Environmental Science and Technology* 49, 5161-5169.

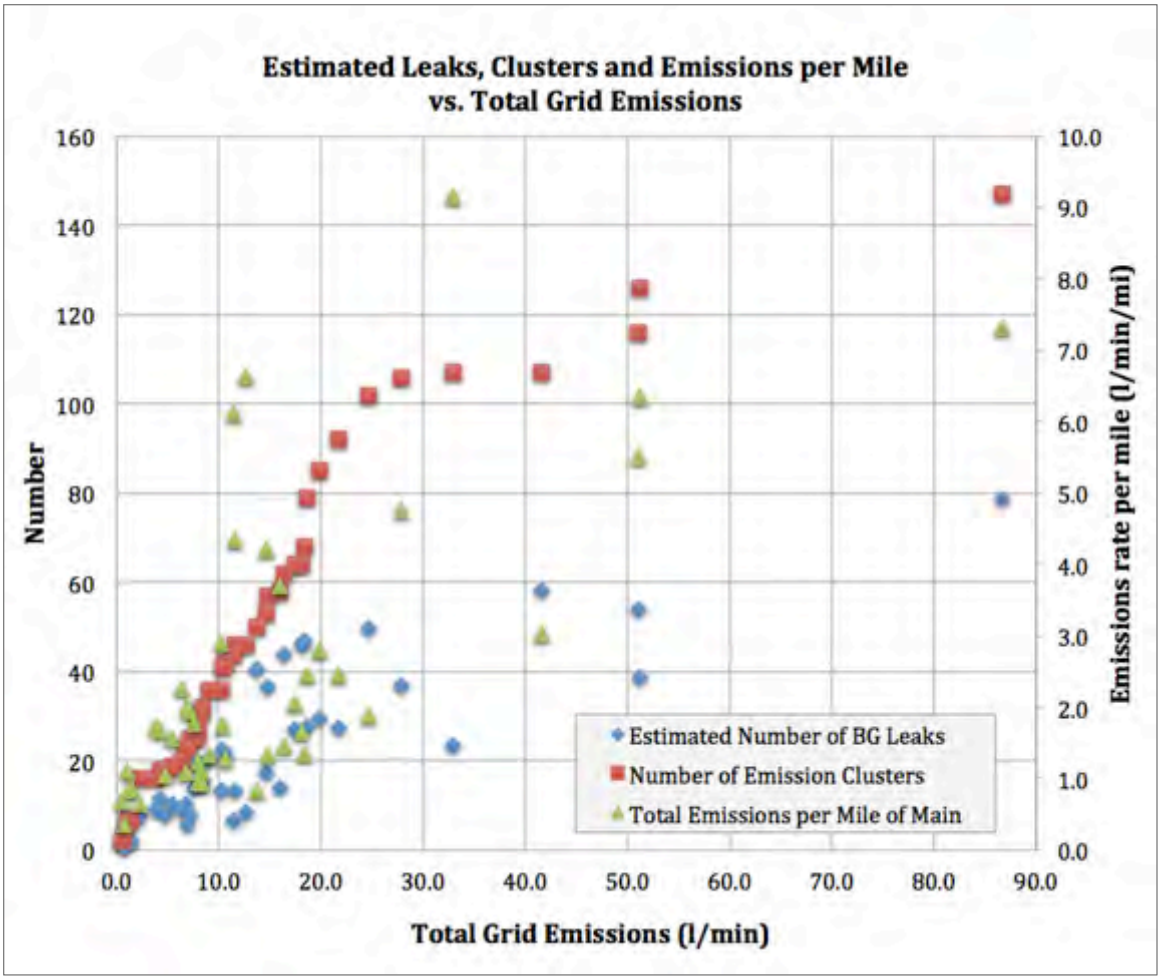


Figure 4. Comparison of total emission rate to per-mile emission rate and per-mile leak number estimation derived from methane mapping data.

The following data is summarized in Appendix 1 for each grid:

- Grid ID
- Miles of UPCI pipe in grid
- Total estimated flow rate (emission rate) (liters/minute)
- Estimated flow rate per mile (liters/minute/mile)
- Rank by total flow rate
- Rank by flow rate per mile
- Total number of emissions clusters within grid
- Total estimated belowground leaks on UPCI mains within grid

The following metrics will be provided or determined by PSE&G after combining the methane emissions results with the existing grid ranking information:

- Hazard Index per mile
- GSMP II UPCI Grid Rank
- Ranked Year of Construction using methane flow rate data
- Planned Year of Construction
- Description of factors contributing to grid bypass decisions (if Planned Year of Construction does not match Ranked Year of Construction)

Summary of PSE&G GSMP II Methane Mapping Project

Some key figures of merit from the data collection and analysis has shown the following summary statistics from the 44 grids:

- Highest emitting grid: 86.6 l/min
- Lowest emitting grid: 0.6 l/min
- Mean grid emissions: 15.3 l/min
- Median grid emissions: 10.5 l/min

The statistics for the emission rate per mile of main were:

- Highest: 9.2 l/min/mi
- Lowest: 0.4 l/min/mi
- Mean: 2.5 l/min/mi
- Median: 1.7 l/min/mi

- Although the total grid emissions trends essentially with the per-mile emission rate, there are exceptions to that trend, also evidenced by visual comparison of the methane maps – there are large variations in both per-mile leak density as well as variability of over two orders of magnitude in leak rates.

- This variability shows the power of the methane mapping technique for providing additional granularity that can be used to maximize methane emissions reductions and/or maximize remediation of the maximum number of belowground leaks through changes to construction priorities based on these methane maps and associated data.

Appendix I: Tabulated Data on GSMP II Grids

Table 1. Detailed statistics for all 44 grids sorted by Grid Rank by Total Grid Emissions. Emissions estimates have a quoted confidence level of 80% (i.e. 10-90% of the distribution). The error estimates are non-symmetric (e.g. Grid 2C-44 has a total grid emission of 86.6 (+23.0 / -15.1) l/min). The terms "flow rate", "emissions" and "emission rate" are synonymous. Mileage is always in terms of miles of UPCI mains.

Grid ID	UPCI Main Pipe Length (mi)	Grid Rank by Total Grid Emissions	Grid Rank by Total Emissions per Mile	Total Grid Emissions (l/min)	Total Emissions Upper Error Bar (l/min)	Total Emissions Lower Error Bar (l/min)	Estimated Number of Belowground Leaks	Number of Emission Clusters	Total Emissions per Mile of Main	Total Emissions per Main Mile, Upper Error Bar (l/min/mi)	Total Emissions per Main Mile, Lower Error Bar (l/min/mi)
2C-44	11.8	1	2	86.6	23.0	15.1	79	147	7.3	1.3	1.9
2H-48	8.1	2	4	51.2	21.6	12.4	39	126	6.4	1.5	2.7
4E-13	9.3	3	6	51.0	7.8	5.2	54	116	5.5	0.6	0.8
1Y-49	13.7	4	11	41.6	6.9	4.7	58	107	3.0	0.3	0.5
2P-51	3.6	5	1	33.0	8.2	5.7	23	107	9.2	1.6	2.3
2R-42	5.9	6	7	27.9	6.5	4.4	37	106	4.8	0.7	1.1
2J-54	13.1	7	20	24.7	3.5	2.3	50	102	1.9	0.2	0.3
2J-46	8.9	8	14	21.8	6.3	4.0	27	92	2.5	0.4	0.7
2J-50	7.1	9	13	19.9	4.2	2.9	29	85	2.8	0.4	0.6
3D-38	7.6	10	15	18.6	4.6	3.0	28	79	2.5	0.4	0.6
1Z-54	13.7	11	28	18.4	2.5	1.7	47	68	1.3	0.1	0.2
3E-37	10.9	12	25	18.1	2.7	1.8	46	64	1.7	0.2	0.2
2L-56	8.5	13	17	17.5	3.3	2.2	27	64	2.1	0.3	0.4
3E-35	11.2	14	27	16.4	2.6	1.7	44	62	1.5	0.2	0.2
2K-54	4.3	15	10	15.9	8.6	5.7	14	58	3.7	1.3	2.0
2J-55	11.1	16	30	14.9	2.4	1.6	36	57	1.3	0.1	0.2
2K-43	3.5	17	9	14.6	4.0	2.9	17	53	4.2	0.8	1.2

2F-53	16.4	18	40	13.7	2.4	1.6	41	50	0.8	0.1	0.1
2R-48	1.9	19	3	12.6	3.9	2.9	9	46	6.6	1.5	2.1
2Y-48	2.7	20	8	11.6	4.4	3.0	13	46	4.3	1.1	1.7
1U-51	1.9	21	5	11.4	9.8	5.5	7	44	6.1	2.9	5.2
2F-48	8.4	22	31	10.7	2.0	1.5	21	42	1.3	0.2	0.2
3F-36	5.9	23	22	10.4	3.0	1.8	23	41	1.8	0.3	0.5
3J-49	3.6	24	12	10.3	3.2	2.1	13	36	2.9	0.6	0.9
2A-02N	6.8	25	29	9.1	1.7	1.2	20	36	1.3	0.2	0.3
1V-59	9.0	26	38	8.3	2.5	1.8	16	32	0.9	0.2	0.3
3B-44	7.5	27	34	8.2	2.1	1.3	16	31	1.1	0.2	0.3
2A-58	8.4	28	37	8.0	2.0	1.3	20	29	1.0	0.2	0.2
2B-59	6.5	29	32	8.0	1.9	1.2	19	26	1.2	0.2	0.3
2N-54	4.3	30	21	7.6	1.8	1.2	14	25	1.8	0.3	0.4
2G-57	3.7	31	19	7.2	2.9	2.0	8	25	2.0	0.5	0.8
2Y-41	3.5	32	18	6.9	6.0	3.9	6	23	2.0	1.1	1.7
3D-45	6.3	33	35	6.8	2.6	1.6	10	23	1.1	0.3	0.4
2P-54	2.8	34	16	6.3	1.8	1.2	9	20	2.3	0.4	0.6
1T-57	3.5	35	26	5.4	1.9	1.2	10	19	1.6	0.3	0.5
3G-47	4.5	36	36	4.7	1.7	1.3	8	18	1.1	0.3	0.4
2C-45	2.6	37	24	4.3	1.1	0.7	12	18	1.7	0.3	0.4
2C-60	2.3	38	23	3.9	1.0	0.7	9	16	1.7	0.3	0.4
3E-30	3.4	39	43	2.2	0.9	0.6	8	16	0.6	0.2	0.3
2C-02N	1.7	40	39	1.5	1.5	0.9	2	7	0.9	0.5	0.8
3F-48	1.5	41	41	1.2	0.6	0.4	3	7	0.8	0.3	0.4
2L-52	1.0	42	33	1.1	0.8	0.5	2	6	1.1	0.5	0.8
3E-48	2.1	43	44	0.8	1.0	0.5	1	3	0.4	0.3	0.5
2C-48	0.8	44	42	0.6	0.2	0.2	2	2	0.7	0.2	0.3

Appendix II: Methane Emissions Maps on GSMP II Grids

In the following pages, methane heat maps are shown for each of the 44 grids along with summary information for each grid. Figure 5 shows these grids on a map.

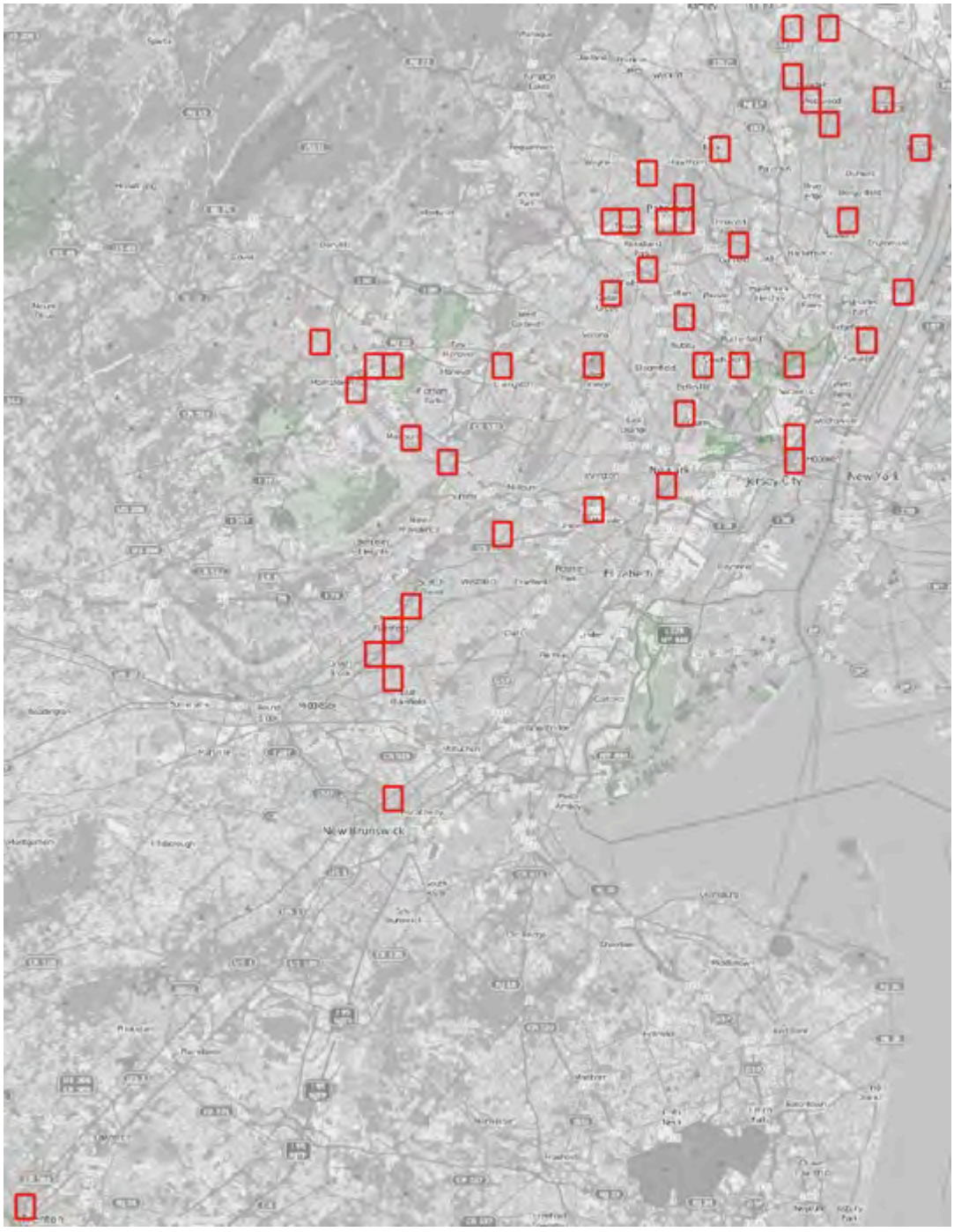
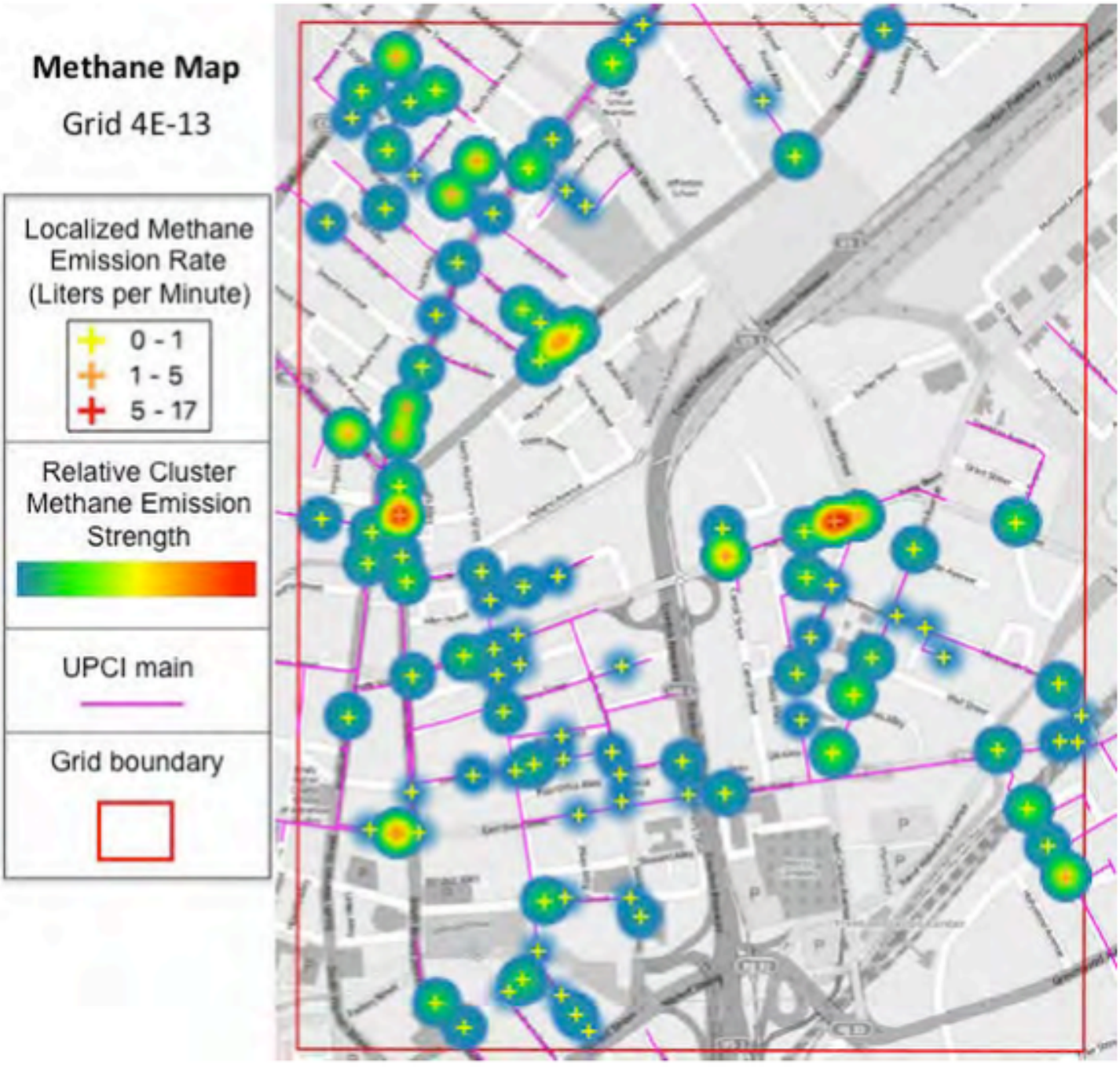


Figure 5. Relative locations of the 44 grids.


Methane Map
Grid 4E-13
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 3E-30

Localized Methane Emission Rate (Liters per Minute)

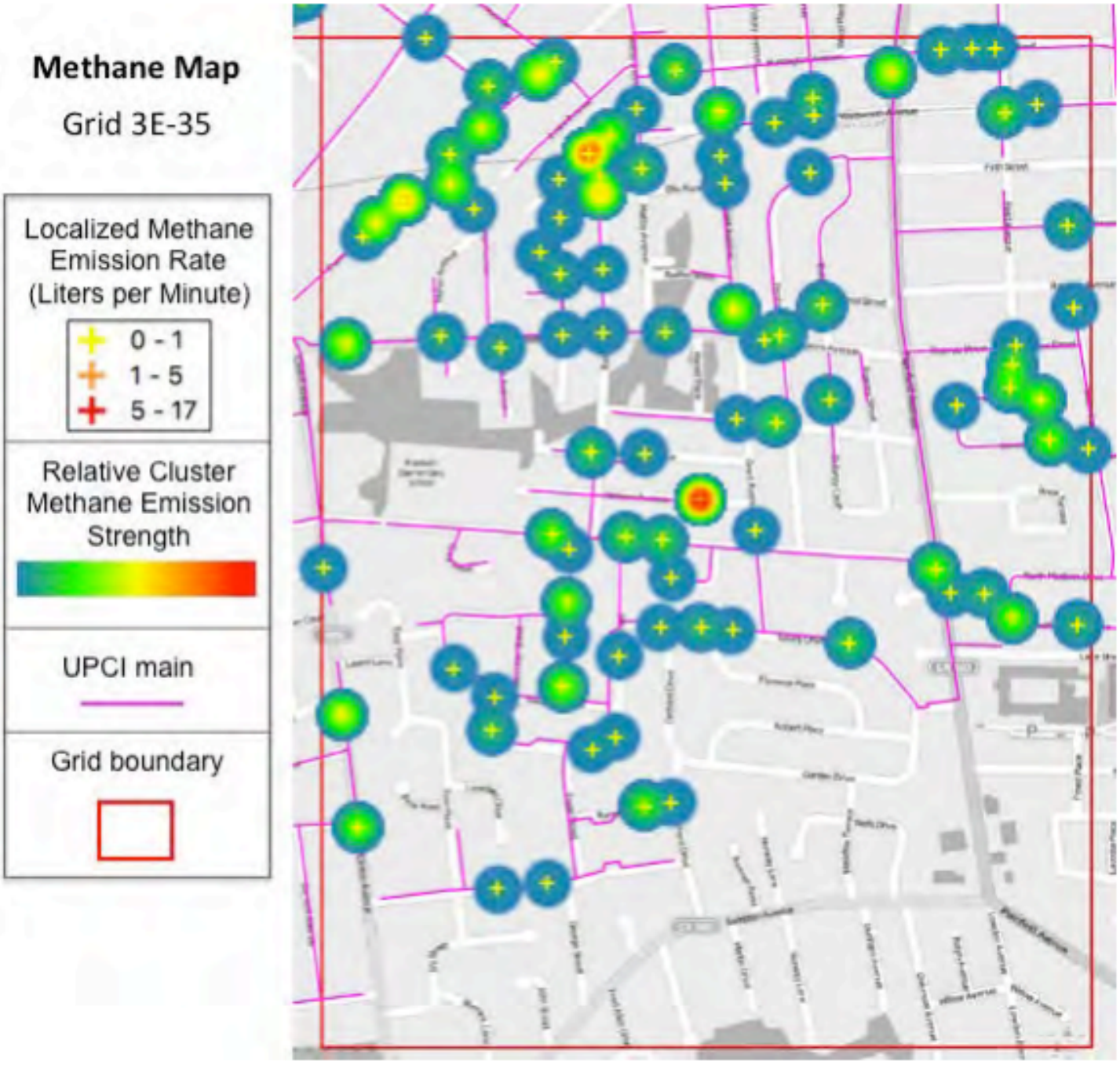
- 0 - 1
- 1 - 5
- 5 - 17

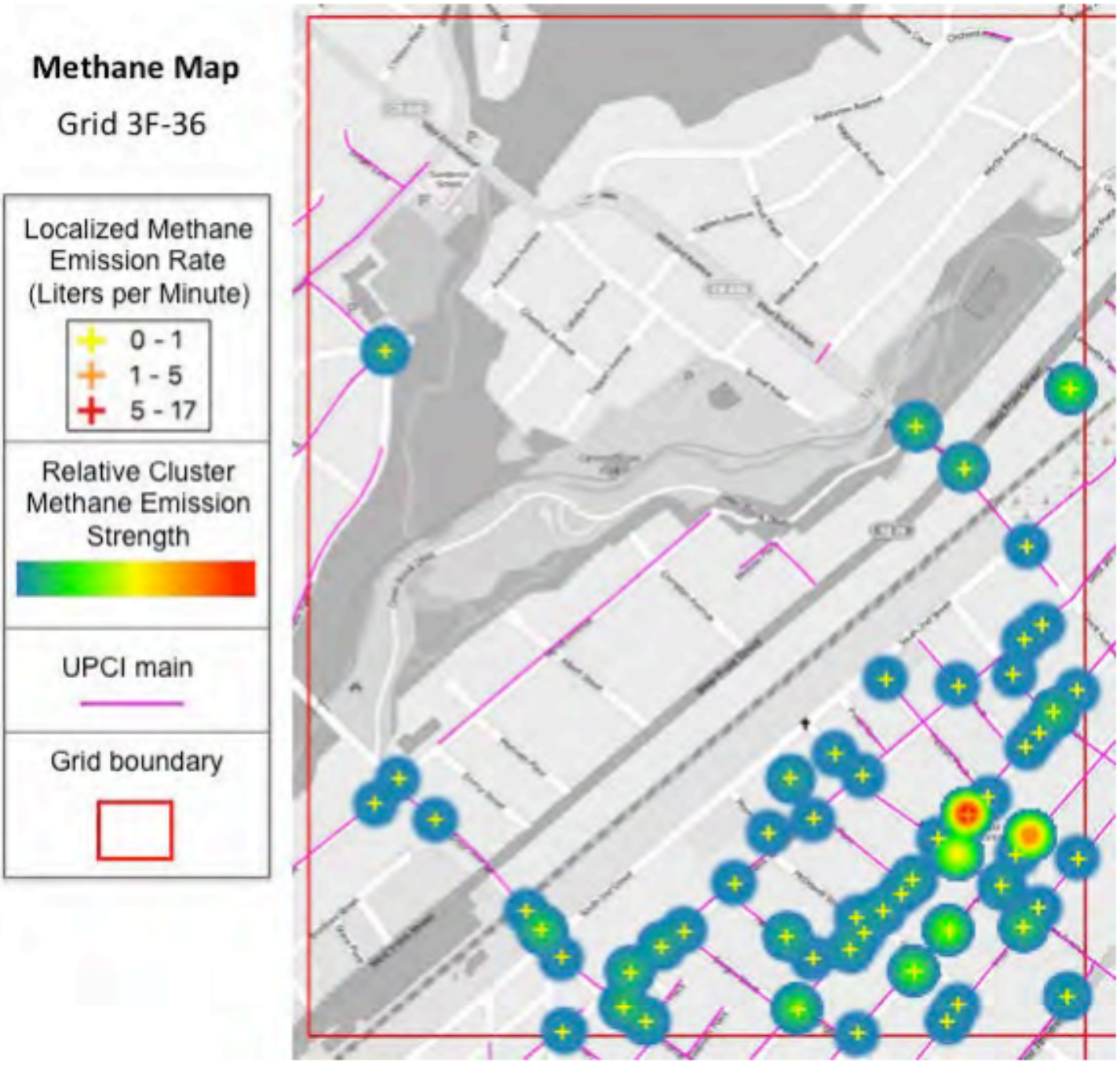
Relative Cluster Methane Emission Strength

UPCI main

Grid boundary




Methane Map
Grid 3E-35
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary


Methane Map
Grid 3F-36
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 3E-37

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 3D-38

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 3J-49

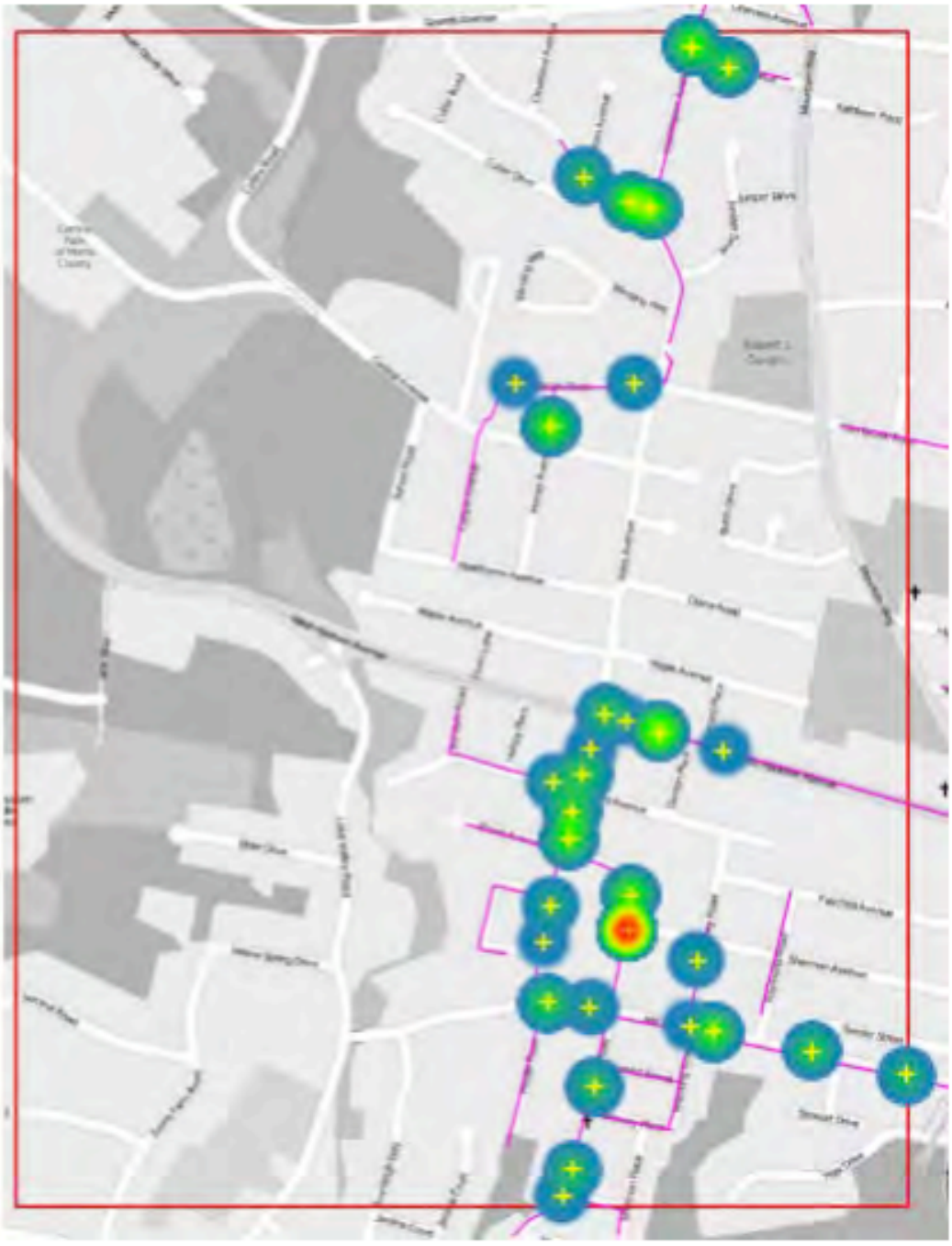
Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 3G-47

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 3F-48

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 3E-48

Localized Methane Emission Rate (Liters per Minute)

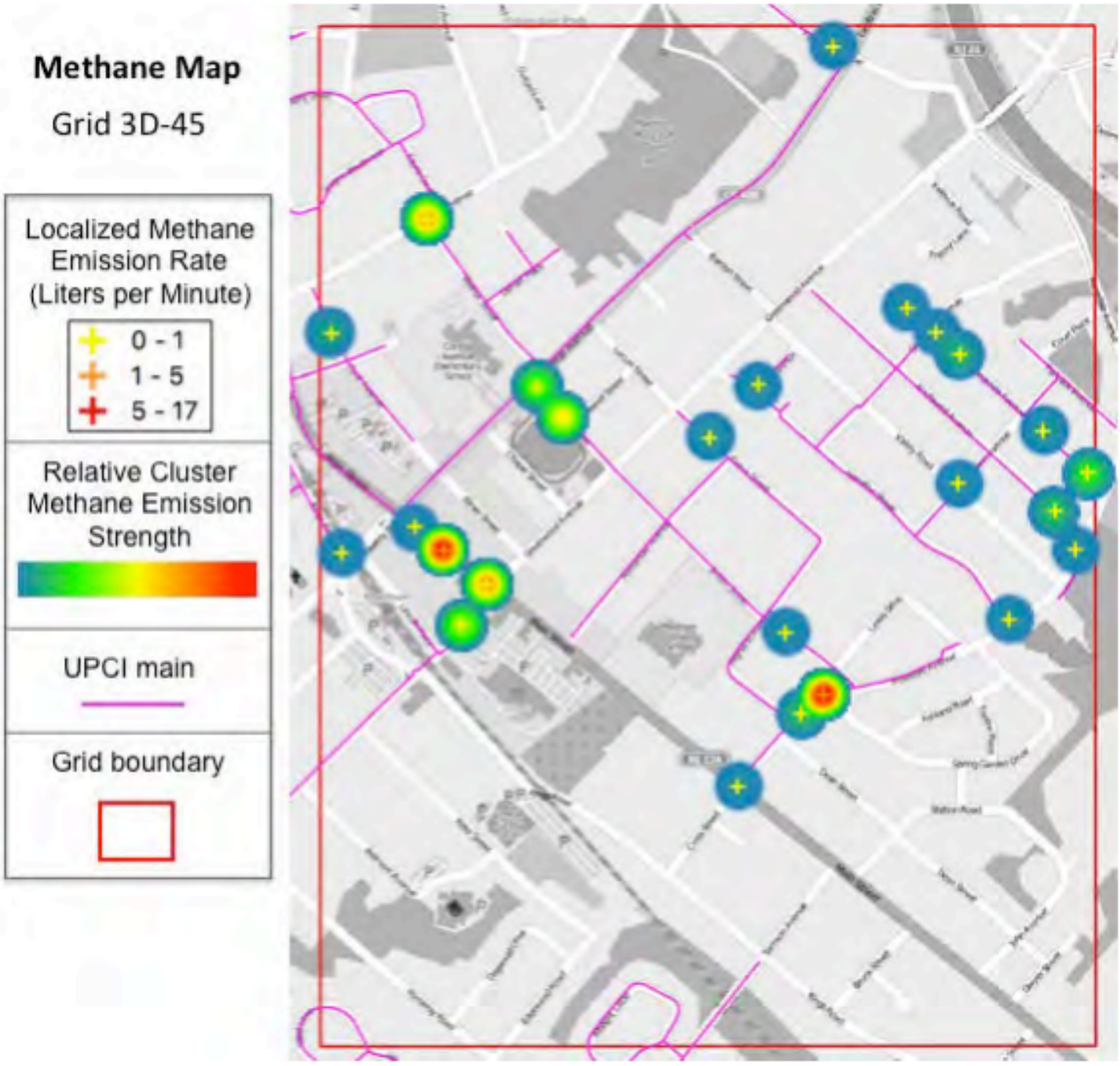
- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary




Methane Map
Grid 3D-45
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 3B-44

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2Y-41

Localized Methane Emission Rate (Liters per Minute)

0 - 1
1 - 5
5 - 17

Relative Cluster Methane Emission Strength

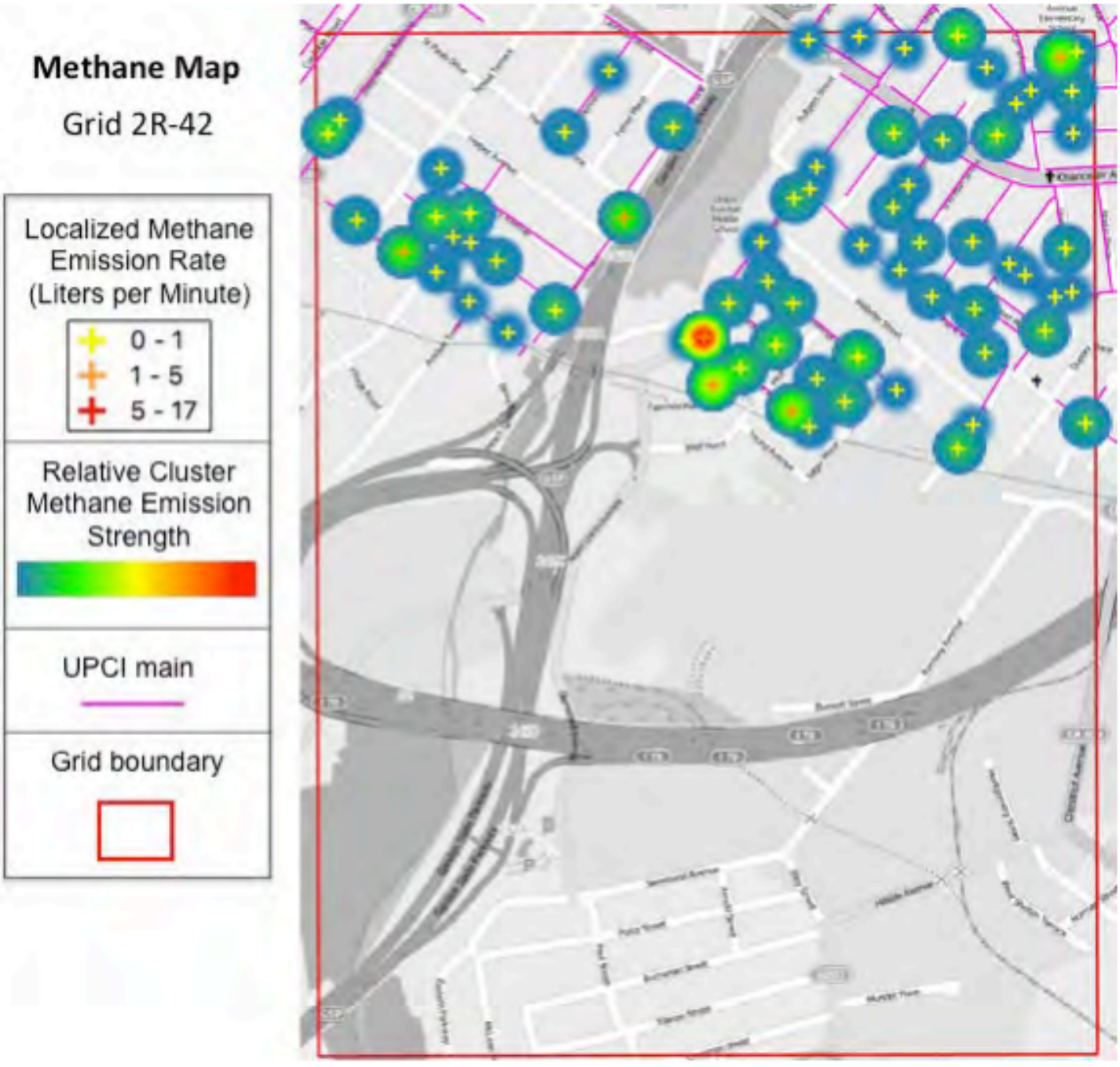
UPCI main

Grid boundary



Methane Map

Grid 2R-42



Methane Map

Grid 2K-43

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2C-44

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2C-45

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2Y-48

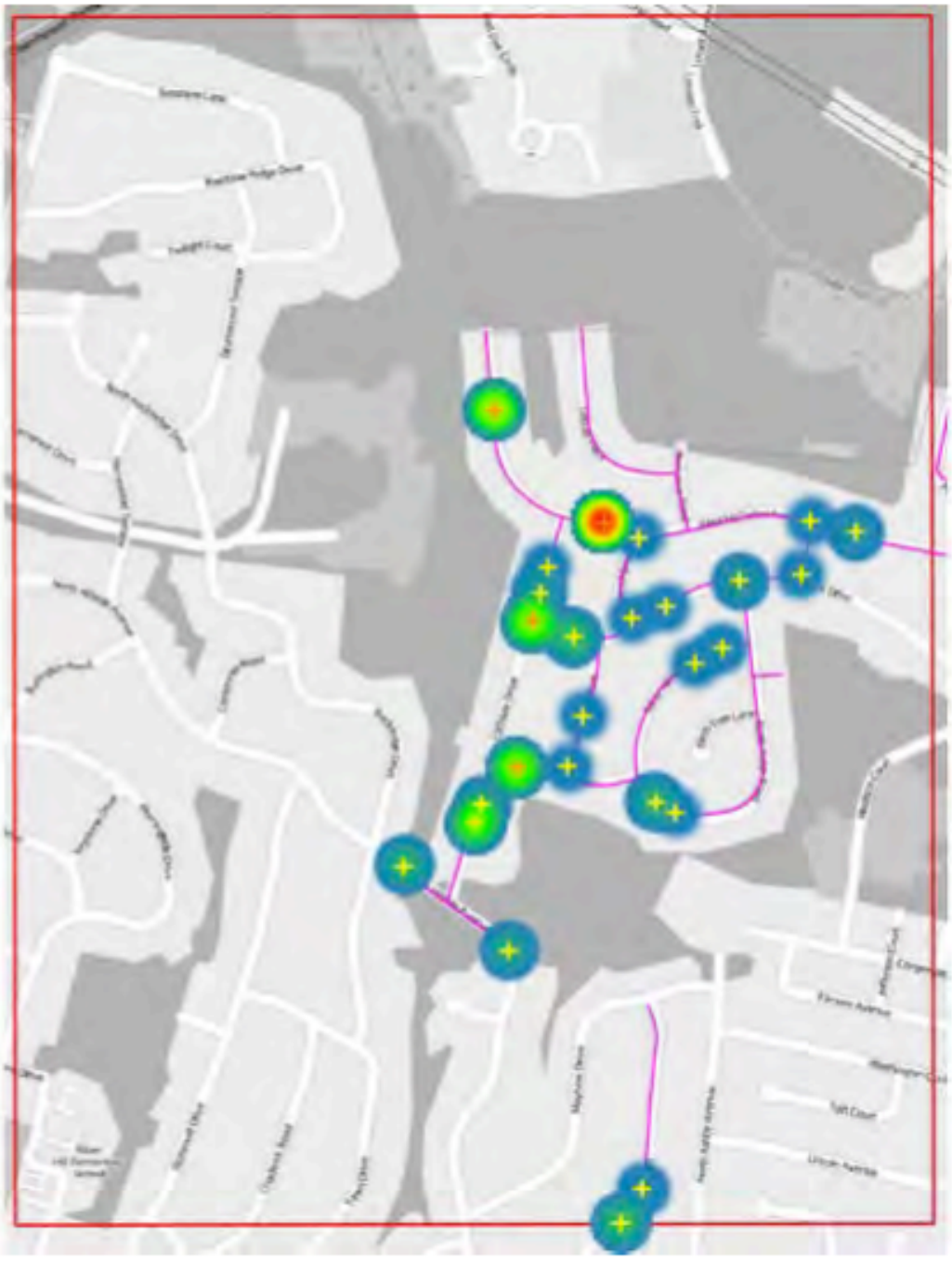
Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2R-48

Localized Methane Emission Rate (Liters per Minute)

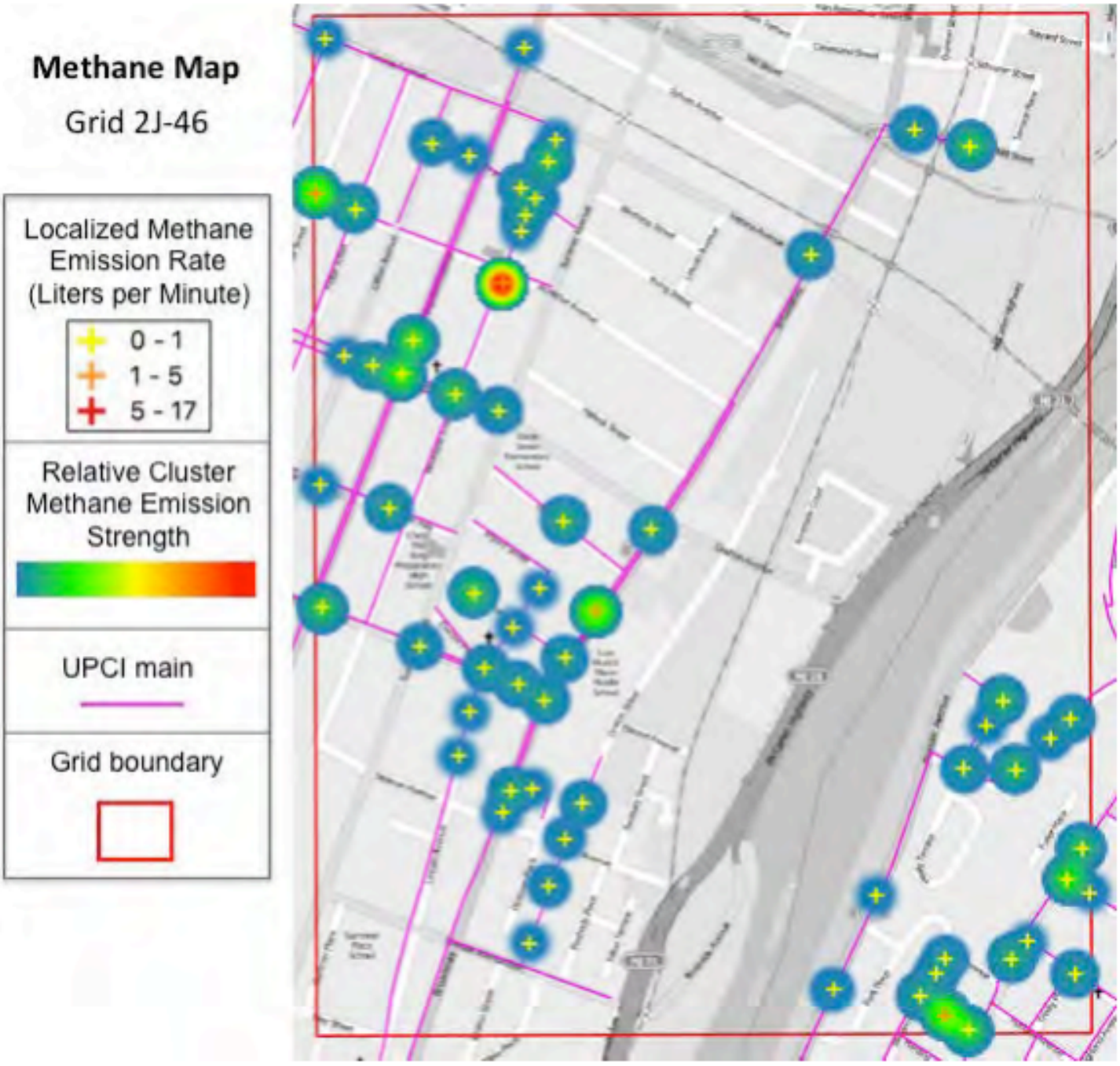
- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary




Methane Map
Grid 2J-46
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 2C-48

Localized Methane Emission Rate (Liters per Minute)

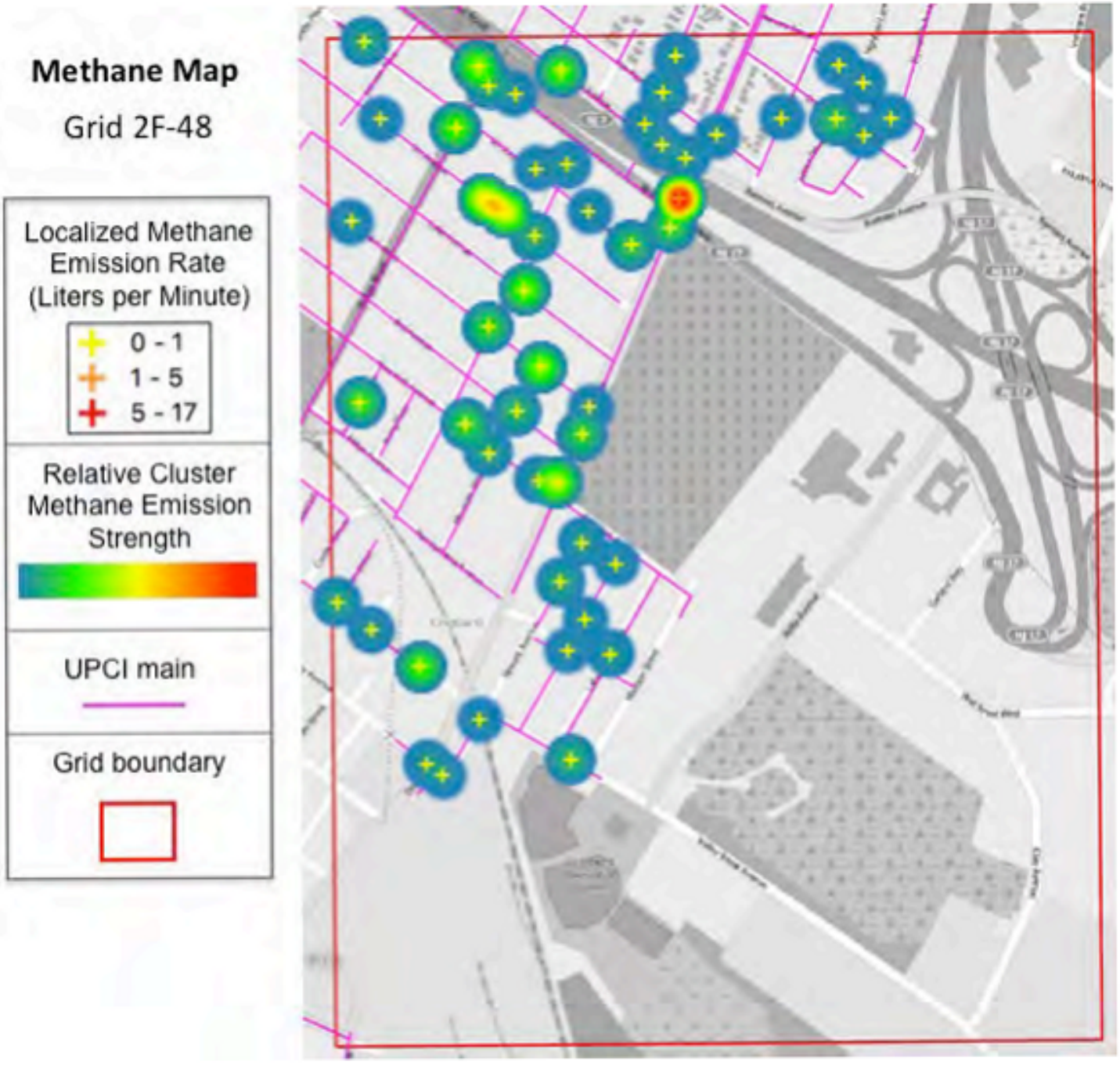
- 0 - 1
- 1 - 5
- 5 - 17

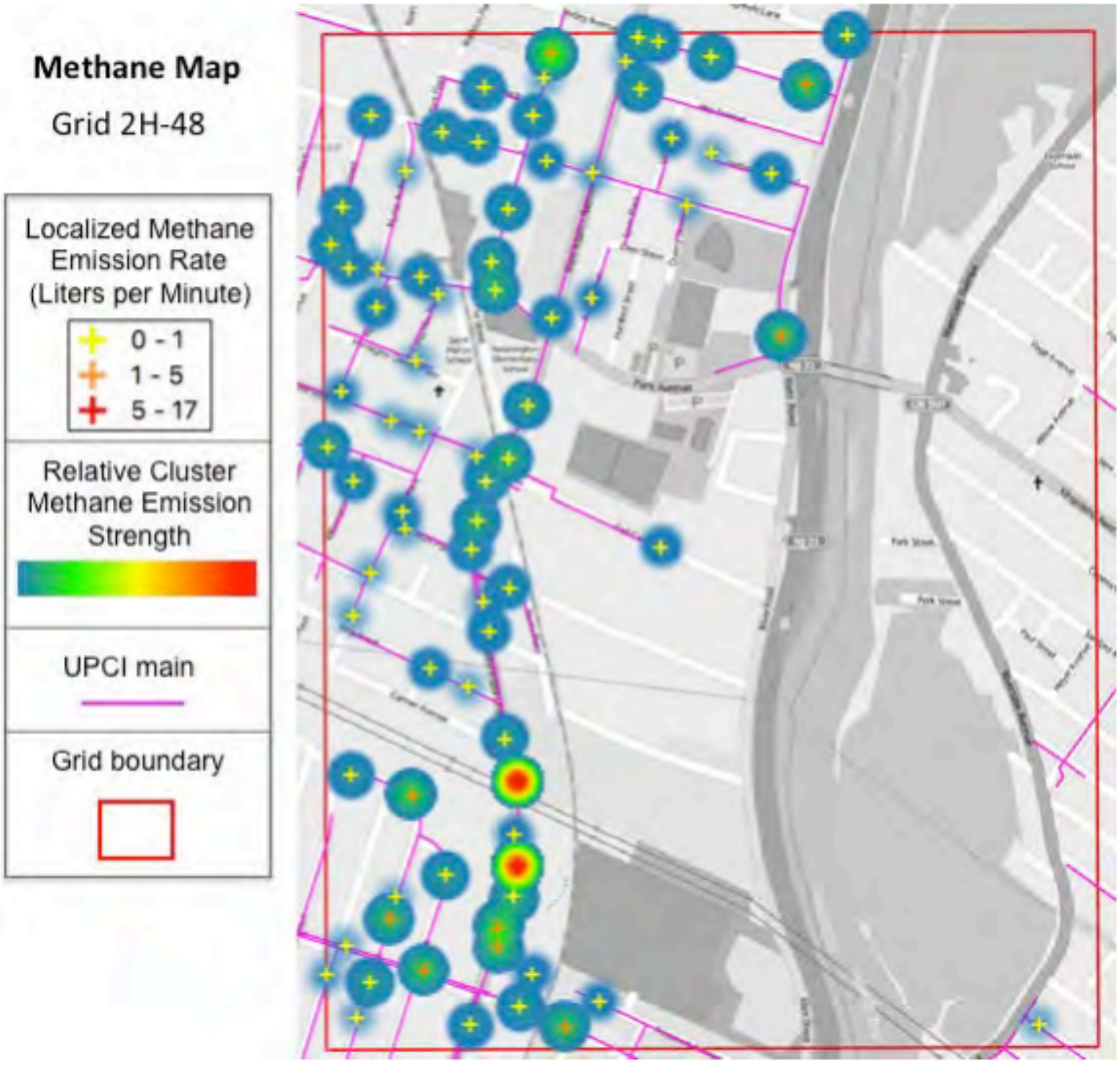
Relative Cluster Methane Emission Strength

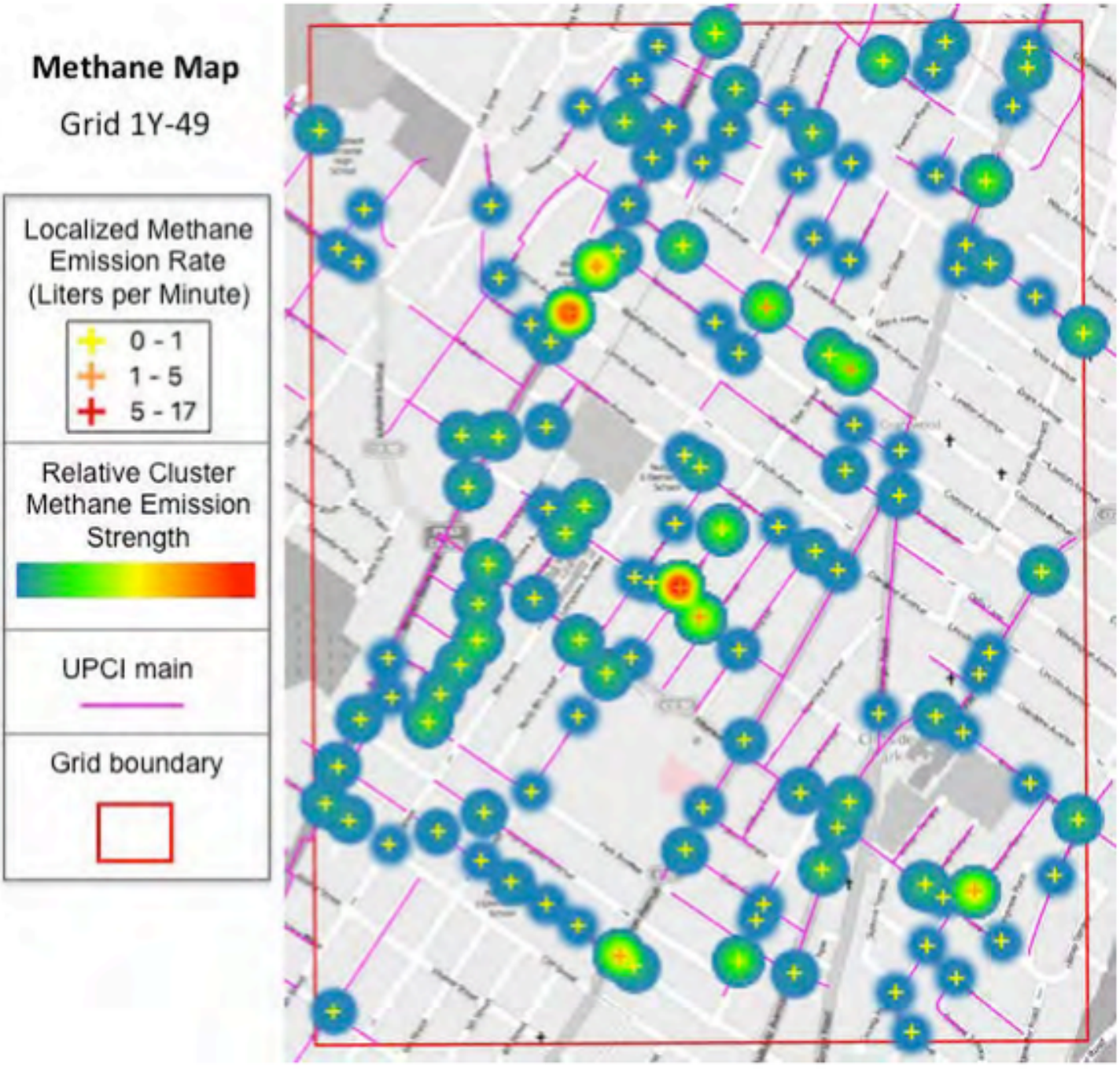
UPCI main

Grid boundary




Methane Map
Grid 2F-48
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary


Methane Map
Grid 2H-48
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary


Methane Map
Grid 1Y-49
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 2J-50

Localized Methane Emission Rate (Liters per Minute)

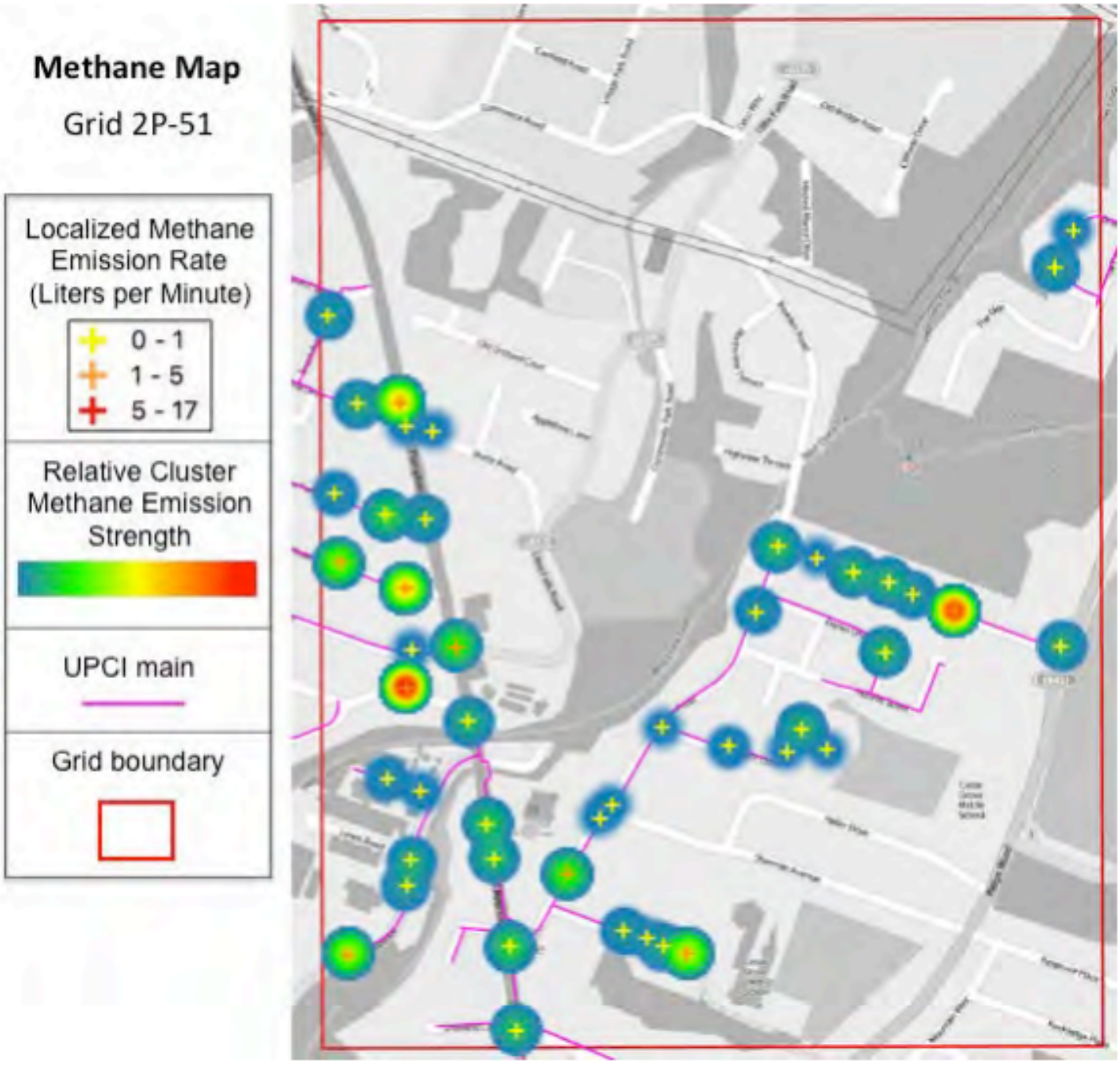
- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary




Methane Map
Grid 2P-51
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 2L-52

Localized Methane Emission Rate (Liters per Minute)

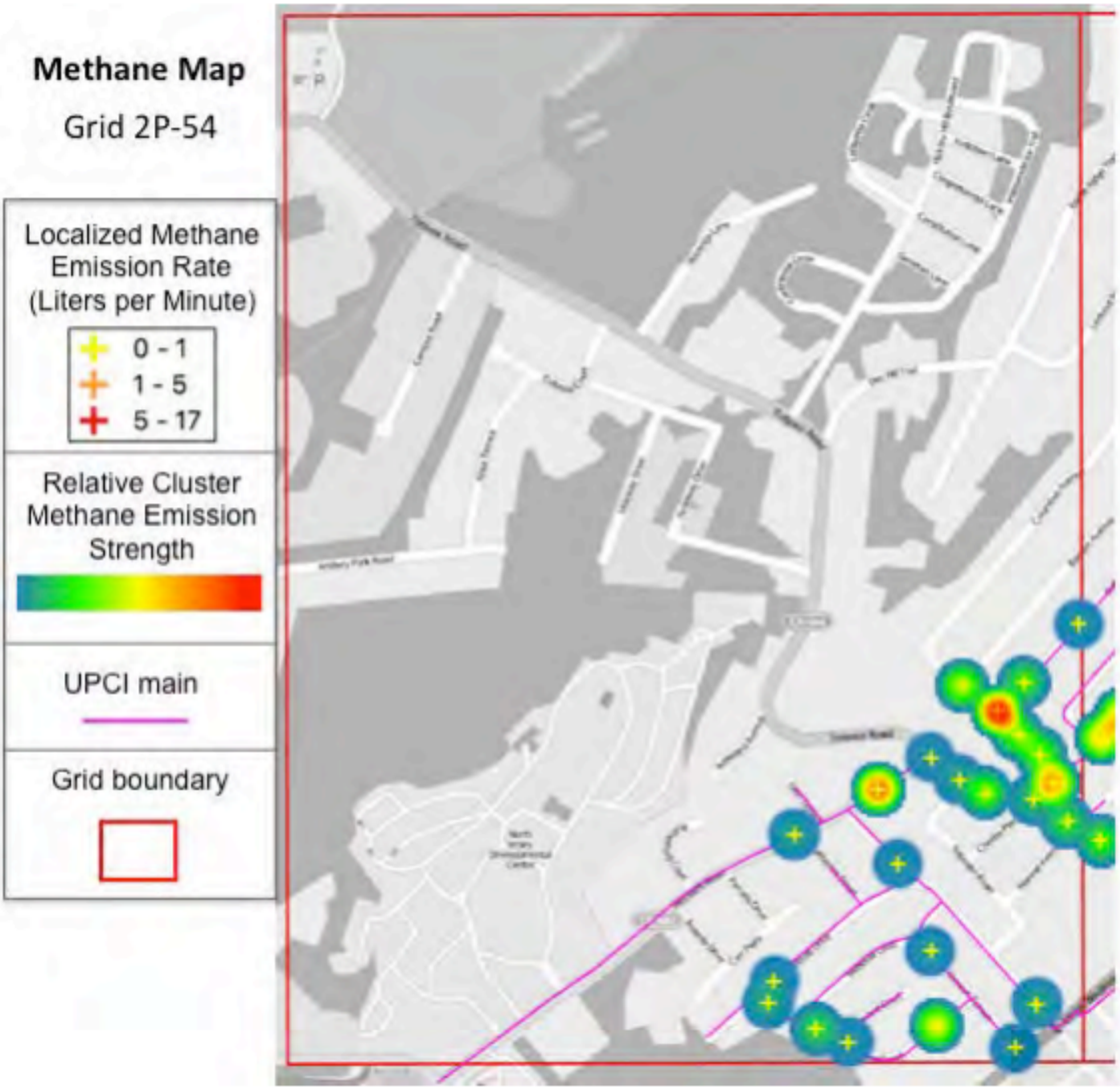
- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

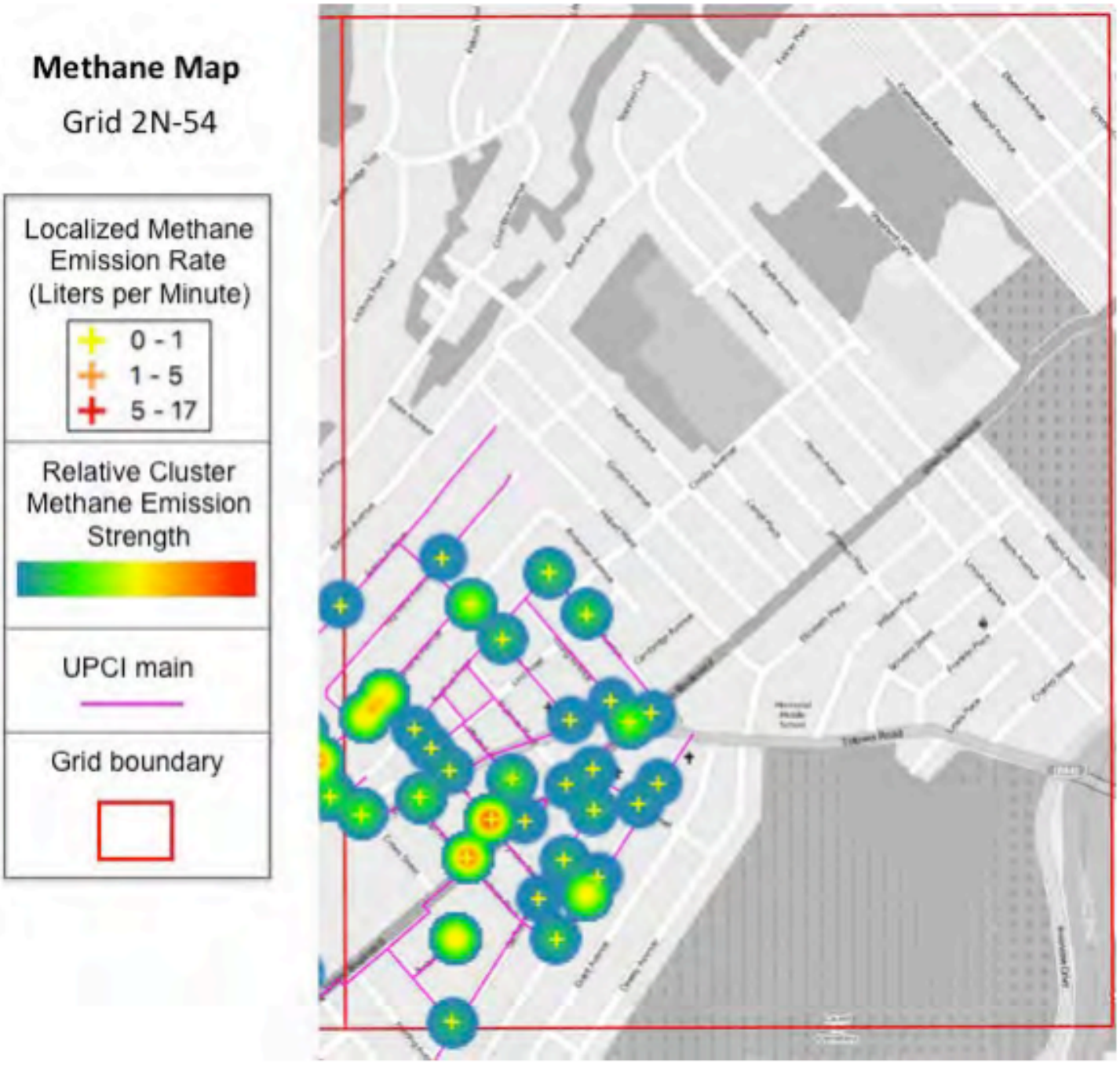
Grid boundary




Methane Map
Grid 2P-54
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 2N-54



Methane Map

Grid 2K-54

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2J-54

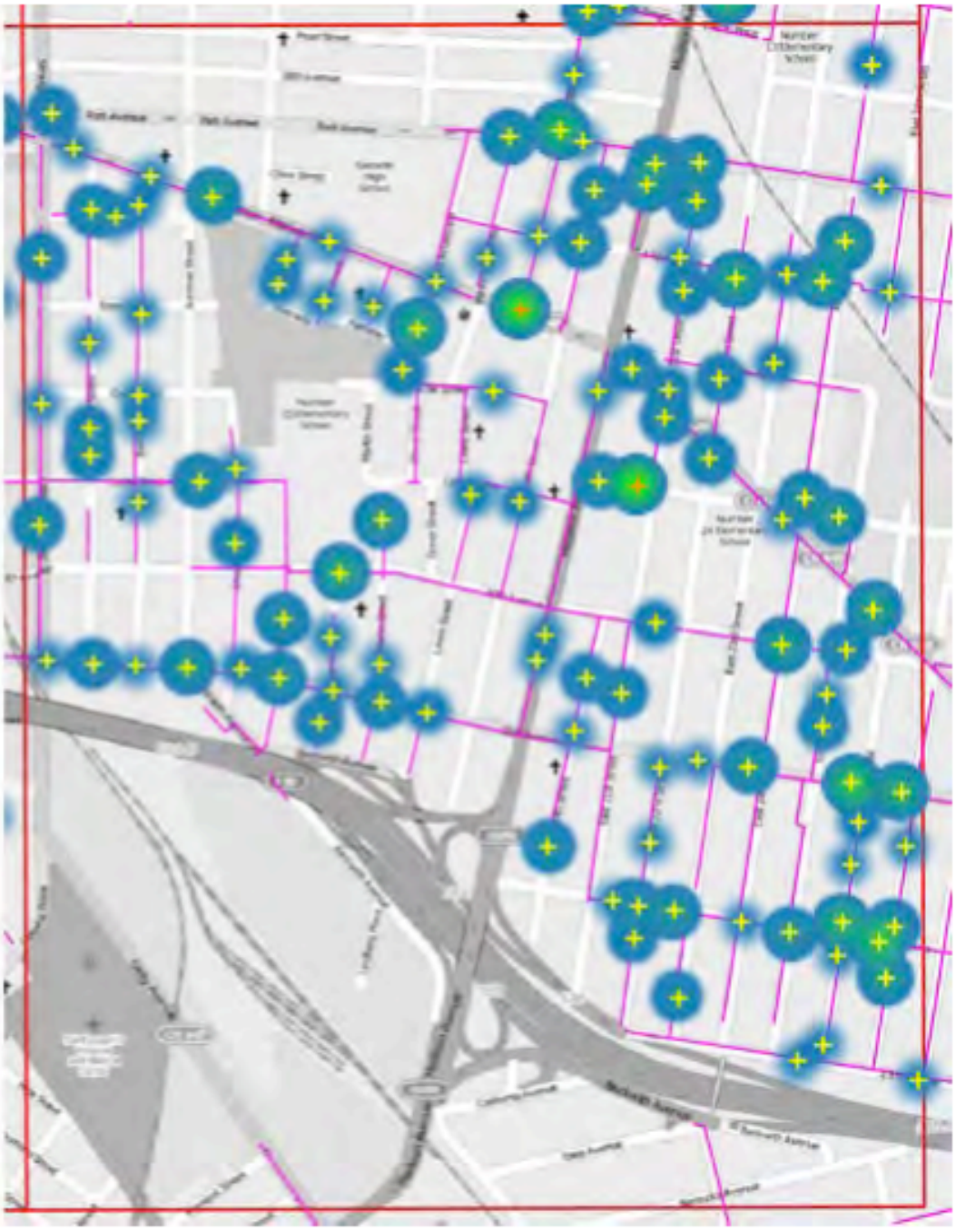
Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2J-55

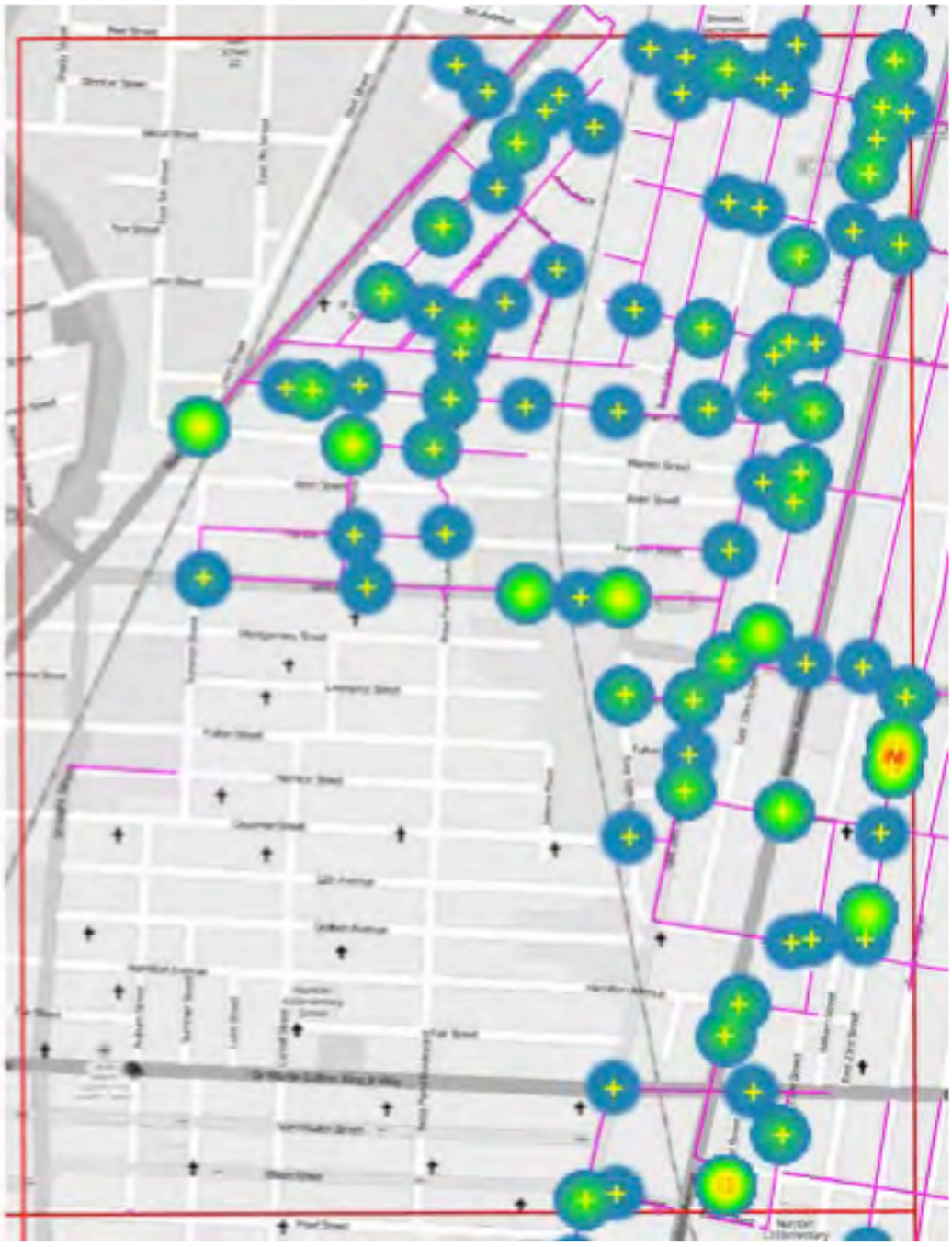
Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2F-53

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 1U-51

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2L-56

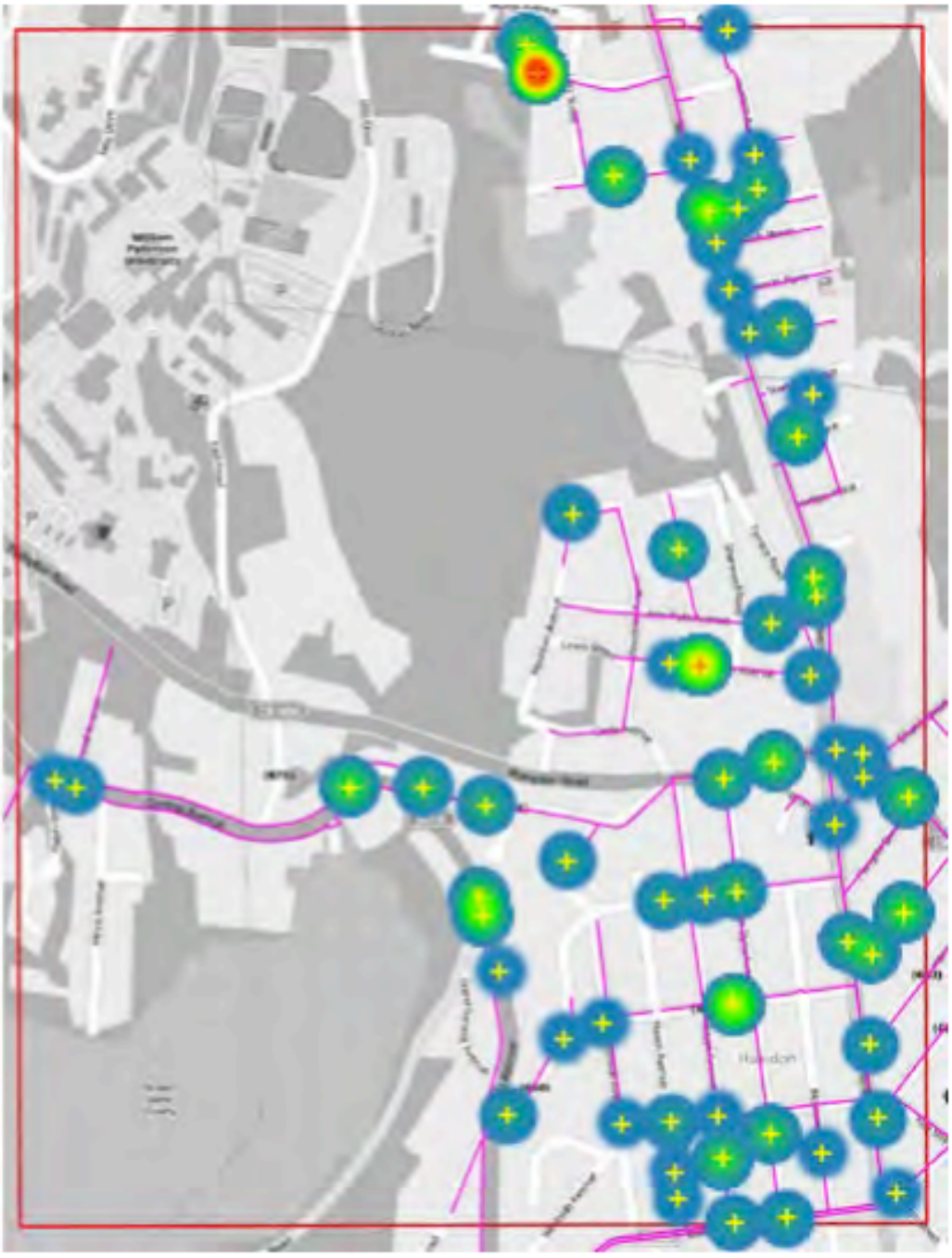
Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

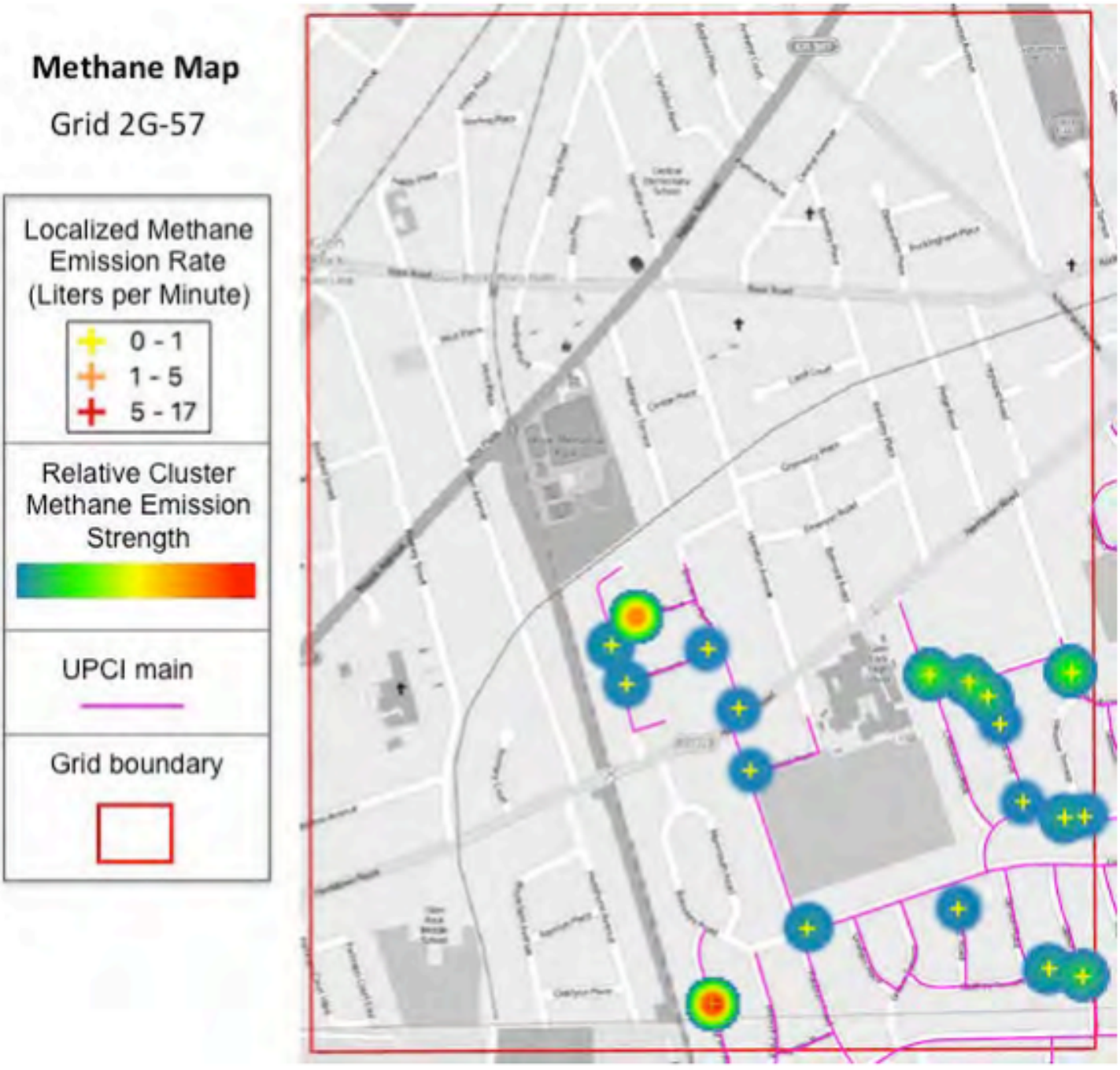
UPCI main

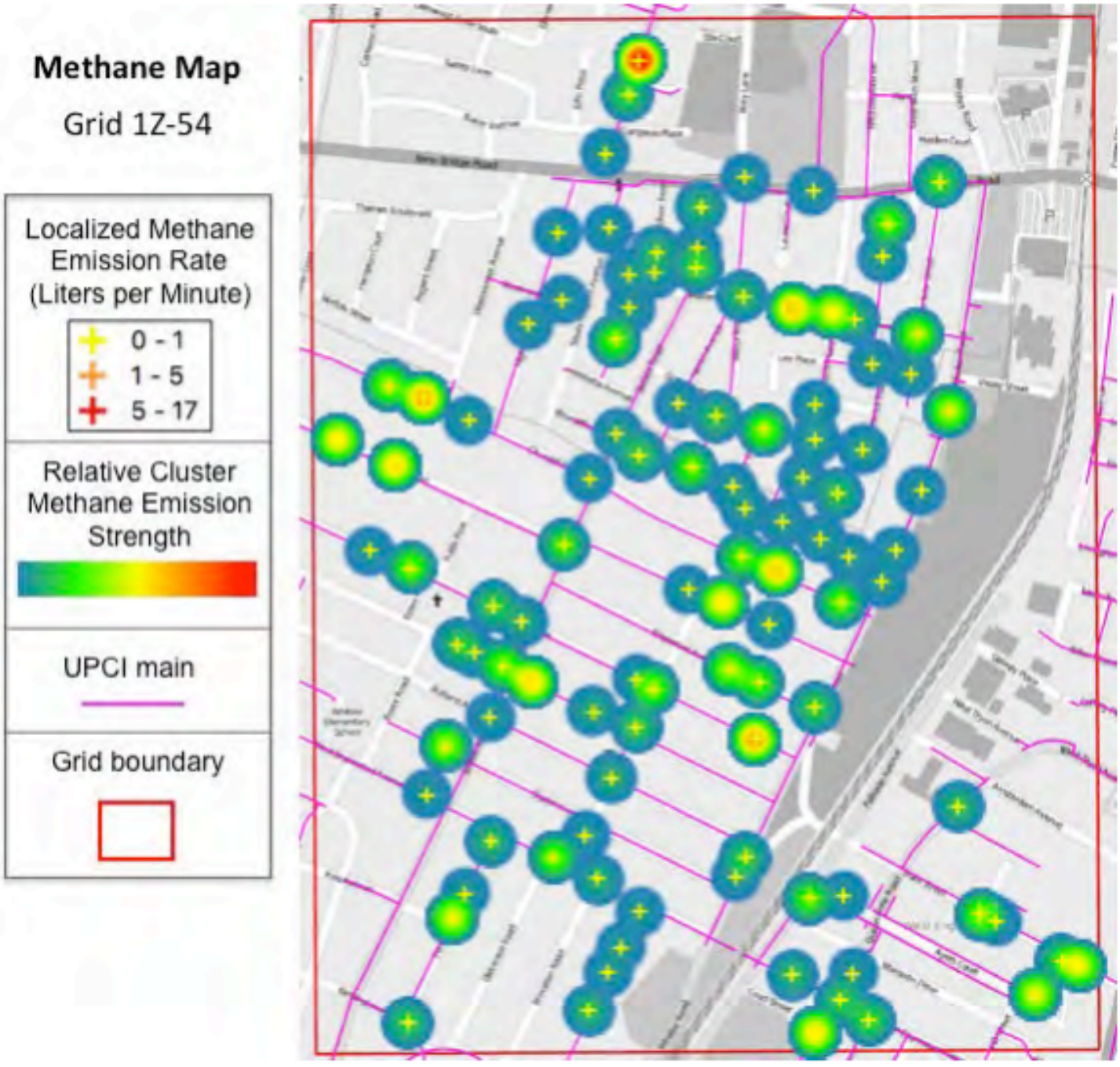
Grid boundary

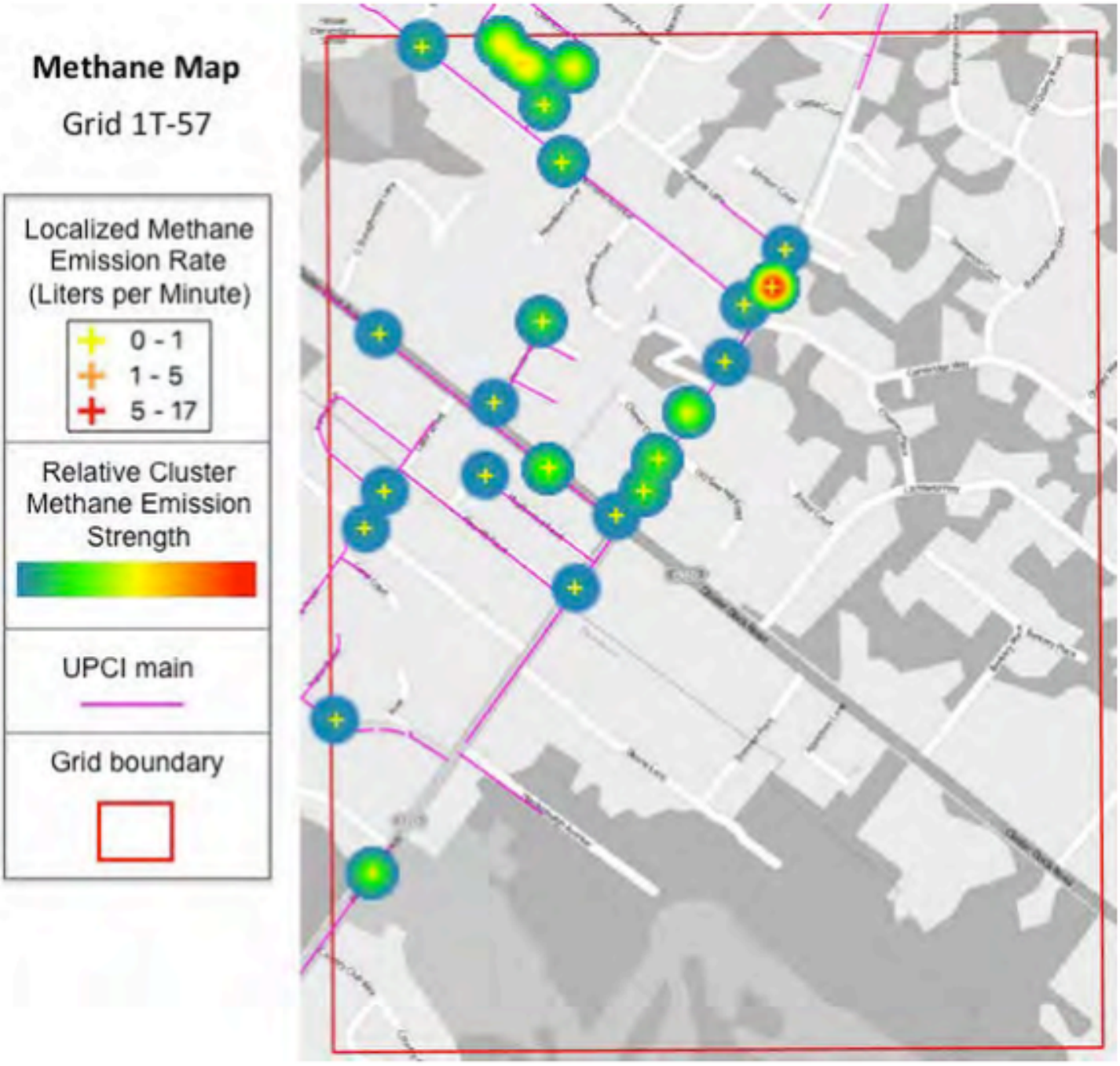


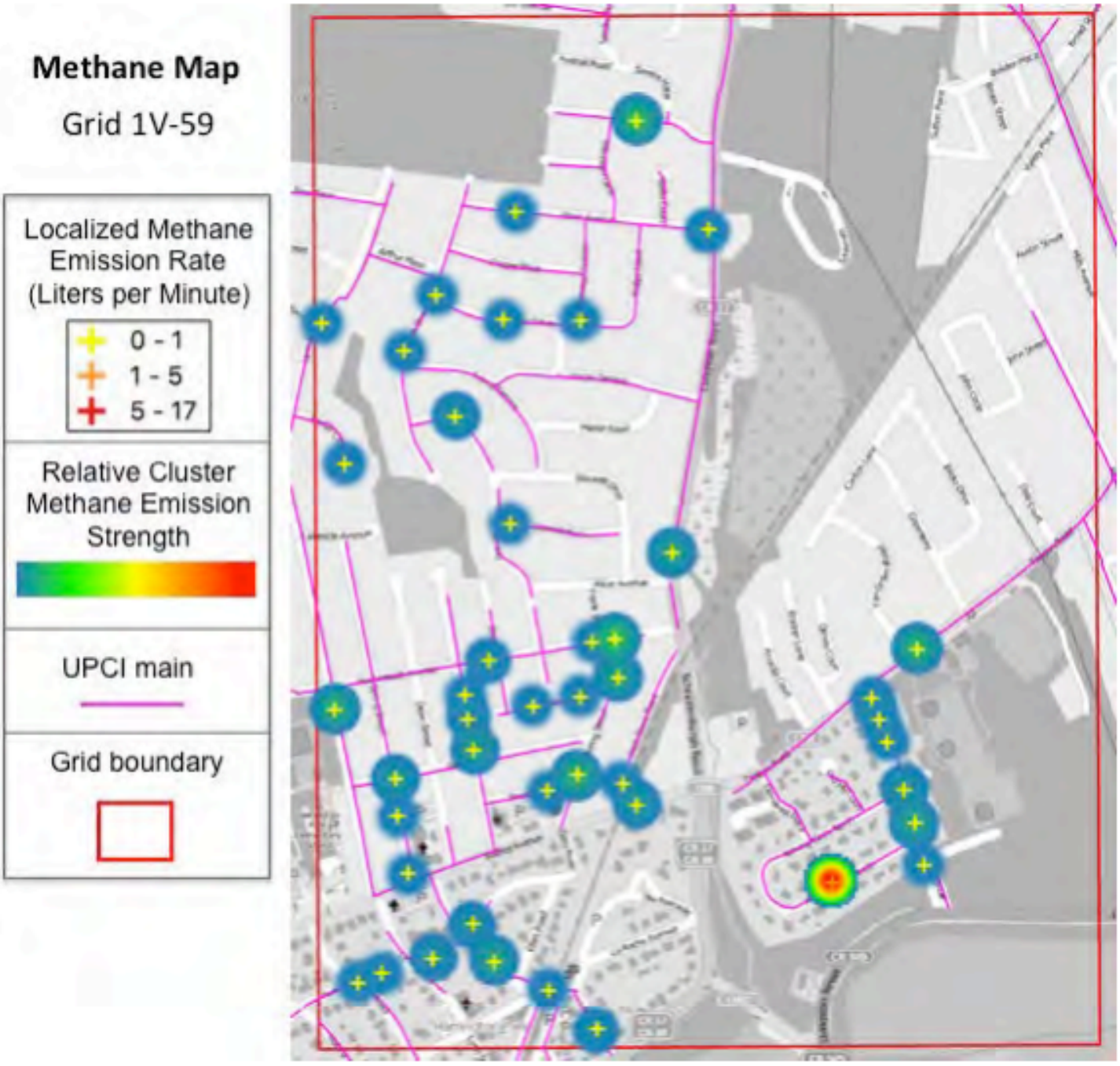
Methane Map

Grid 2G-57



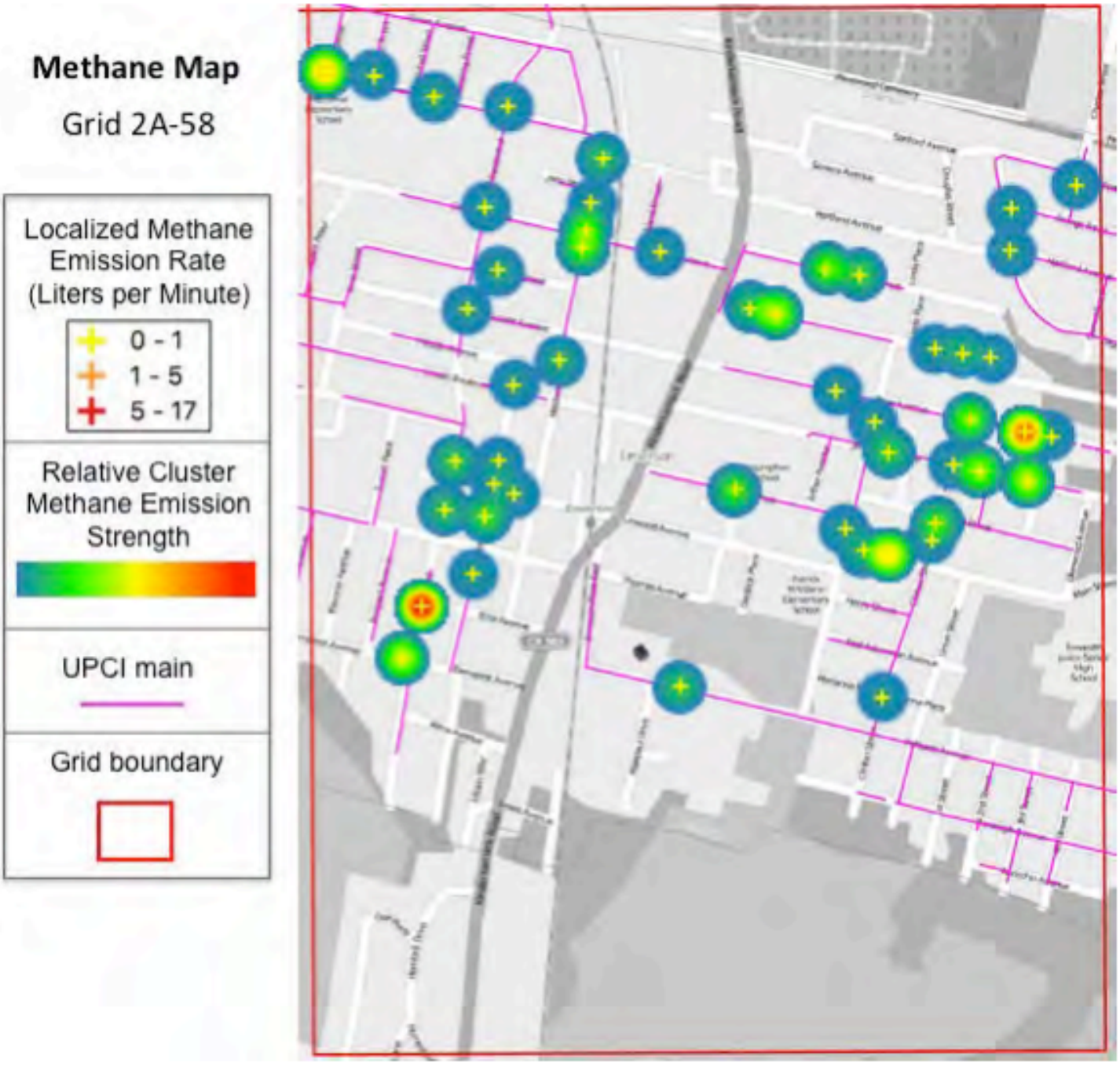

Methane Map
Grid 1Z-54
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary


Methane Map
Grid 1T-57
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary


Methane Map
Grid 1V-59
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

Methane Map

Grid 2A-58



Methane Map

Grid 2B-59

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2C-60

Localized Methane Emission Rate (Liters per Minute)

- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary



Methane Map

Grid 2C-02N

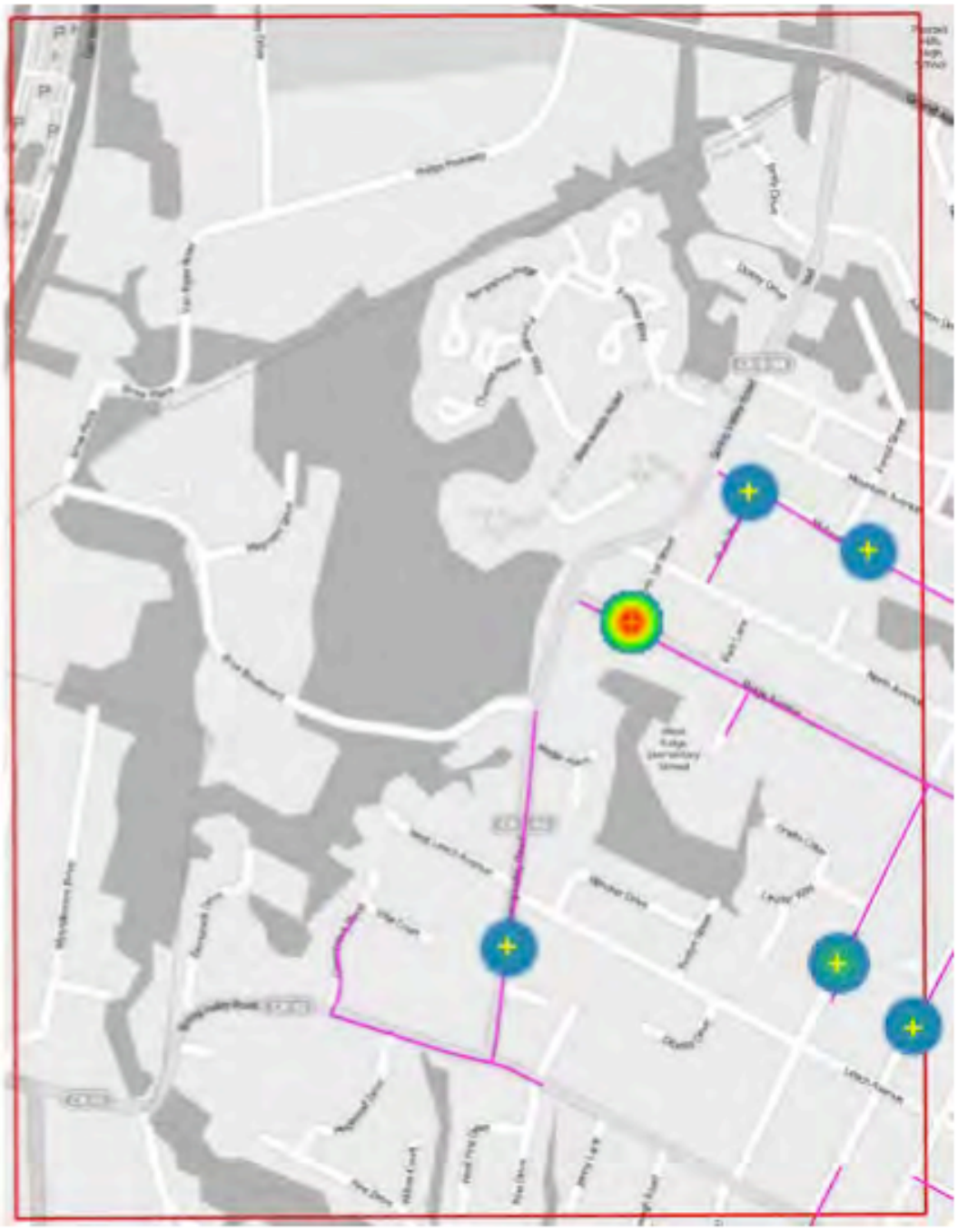
Localized Methane Emission Rate (Liters per Minute)

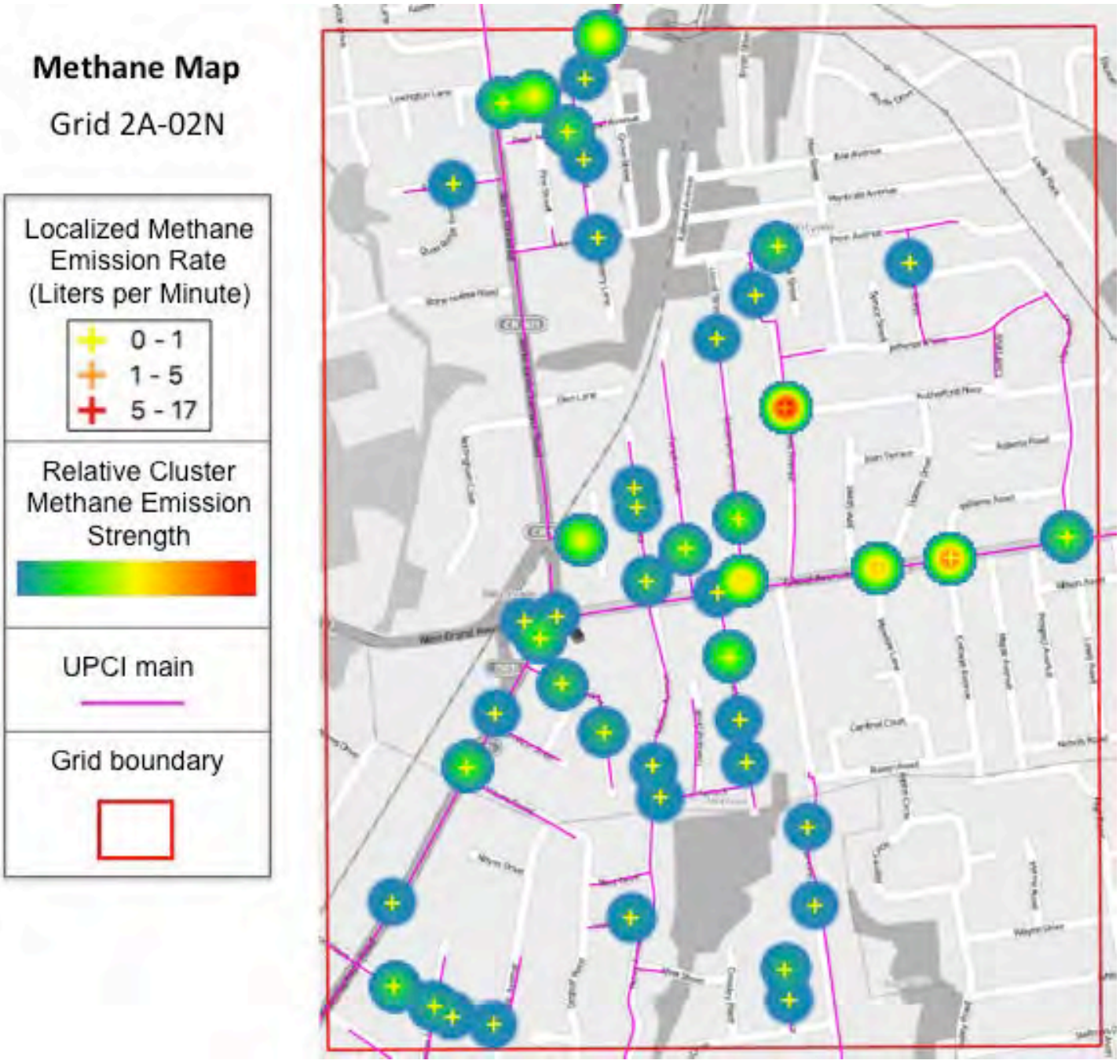
- 0 - 1
- 1 - 5
- 5 - 17

Relative Cluster Methane Emission Strength

UPCI main

Grid boundary




Methane Map
Grid 2A-02N
Localized Methane Emission Rate (Liters per Minute)
0 - 1
1 - 5
5 - 17
Relative Cluster Methane Emission Strength
UPCI main
Grid boundary

GSMP II UPCI Grid Ranking

A grid ranking process has been developed based on the Company's Hazard Risk Index Model. The approach is similar to the hazard ranking method used in GSMP I. PSE&G targets the replacement of its riskiest gas assets through the use of a ranking methodology that prioritizes main segments with the highest risk, through the use of the Hazard Index. The Hazard Index is based on a predictive model constructed from leak history "environmental factors" that include: building setback, number of underground utilities, demographic area (urban, suburban, rural), building types (industrial, commercial, or residential), and asset information (pipe diameter, operating pressure). Through the "weighted leak history" factor, past main breaks are considered and weighted based on how recently they occurred. Each map grid is evaluated by adding the hazard indexes for the individual utilization pressure segments within the grid and dividing them by the total miles of utilization pressure cast iron in the grid, arriving at a hazard index per mile for each map grid. Consistent with the hazard index per mile results, grids are ranked by highest to lowest and then placed into A, B, C and D priority grids categories. Grids with a Hazard Score over 15 are treated as the highest priority (A). B grids have a score between 15 and 10, C grids have a score between 10 and 5 and D grids have a score lower than 5.

Per the GSMP II Stipulation, PSE&G retained the services of Picarro to conduct and complete a methane leak survey of approximately 280 miles of UPCI located within the highest ranked B grids during the Fall of 2018. The 280 miles of main correlated to 44 grids that were surveyed. Consistent with the approach for GSMP I, an "Estimated Flow Rate per Mile (Liters/minute/mile)" was determined for each of the surveyed grids. Once the results for the 44 grids were determined, a discussion between PSE&G and the Environmental Defense Fund (EDF) occurred on Dec 4th 2019 to determine a threshold for accelerating the subset of B grids with significant methane emissions. A value of 4.5 L/min/mi was agreed to verbally at this time. Having not received additional feedback from the EDF, PSE&G moved forward with this value to sub-prioritize the surveyed grids. Per the stipulation, these grids were ranked as the highest priority work after the A grids. Planning discussions with municipalities occurred for all grids accelerated by the methane mapping survey. In a few isolated cases, factors like project feasibility, cost and construction efficiency altered the outlined prioritization and has been documented.





Exhibit __ (JF-4):

PSE&G Presentation "Replacement Main Prioritization: A Practical Application of Using Risk and Methane Emissions" (May 2, 2019)

Replacement Main Prioritization

A Practical Application of Using Risk and Methane Emissions

MAY 2, 2019

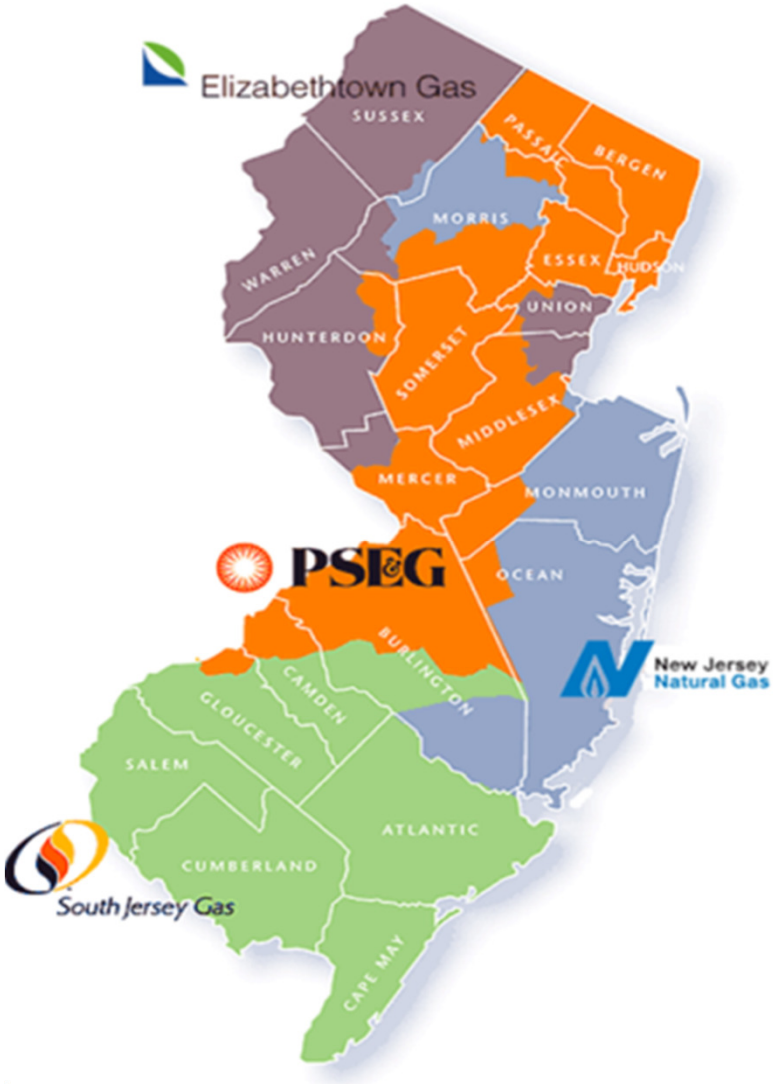
WADE MILLER, PE- DIRECTOR OF GAS TRANSMISSION & DISTRIBUTION ENGINEERING
ANDREW ARGESKI, PE- GAS PLANNING & DESIGN MANAGER

PSEG

We make things work for you.

We have the **energy** to make things work ... for you.

Getting to know PSE&G



- 6th Highest Gas Utility in US sales
- Serves 10 of the top 15 cities in NJ
- ~2,400 employees
- 12 District Headquarters
- 17,955 miles of gas distribution main
- 57 miles of gas transmission main
- 1.2 million gas services
- 1.8 million gas customers
- Sales volume growth: 1% per year

What is the Gas System Modernization Program (GSMP)?

- Accelerated cast iron and unprotected steel main and service replacement program
- Upgrades legacy low (utilization) pressure systems to medium pressure
- Relocates inside meter sets to outside
- Installs excess flow valve (EFV) safety devices
- Supports DOT focus on replacing the highest risk, most leak prone facilities



Continued replacement at these levels would take 25 years to replace/rehabilitate all the cast iron and unprotected steel

Gas System Modernization Program

- PSE&G currently operates and maintains over 4,400 miles of cast iron and unprotected steel gas distribution main.
- The program provides for investment and clause recovery of Utilization Pressure Cast Iron (UPCI) and Unprotected Steel replacement main, services, and associated uprating of plastic and protected steel in targeted areas
  - GSMP I started in 2016 (3 year term - $900M)
  - GSMP II started in 2019 (5 year term – $1.9B)
- Stipulated Base CapEx spend requirement associated with the program approval
  - Includes High Pressure Cast Iron (HPCI), UPCI, unprotected steel main and service replacement
  - Includes program and stipulated base inside meter set relocations
- Total ~170 miles of main replacement per year in Program and Stipulated Base
- The first two approvals are the beginning phases of a long-term 25 year replacement strategy for cast iron and unprotected steel mains
- Benefits:
  - Methane emission reduction is estimated at 30,000 metric tons of CO2 equivalent per year*
  - Medium pressure system allows usage of high efficiency appliances by customers
  - Includes installation of excess flow valve safety devices where applicable

* EPA SUBPART W METHODOLOGY.

GSMP Stipulation

The replacement of mains in the Program shall follow the prioritization based on the grid based Leak Hazard Indices developed by PSE&G using its Hazard Assessment model.

"...Recognizing that considering methane emission flow volume (i.e., emission size) as part of prioritization will reduce the amount of natural gas lost from emissions to the benefit of customers, and reduce the environmental impacts of such emissions, the Signatories agree that for grids with comparable Hazard Index/Mile, available methane emissions survey data estimating flow volumes, as prepared by the Environmental Defense Fund using Program plans, system information and maps provided by PSE&G, will be used, as appropriate, in sub-prioritizing replacement activities…"

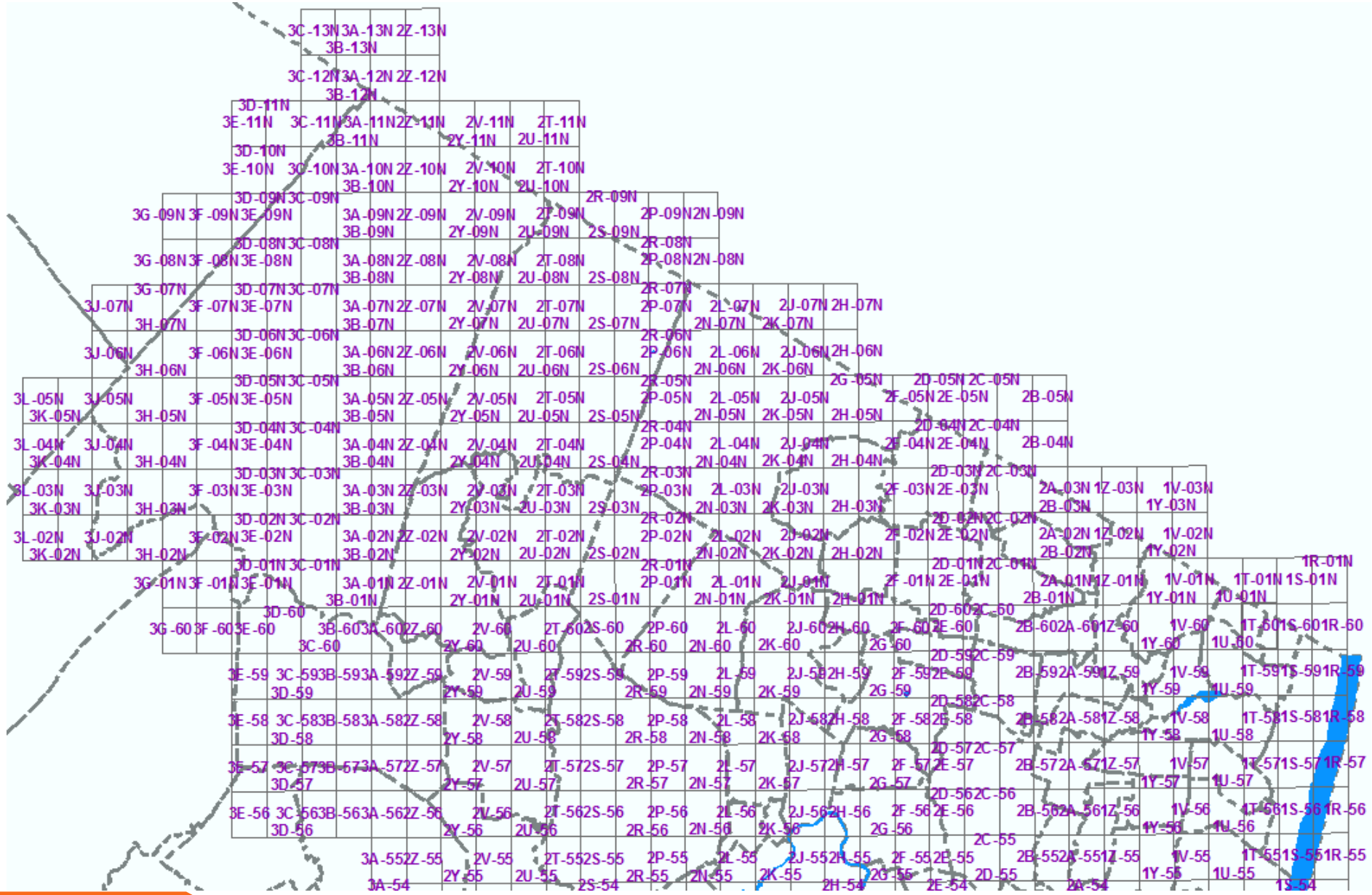
Accelerated UP Cast Iron (UPCI) Replacement

- Goal - Replace priority areas most efficiently
  - Highest potential hazard
  - Contiguous area for construction efficiency

- Map grid system utilized
  - 1 square mile area
  - 1 – 20 miles of low pressure cast iron per grid
  - Similar environmental conditions



PSE&G Grid Mapping System



Prioritization of UPCI Replacement Main

- Hazard Index (HI) rankings used to express and compare relative hazard for main segments having a history of breaks.
- Factors used in the calculation
  - Hazard Index = Weighted Break History (WBH) x Environmental Index (E)
  - WBH = The sum of the factors multiplied by the number of annual break repairs for each period (factors higher for recent breaks)
  - Environmental Index evaluates the environmental conditions at the main segment location that may affect the relative hazard of a break and is based upon the following factors
    - Building Density
    - Operating Pressure
    - Building Occupancy
    - Underground Utility
    - Building Set-back
    - Nominal Pipe Size
- Mileage is based upon total low pressure cast iron mileage in grid



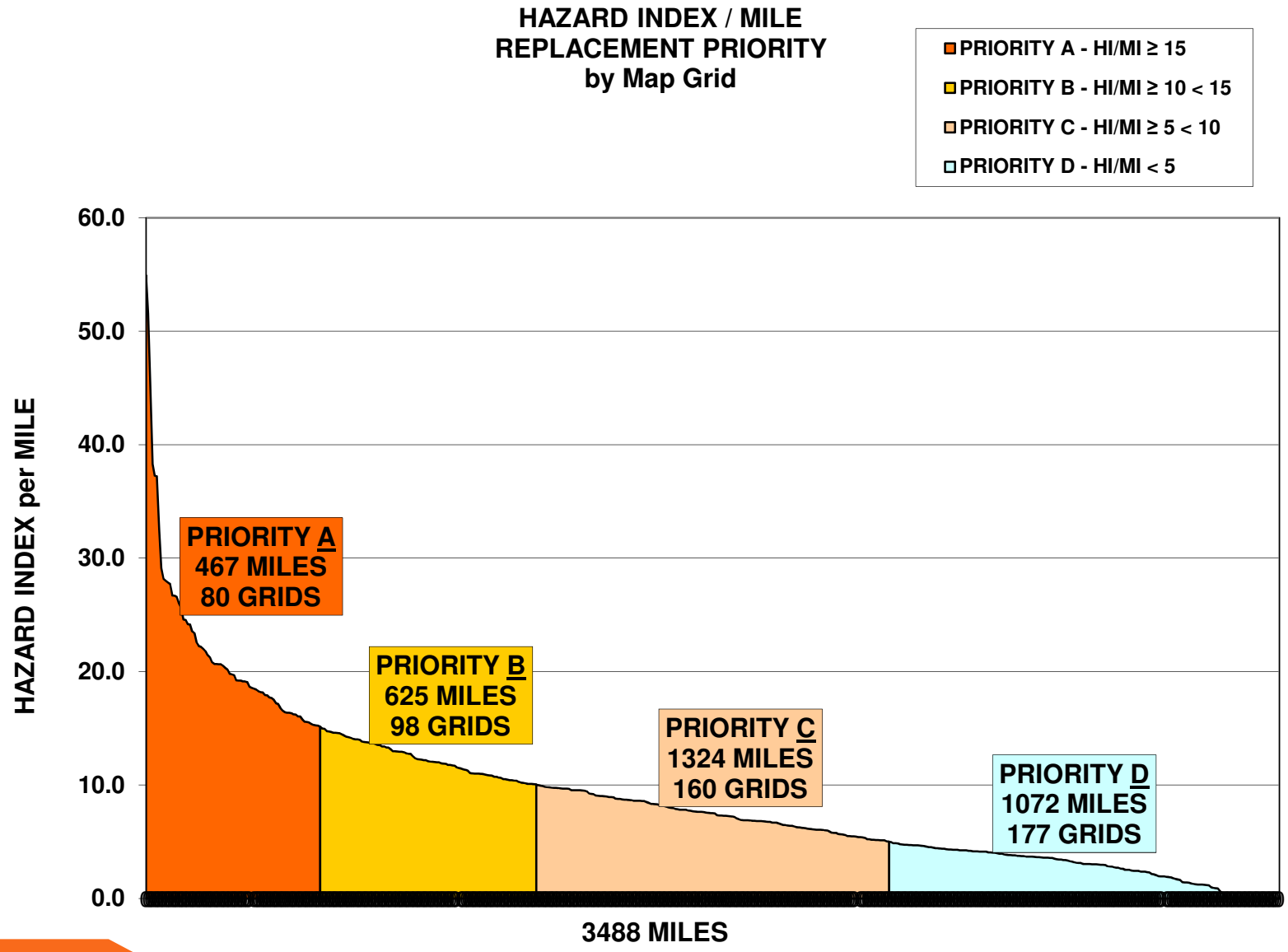




Prioritization of UPCI Replacement Main (cont'd)

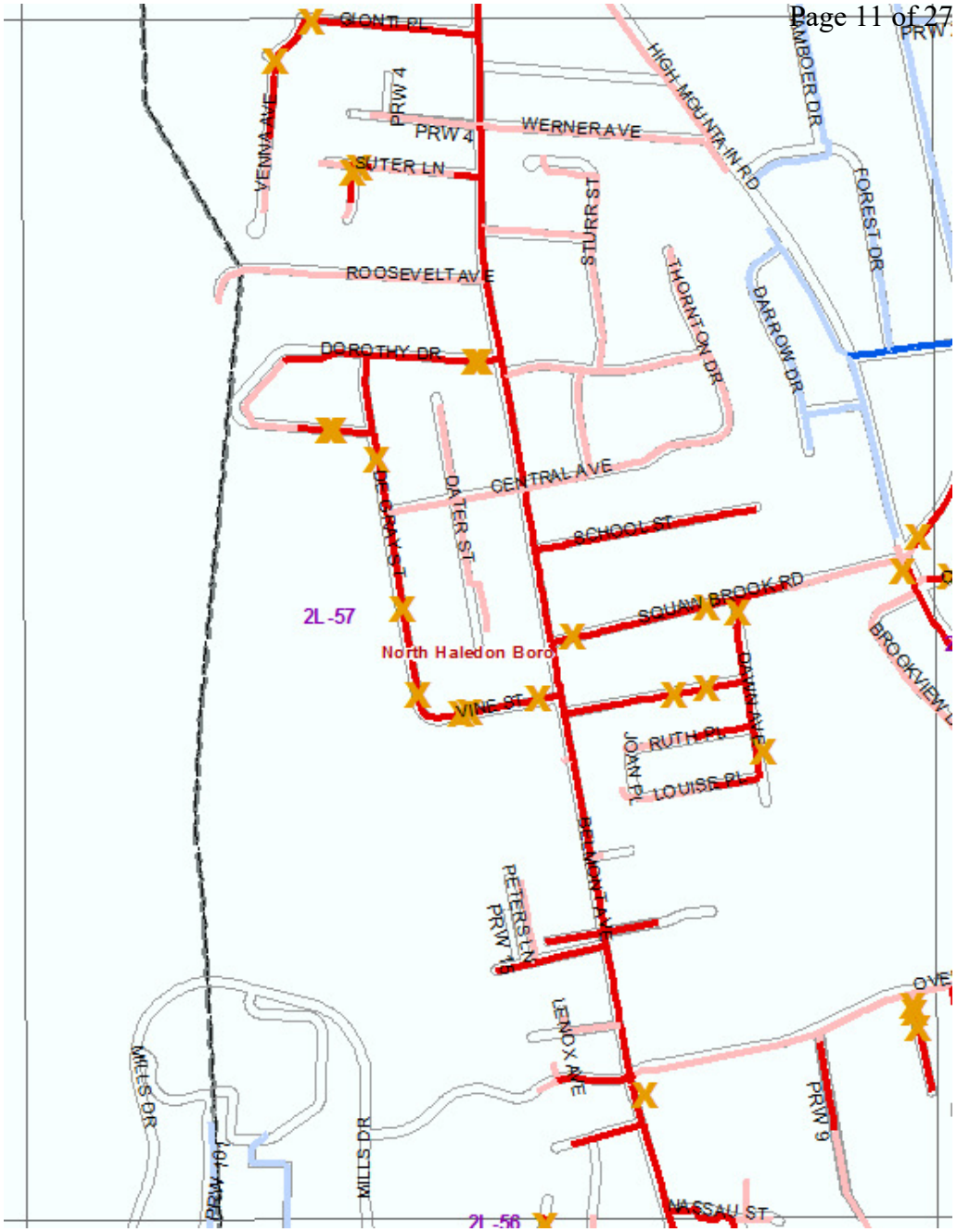
- Mains with break history - Hazard Index
- Individual segments within a grid are summed to obtain total hazard index for the grid
- Miles of UPCI main in grid are summed
- Hazard score divided by miles gives HI/Mi score
- Map Grids ranked by HI/Mi

GSMP I - UP Cast Iron Main Prioritization



Grid 2L-57 (Rank 2)
UP CI = 3.8 miles
HI/MI = 45.4

- UP Cast Iron
- EP Cast Iron
- UP Plastic and Steel
- EP Plastic and Steel
- Regulator
- CI Break



Hazard Index – Grid 2L - 57

District	Street	Municipality	Install Year	Main Size	Main Type	Pressure	Segment Length	B	P	O	U	S	Last Repair Date	Number of Breaks	WBH	BPOU/S	Env Index E	Hazard Index	Wall Map Grid
DGOK	VINE ST	Haledon Boro	1900	6	CI	UP	700	4	1	4	2	1.5	1/8/2014	3	12	21	2.3012	27.615	2L-57
DGOK	DE ROON AVE	Haledon Boro	1900	4	CI	UP	458	8	1	4	3	1	4/4/2012	3	9	96	3.0307	27.2765	2L-57
DGOK	MORNINGSIDE AVE	North Haledon Boro	1953	4	CI	UP	929	8	1	4	3	1	1/28/2013	2	8	96	3.0307	24.2458	2L-57
DGOK	BELMONT AVE	North Haledon Boro	1927	8	CI	UP	460	8	1	15	4	1	11/6/2013	1	5	480	4.5596	22.7982	2L-57
DGOK	DE GRAY ST	Haledon Boro	1955	6	CI	UP	1037	4	1	4	3	1.5	2/14/2013	2	7	32	2.5705	17.9933	2L-57
DGOK	DAWN AVE	Haledon Boro	1951	6	CI	UP	426	8	1	4	3	1	1/14/2011	1	3	96	3.3	9.8999	2L-57
DGOK	GIONTI PL	North Haledon Boro	1928	6	CI	UP	885	4	1	4	2	3	2/11/2014	1	5	11	1.841	9.205	2L-57
DGOK	SQUAW BROOK RD	North Haledon Boro	1937	6	CI	UP	962	4	1	4	3	1	1/12/2009	2	2	48	2.8397	5.6794	2L-57
DGOK	MEADOW PL	North Haledon Boro	1954	4	CI	UP	187	4	1	4	4	1	2/22/2010	1	2	64	2.7615	5.523	2L-57
DGOK	DOROTHY DR	North Haledon Boro	1964	4	CI	UP	276	4	1	4	3	1	3/25/1999	2	2	48	2.5705	5.141	2L-57
DGOK	HIGH MOUNTAIN RD	North Haledon Boro	1900	8	CI	UP	109	2	1	4	3	1.5	12/30/2010	1	2	16	2.3012	4.6025	2L-57
DGOK	SUTER LN	North Haledon Boro	1954	4	CI	UP	93	4	1	4	1	1.5	2/20/2010	1	2	11	1.5718	3.1435	2L-57
DGOK	DAWN AVE	North Haledon Boro	1951	6	CI	UP	267	4	1	4	3	1	12/8/2003	1	1	48	2.8397	2.8397	2L-57
DGOK	DOROTHY DR	North Haledon Boro	1957	6	CI	UP	682	2	1	4	2	3	1/8/2001	2	2	5	1.3807	2.7615	2L-57
DGOK	VENNA AVE	Haledon Boro	1929	6	CI	UP	325	2	1	4	2	3	2/2/2009	1	1	5	1.3807	1.3807	2L-57
														Total Hazard Score				170.1051	
														Total CI Miles in Grid				3.75	
														Hazard Index Per Mile				45.36	

Top 20 Hazard Index/Mile

UPCI GRID	UPCI MILES	2014 HAZARD INDEX	HAZARD INDEX/MILE	HI/MILE RANK
2A-48	1.0	55.0970	54.9	1
2L-57	3.7	170.2419	45.4	2
2K-45	5.0	185.4933	37.3	3
2Z-41	1.2	43.9937	37.2	4
2K-44	3.0	109.7977	36.7	5
2B-46	2.9	103.7972	36.2	6
2K-55	11.1	360.4543	32.5	7
2J-51	10.1	294.1113	29.1	8
2D-58	3.1	87.5603	28.2	9
2A-45	2.4	66.1032	28.0	10
2K-57	4.1	115.1842	27.9	11
2L-58	1.7	48.0314	27.7	12
3D-46	2.1	55.6910	26.6	13
3J-50	1.4	37.6969	26.0	14
1Z-47	7.7	200.3936	25.9	15
3C-25	1.4	35.9431	25.6	16
2H-50	6.6	162.3633	24.8	17
2L-51	8.1	194.9827	24.2	18
2H-45	3.6	87.6968	24.2	19
2L-43	7.1	167.2065	23.6	20

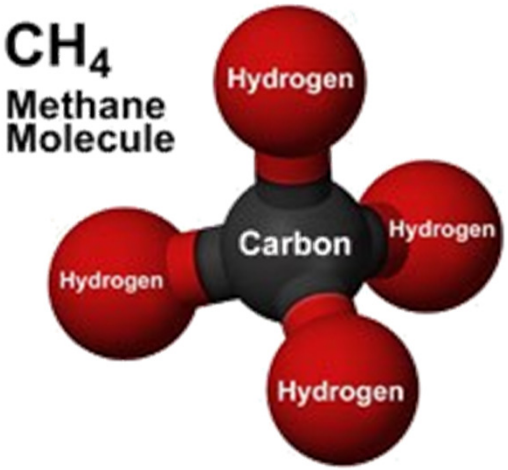
Methane as a Greenhouse Gas

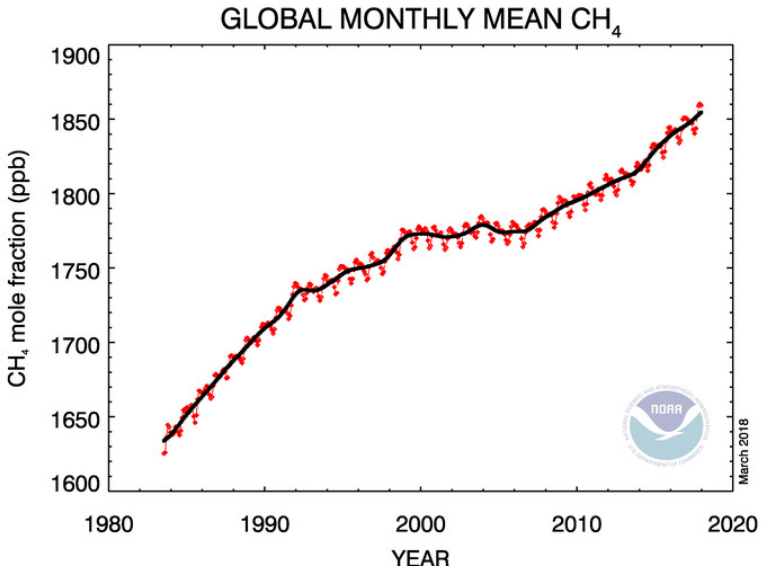
- Methane has 84 times the warming effect of carbon dioxide over a 20 year period

- EDF estimates that about 25% of the manmade global warming we're experiencing today is caused by methane emissions

Typical Composition of Natural Gas

Methane	CH_4	70-90%
Ethane	C_2H_6	0-20%
Propane	C_3H_8	
Butane	C_4H_{10}	
Carbon Dioxide	CO_2	0-8%
Oxygen	O_2	0-0.2%
Nitrogen	N_2	0-5%
Hydrogen sulphide	H_2S	0-5%
Rare gases	A, He, Ne, Xe	trace





Working with the EDF

- In advance of GSMP I, PSE&G engaged the Environmental Defense Fund (EDF) to quantify methane emissions in our service territory to consider in the prioritization of the work
- Mapping was performed over a six month period
- Study was done at no cost to PSE&G
- PSE&G followed the EDF equipment with its own optical methane leakmobile to compare data



EDF ENVIRONMENTAL DEFENSE FUND®
Finding the ways that work

EDF Overview - Continued

- The EDF partnered with Google and Colorado State University on a nationwide program to detect and map methane leaks from natural gas distribution systems
- A Google street-view car, equipped with state of the art methane and meteorological sensors, was driven repeatedly along streets with natural gas pipelines to map emissions
- Urban areas have been mapped across the country (Birmingham, Boston, Burlington, Chicago, Dallas, Indianapolis, Jacksonville, Los Angeles, Mesa, Pittsburgh, Staten Island, and Syracuse)
- The same technology used to map these cities was also used for the PSE&G project

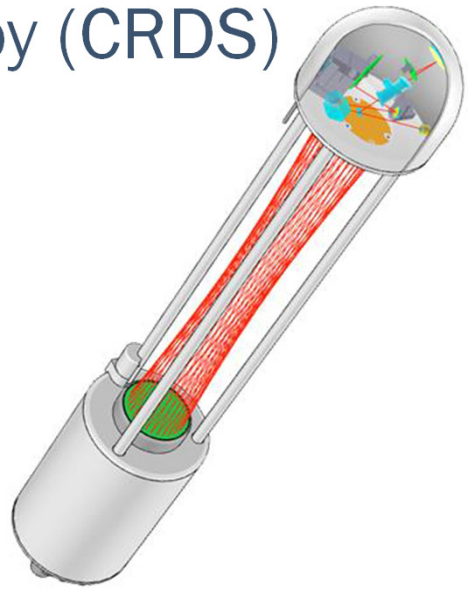


Google

Colorado State University

What Technology Was Used?

- Advanced GPS technology and anemometer
- Open path, Cavity Ring-Down Spectroscopy (CRDS) LiCor analyzer
  - High data collection rate
  - No pumps (closed path CRDS)
  - The longer the laser path, the better the sensitivity in detecting molecular signatures
  - Equipment uses a series of mirrors within the sample cavity to reflect the laser path from a distance of 25 cm to over 20 km



Methane Quantification Data

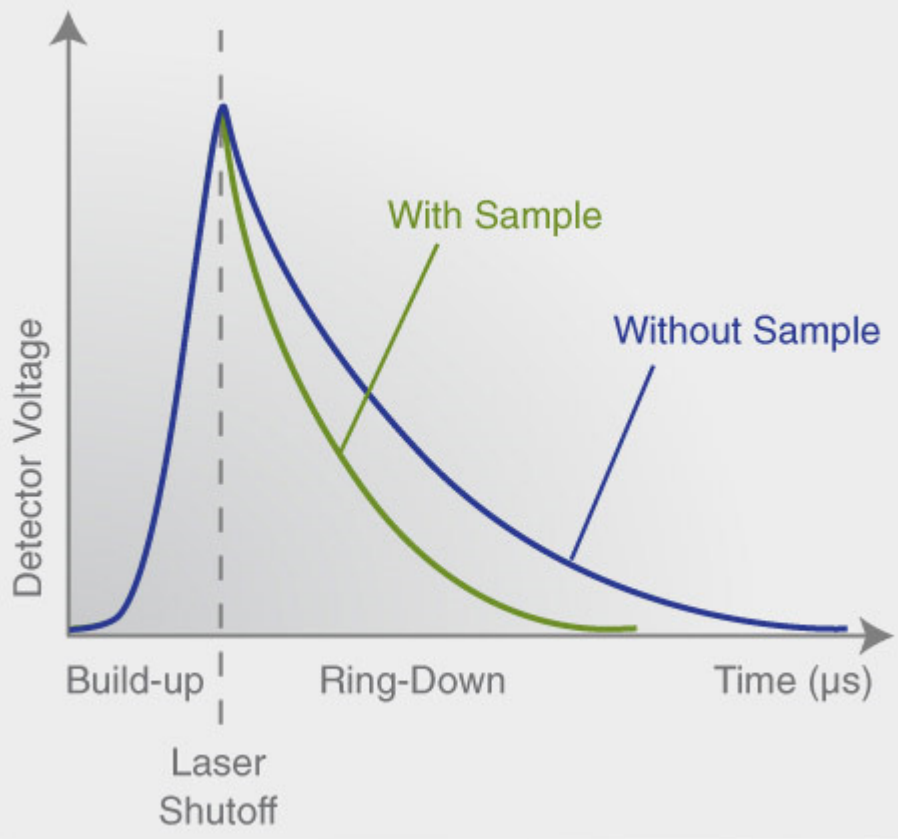
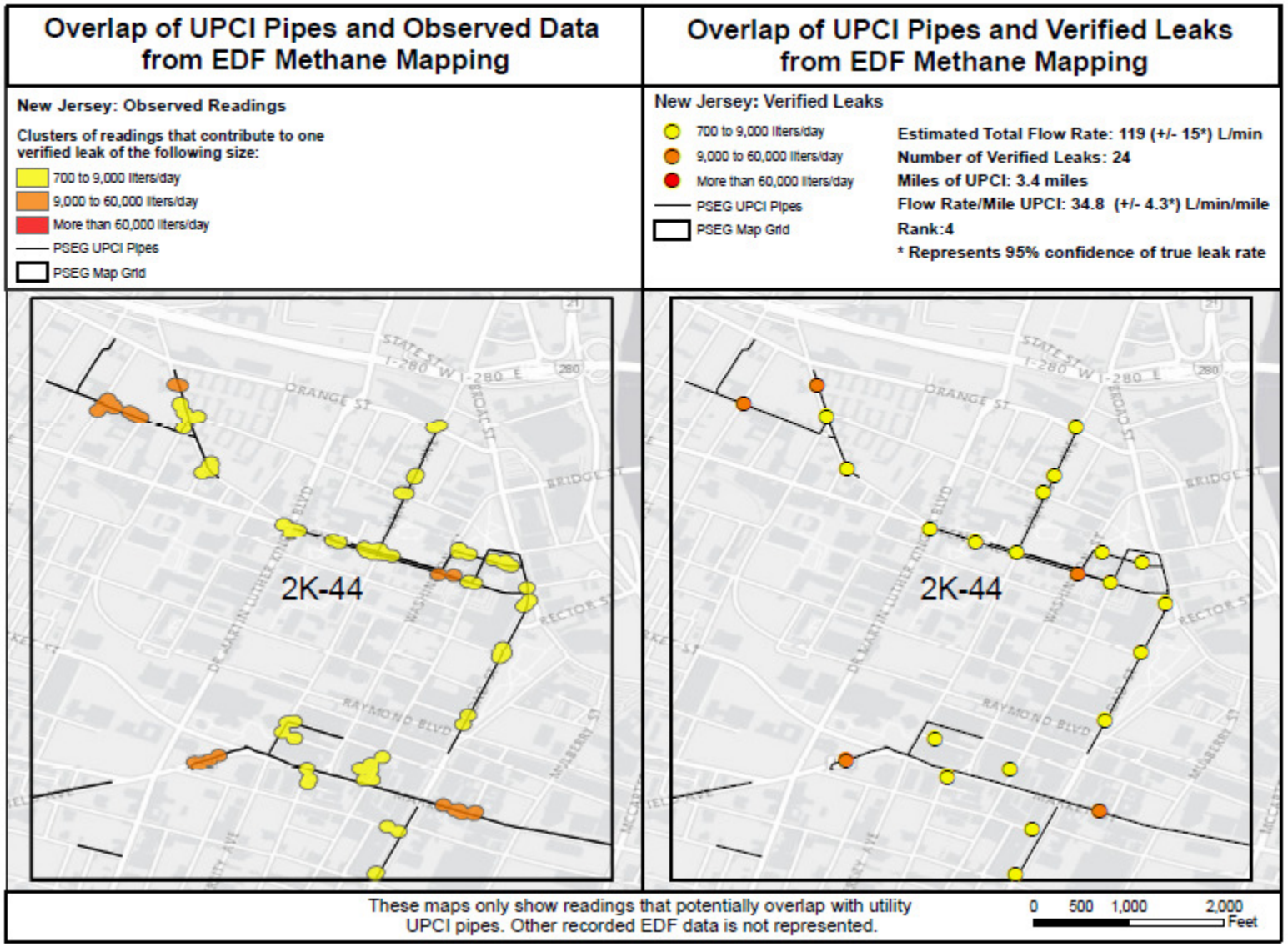


Fig 1. Ring Down Graph. Adapted from Picarro. Retrieved from Picarro.com

- Different gases absorb light (laser) at specific rates
- Normal atmospheric air has a certain decay pattern as the laser fades inside the sample chamber (blue graph)
- When a gas like methane is in the sample, it absorbs light at a different decay rate than the control (green graph)
- The laser wavelength and difference in decay rates is used to quantify methane by analyzing the sample data stream through a series of algorithms
- Wind and precipitation are factors in sampling

Readings vs Indications



Using the Results in GSMP I

- Hazard Index per Mile (HI/Mi) still primary risk ranking tool
- Any grid with HI/Mi > 25 is highest priority
- Where HI/Mi is comparable (< 25), EDF data used to help **sub-prioritize** by leak rate of liters per minute per mile of UPCI pipe in the grid (L/Min/Mi)
  - Grids with outlying leak rates of >10 L/Min/Mi take highest priority
  - Grids with leak rates of <10 L/Min/Mi as well as non-surveyed grids take secondary priority
- Grids are evaluated for construction efficiencies and logistics as well as permitting and municipality conflicts prior to setting the final prioritization
- Results reviewed with EDF and submitted to the NJ Board of Public Utilities

Reduction in Emissions

- Outlier grids (>10 L/min/mi) were looked to be moved up in schedule where possible
- Mains retired earlier than originally planned stop emitting methane faster
- By accelerating high emissions grids, PSE&G was able to reduce total grid emissions by 83% early in the program.
- To achieve the same emissions reductions, 35% less main abandonments were needed vs if PSE&G followed strictly by hazard ranking.
- The accelerated grids the company prioritized for upgrades accounted for more than 37% of the emissions but only 9% of the mileage on which leak rates were measured.

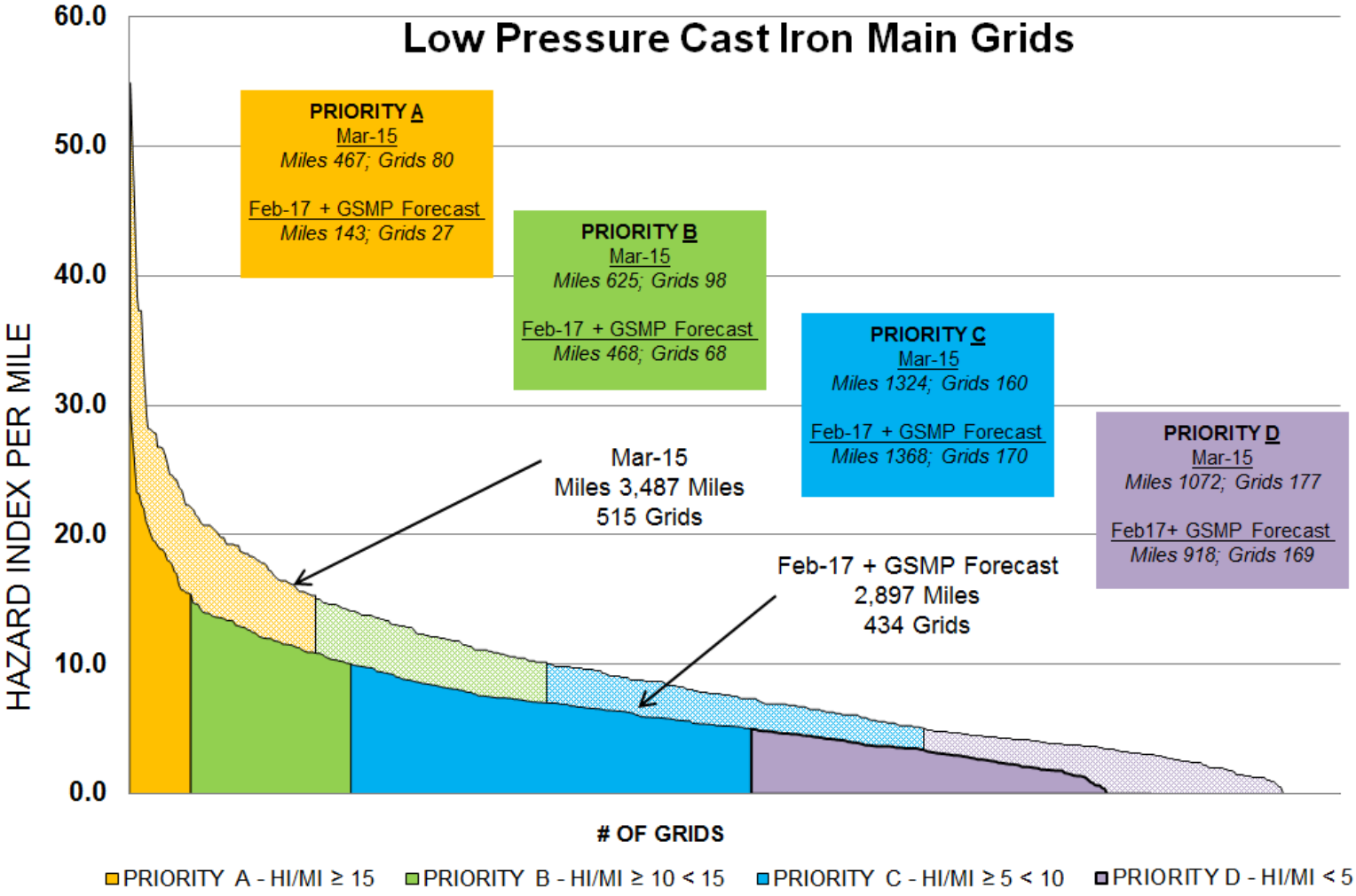


Continuing the Program into GSMP II

- GSMP II filed in 2017 and approved in Spring 2018 as a five year extension
- Hazard Index and methane mapping to be used again to prioritize grids
- Picarro was chosen to map 44 “B Grids” of similar HI/mi that covered the 280 miles agreed to in the stipulation



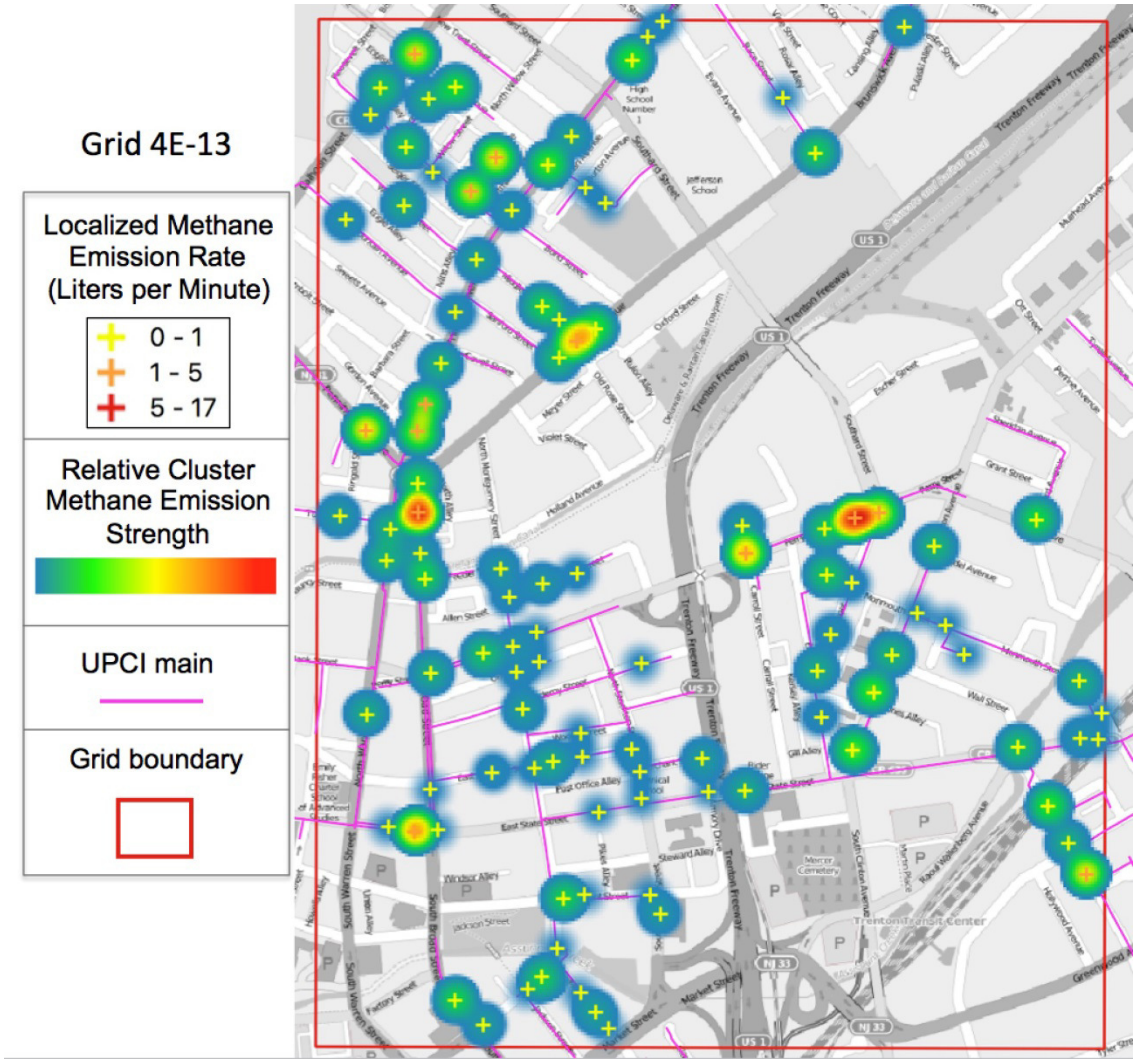
Reduction in Risk and Methane Mapping



Continuing to address the highest hazard main segments

Methane Quantification Survey

- Areas require 3 passes on each side of the street for proper sampling (95% statistical confidence interval)
- Indications are run through an algorithm with wind, vehicle speed, ethane content and other factors, leak rates are determined
- Heat maps can show areas of high emissions and calculated leak rates

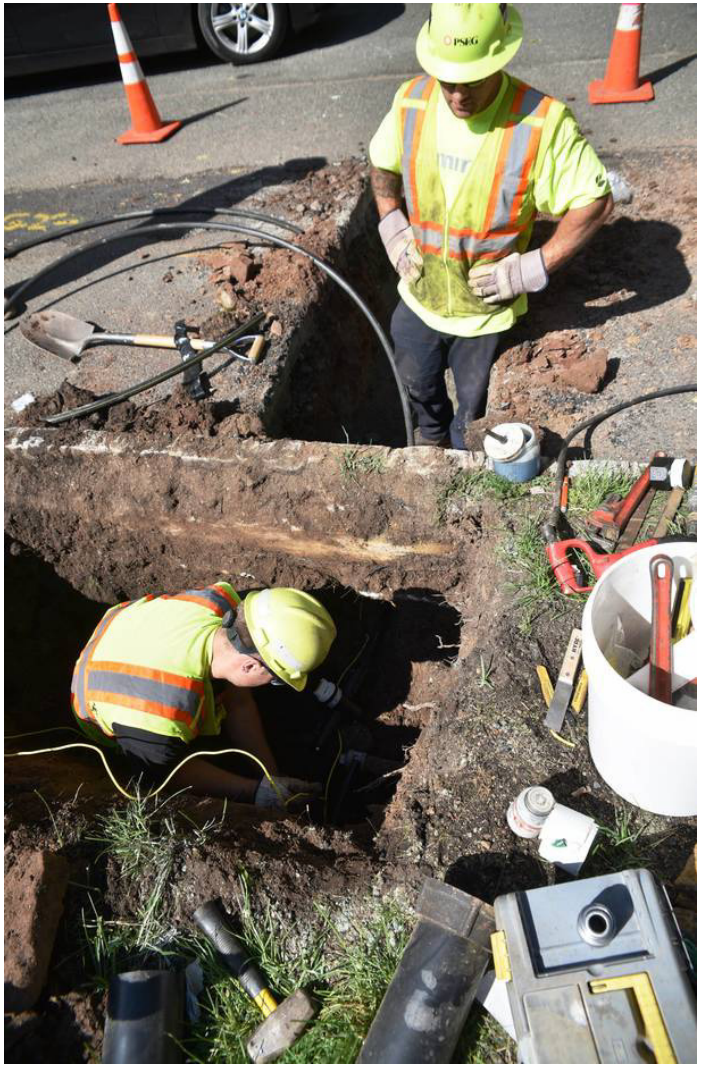


Using GSMP II Results



- Discussion with EDF after data collected to set prioritization
- Threshold of 4.5 L/min/mi used for accelerating grids that were surveyed (down from 10 L/min/mi in GSMP I)
- 6 grids accelerated
- If retired sooner than “as is” plan, they account for 41% of the methane loss in only 16% of the grids surveyed
- Construction beginning in Spring of 2019

Key Takeaways



- Hazard Ranking and safety are highest priority
  - Hazard Rank and Leak Volume do not necessarily correlate
- Methane Emissions sub prioritization useful for areas of relatively equal hazard
  - Better for the environment
  - Less chance of non-hazardous leaks getting worse
  - Fewer potential customer calls/complaints
- Other LDC's and PUC's continue to discuss best applications for the technology's use

Questions?



Exhibit __ (JF-5):

Kerans, Mike, Picarro Surveyor Leak Detection Study – Sacramento Side-by-Side Study (2012)

Picarro Surveyor™ Leak Detection Study

Sacramento Side-by-Side Study

3/22/2012 – 4/17/2012

Principal Author: *Mike Kerans - Pacific Gas & Electric, Sr. Gas Engineer*

Contributing Authors: *Timothy Clark - The Spear Group, Pacific Gas & Electric's Project Lead*
Stephen Conley - University of California, Project Scientist
Eric Crosson - Picarro, Chief Technical Officer
Mike Kerans - Pacific Gas & Electric, Sr. Gas Engineer
Mark Piazza - Pipeline Research Council International, Program Manager

ABSTRACT. The Sacramento Side-by-Side Study was conducted from March 22, 2012 through April 17th, 2012 and designed to assess the capabilities and accuracy of the vehicle mounted Picarro's Surveyor™ methane detection system and P- Cubed Software which is designed to locate natural gas pipeline leaks emanating from distribution systems. The Study was conducted in Pacific Gas & Electric's Sacramento Gas Division. Its primary purpose was to compare the performance of Picarro's Leak Detection System against traditional ground based leak survey methods. This report highlights the leak survey results of 16 gas facility maps containing 4,195 gas services and 56 miles of distribution main. The report compares the survey rate, ability to locate gradable and non-hazardous gas leaks while assessing the survey coverage of both methods.

1. INTRODUCTION.

Preceding the Sacramento side-by-side study Pacific Gas & Electric conducted two controlled tests at its Livermore Training Academy[1] and a similar leak survey training facility located in Southern Nevada[2]. The controlled tests were small scale conducted in training facilities where the leaks could be operator controlled, leak locations were known (by both Picarro and test administrators in Livermore and only test administrators in Nevada), and where additional atmospheric data could be collected. These controlled tests established a "Proof of Concept" showing that the Picarro instrument could detect natural gas leaks in simulated natural gas distribution systems from a variety of leak sources both above and below ground and at various concentrations.

Analysis of the results from the control tests was utilized to further refine the Picarro leak detection algorithms and software which would be used for field testing in Pacific Gas & Electric's Diablo[3] division. Sacramento Division was selected as a second side-by-side field study to test the leak detection ability of the Picarro Surveyor™ against traditional leak detection instruments currently used by PG&E.

By comparison the proposed methods and instruments developed by Picarro make use of a vehicle mounted cavity ring down spectrometer fitted with a sonic anemometer, a GPS device to measure methane concentrations, and various atmospheric conditions relative to a fixed position to sense and locate natural gas leaks. The Picarro instrument utilizes a cavity ring down spectrometer and can detect methane in the parts per billion (ppb) range allowing the survey method to be conducted primarily in a vehicle driven at common traffic speeds (see below in FOV section for more explanation on vehicle speeds). However, the Picarro instrument samples gas every second in comparison with the traditional tools which sample continuously. Additionally, the higher sensitivity and associated P-Cubed software does not require the surveyor to have detailed knowledge of buried mains and services as Picarro's software generates a Leak Indication Search Area (LISA) once a plume of methane is detected (see *Figure 1.0).* Moreover, the Picarro method and instrument differs from current methods as Picarro's software generates a Field Of View (FOV) plot showing which mains and services have been surveyed by the instrument, a feature not widely available with current methods and instruments. Also, the speed of the vehicle has a direct affect on the FOV. The coverage is lessened the faster the vehicle is driven.

A LISA is a shown below in *Figure 1.0* as indicated by the yellow wedge. By using a set of GPS coordinates and a corresponding LISA to investigate, the Piccaro method requires a natural gas leak be investigated by the field surveyor, field graded, and recorded or repaired, since the Picarro Surveyor™ cannot pinpoint or grade leaks. TFOV plot shown in *Figure 1.0* is illustrated by the green and tan areas and is representative of the region which the Picarro was surveying. By detecting methane plumes at distance and targeting them, the Picarro method and instrument show a shift in leak survey methodology compared to current methods and instruments.

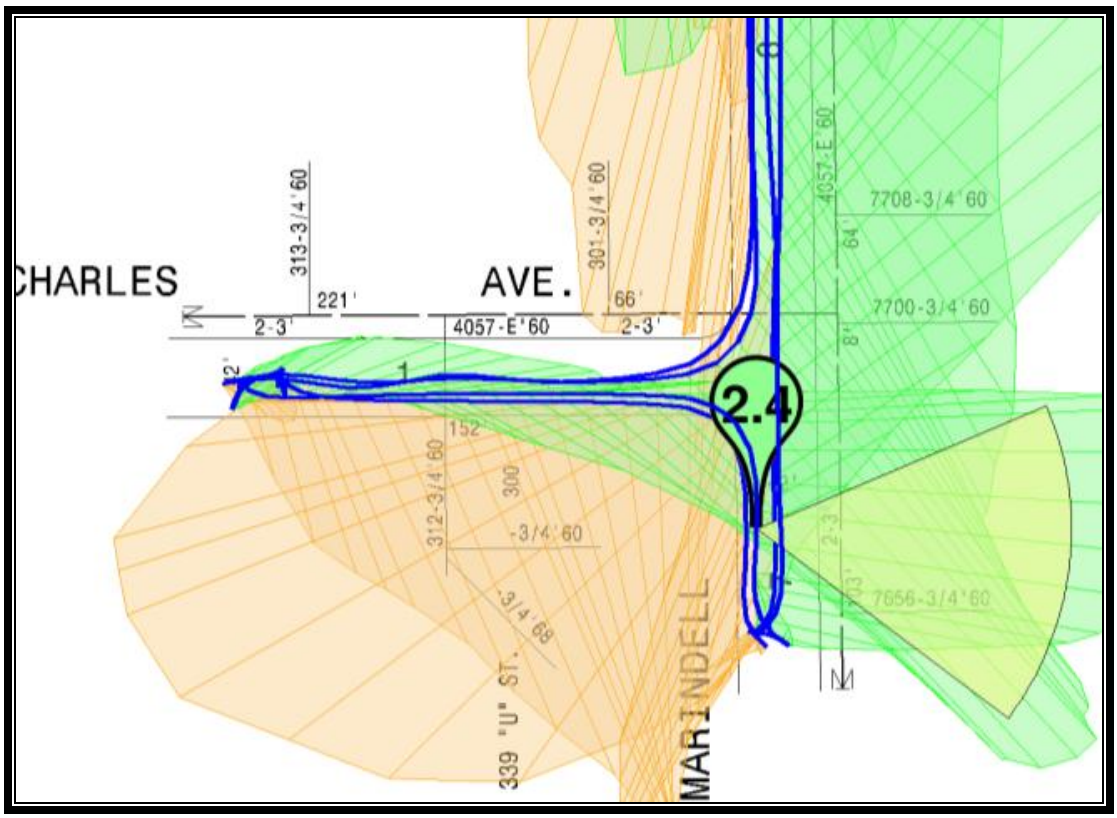


Figure 1.0 – Leak Indication Search Area (LISA)

As the Picarro Surveyor™ methods and instruments represent a new model by which gas utilities can leak survey distribution systems it is critical for Pacific Gas & Electric to fully evaluate Picarro's leak survey system while understanding the benefits and impacts of potentially adopting this technology. To further evaluate Picarro's technology, the Sacramento Side-by-Side Study was designed to utilize results and recommendations from the previous Diablo Side-by-Side Study and compare the results of traditional ground survey to the Picarro Surveyor™ and associated P-Cubed Software. The Study's primary focus will be on each method's ability to sense and locate leaks as well as the survey rates and FOV, or coverage, of traditional and Picarro leak surveys.

2. METHOD. Pacific Gas & Electric's Sacramento Gas Division is located in California's Central Valley region, a vast flat river valley situated between the coastal mountain range and Sierra Nevada mountain range. The Sacramento Division gas operation services approximately 545,000 natural gas customers, a significant portion of PG&E's total gas customer base. The Sacramento Division, specifically the 16 gas facility plat maps, were chosen due to their flat level land and variety of suburban and rural regions. Additionally, the combination of the Sacramento Division's large number of customers and variety in suburban and rural regions are representative of much of PG&E's service territory.

The Sacramento valley in the Sacramento Division has an elevation ranging from 30 to 150 feet and is surrounded by rolling grassland, farmland, and urban sprawl in the form of housing and light commercial. The small farms in the vicinity of the area surveyed vary from small livestock operations to mixed produce production while the suburban areas represent common subdivisions for the region. *Figure 2.0* is an aerial view of the operating Division, specifically the 16 plat maps which were surveyed during the side-by-side test.

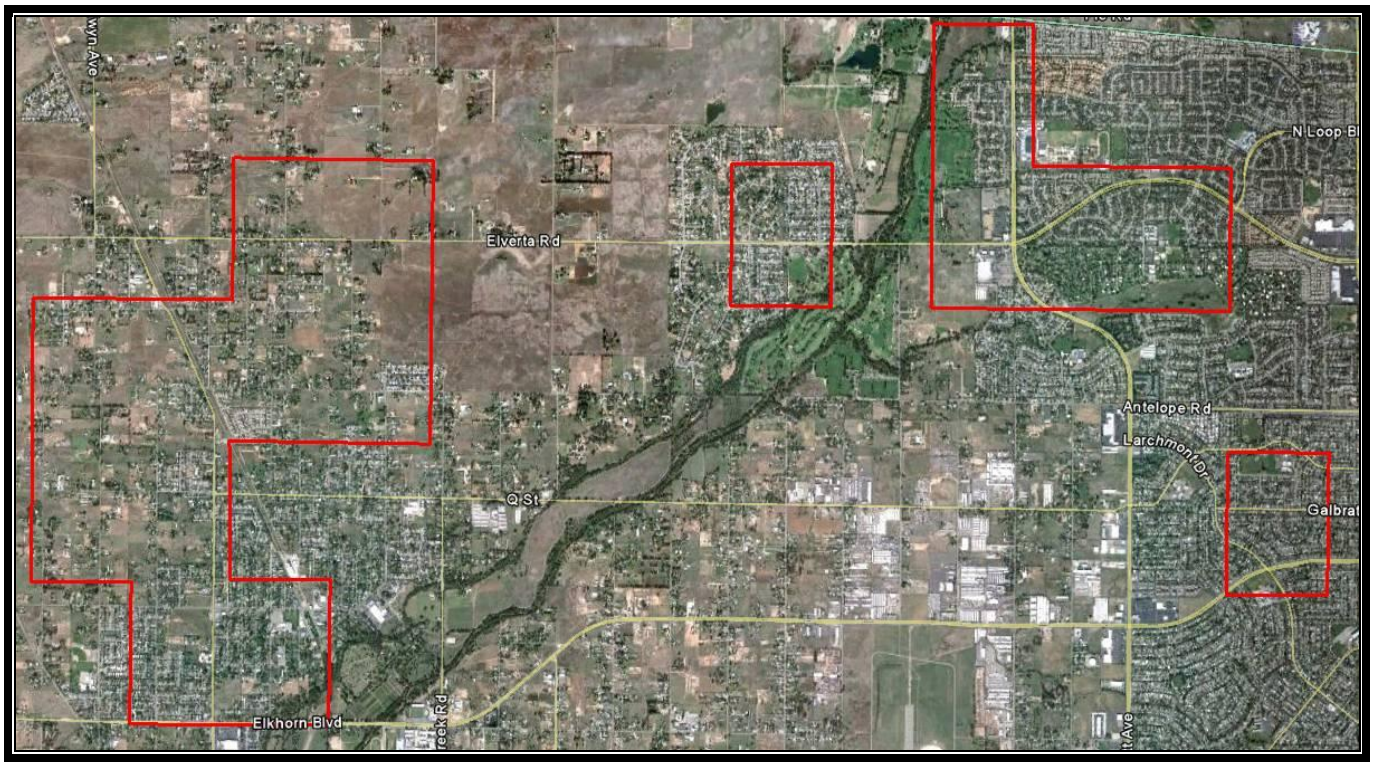


Figure 2.0 – Sacramento Valley (area surveyed in red)

The side-by-side testing method took measurements of three key metrics to evaluate the effectiveness of current survey methods and instruments to the Picarro Surveyor™ and P-Cubed software. These three metrics were:

1. Total leaks found.
2. System survey coverage.
3. Time to complete leak survey.

To ensure the Local Leak Survey Technicians remained unbiased in their survey methods and techniques as each surveyed the 16 plat maps in a routine 5-year leak survey cycle, they were not aware that the 16 plats surveyed would be part of a side-by-side study. The traditional leak survey of the 16 plats occurred from 1/12/2012 - 2/21/2012 and recorded all findings in accordance with PG&E Standard Operating Procedures. Additionally, as part of the traditional leak survey process the local Mapping Office provided an "Open Leaks Report" and plotted all known existing leaks on the corresponding 16 plat maps prior to local operations conducting the traditional leak survey while Picarro drivers were not provided the same plat maps with existing leak information shown.

Following the traditional leak survey the Picarro Surveyor™ (installed in a Chevrolet Sierra Hybrid pickup truck) drivers were provided with unmarked versions of the 16 plats to be leak surveyed and were instructed to survey (drive) the maps from 3/22/2012 - 4/17/2012 (start of Picarro survey began approximately a month after the traditional survey was completed). Drivers were instructed to

record their time driving and note any large leak indications. Upon completion of the Picarro field surveys Picarro LISA plots were generated to allow Local Leak Survey Technicians to investigate and grade leaks found. Leak results were recorded and repaired according to PG&E Standard Operating Procedures.



Figure 3.0 – Chevrolet Sierra Hybrid used for the Picarro survey

Atmospheric weather conditions affect the Picarro Surveyor's™ performance significantly and wind speeds greater than zero are required. During the weeks of testing the weather, as measured from the nearby airport, was reported to be generally calm with very little precipitation and temperatures and winds averaging 54 °F and 5 mph.

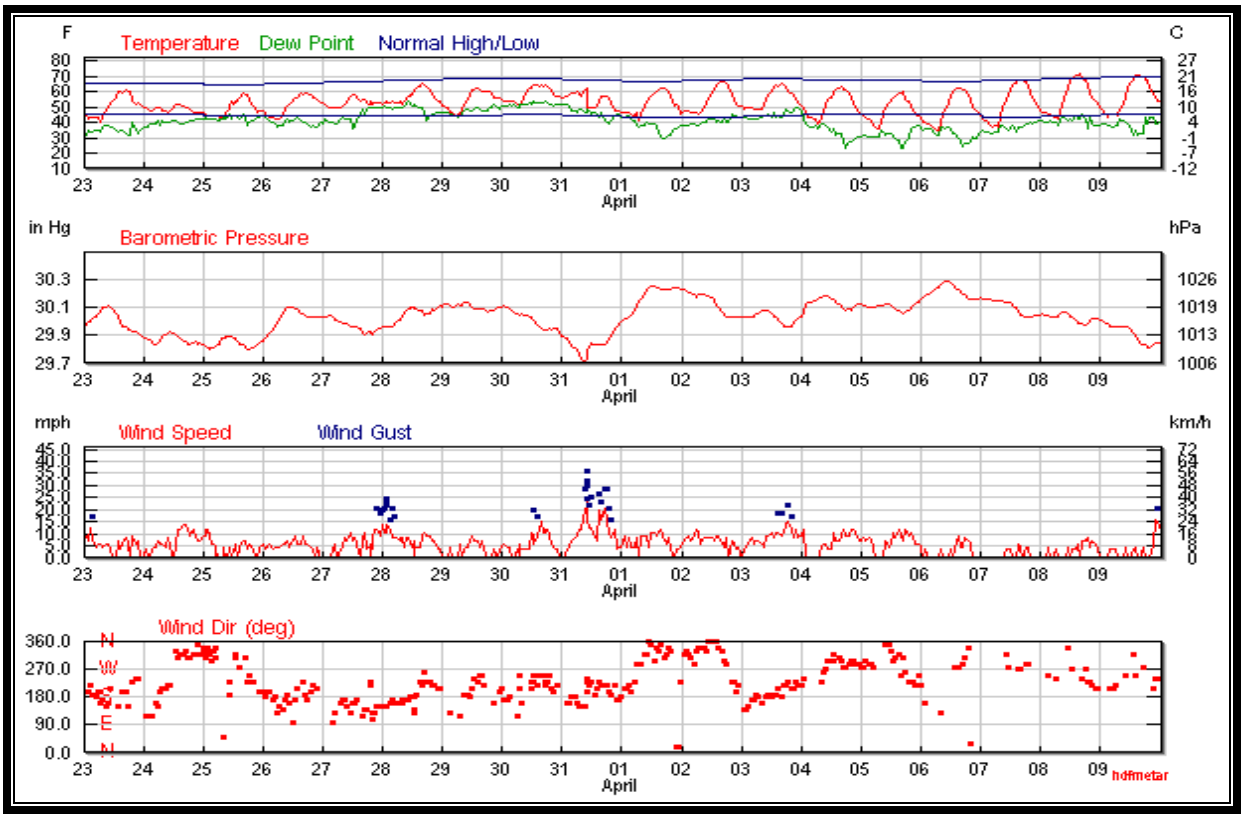


Figure 3.1 - KSAC Weather Station data (3/23/2012 – 4/9/2012) from Weather Underground

The Sacramento side-by-side study benefited and implemented Picarro methods gleaned during the Diablo side-by-side study which increased the effectiveness of the Sacramento Picarro survey. For the Sacramento Picarro survey drivers were instructed to conduct a drive in the early morning from 3:00 am - 10:00 am and a second late evening drive from 7:00 pm - 3:00 am. Additionally, to maximize the use of street space and to ensure complete coverage, drivers were instructed to drive both sides of all streets.

3. RESULTS. *SURVEY COVERAGE*. The traditional ground survey was conducted first and traditional leak survey coverage was confirmed to be 100% of mains and services with all leaks found, graded, or repaired. Following the completion of both (the early morning and late evening) Picarro field surveys, analysis of Picarro's leak survey data was conducted by reviewing FOV plots generated by Picarro's software.

As learned from the Diablo side-by-side study both the night and daytime FOV plots were displayed on a single plot in order to determine the coverage by the Picarro Surveyor™. Shown in *Figure 4.0*, are the Picarro Survey results from the daytime and nighttime runs for Sacramento Plat "G".

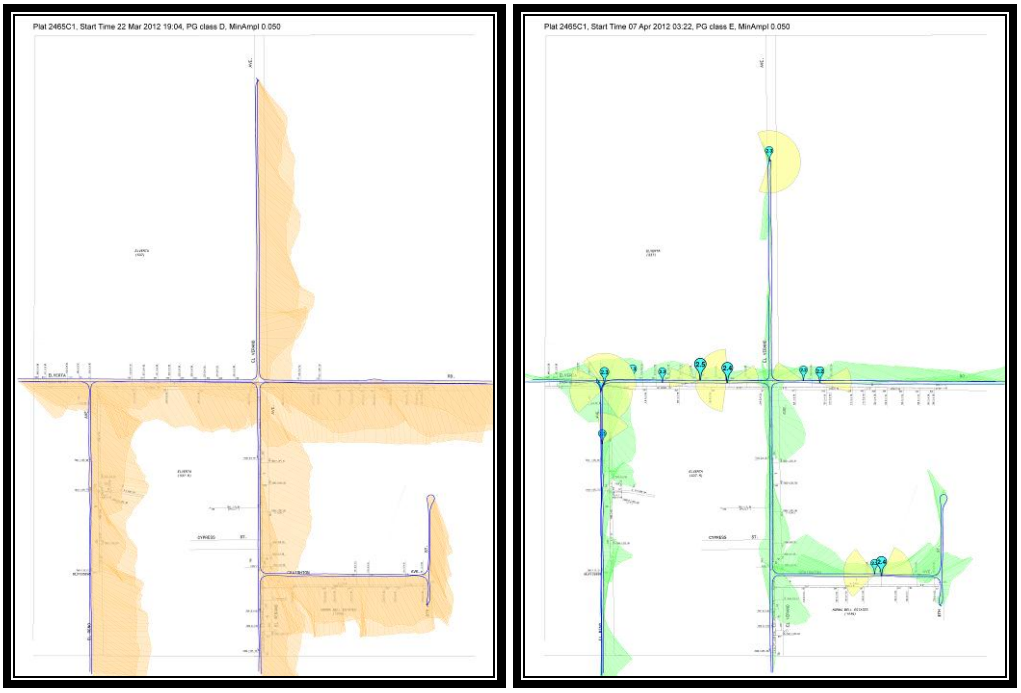


Figure 4.0 – Sacramento plat maps G Evening & Morning Picarro Runs

A combination of the evening and morning Picarro runs shown in *Figure 4.1* illustrate the full coverage potential of the Picarro Surveyor during this study. There is a very distinct change in direction of the wind in these plots as shown by the evening FOV (tan) which provided coverage to the south as compared to the morning FOV (green) which gave coverage to the north. By combining an evening and morning Picarro Surveyor™ run, the leak survey can take advantage of the natural shifts in wind directions.

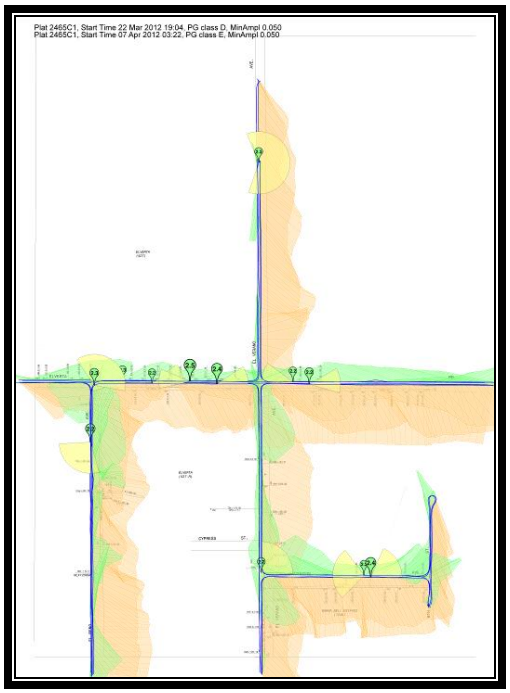


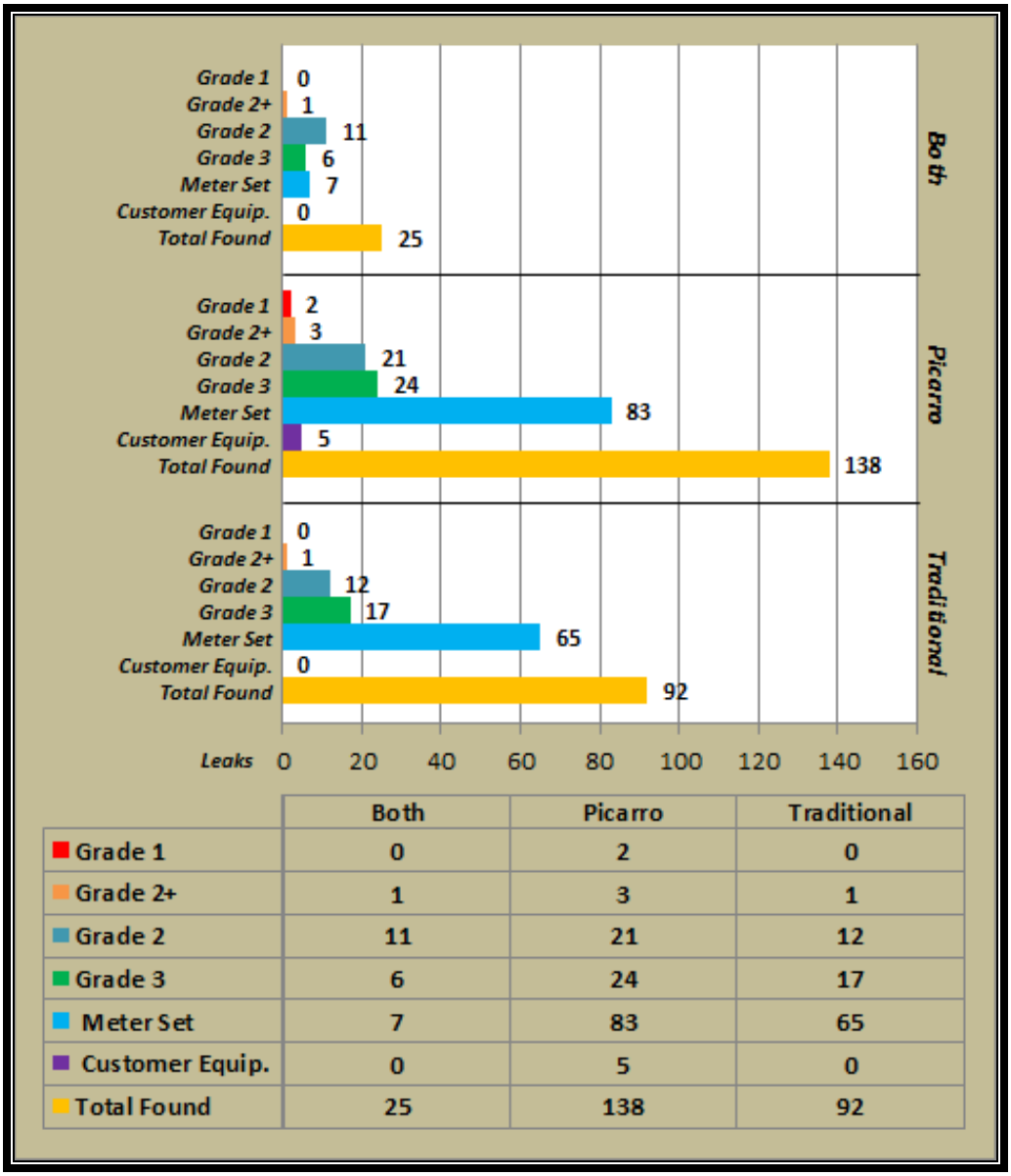
Figure 4.1 – Sacramento Plat Map G Evening & Morning Run Combined

A summary of survey coverage for all 16 Sacramento plat maps is shown in *Table 1.0*. After combining Picarro's evening and morning runs, the overall service count and miles-of-main coverage were found to be 88% coverage of services and 97% of mains. The percentage of mains and services not covered by Picarro's FOV plot would need to be leak surveyed by traditional leak survey technicians in order to obtain 100% coverage of all gas facilities and fulfill regulatory requirements.

PG&E Plat Map Data			Picarro - 1st Run				Picarro - 2nd Run				Picarro Combined Runs			
Plat	Services	Miles of Main	Services	% Service	Miles of Main	% Miles of Main	Services	% Service	Miles of Main	% Miles of Main	Services	% Service	Miles of Main	% Miles of Main
A	48.0	0.849	27.0	56%	0.770	91%	31.0	65%	0.849	100%	37.0	77%	0.849	100%
B	126.0	3.030	84.0	67%	3.030	100%	103.0	82%	3.030	100%	107.0	85%	3.030	100%
C	42.0	1.154	24.0	57%	1.154	100%	17.0	40%	1.154	100%	28.0	67%	1.154	100%
D	302.0	4.046	206.0	68%	3.888	96%	216.0	72%	3.987	99%	266.0	88%	3.998	99%
E	511.0	8.286	452.0	88%	7.975	96%	433.0	85%	7.955	96%	502.0	98%	8.244	99%
F	353.0	3.475	237	67%	3.098	89%	211.0	60%	3.019	87%	284.0	80%	3.337	96%
G	64.0	1.110	37.0	58%	1.110	100%	41.0	64%	1.084	98%	56.0	88%	1.110	100%
H	4.0	0.106	3.0	75%	0.106	100%	2.0	50%	0.106	100%	3.0	75%	0.106	100%
I	197.0	3.152	146.0	74%	3.147	100%	124.0	63%	2.642	84%	172.0	87%	2.983	95%
J	191.0	2.759	142.0	74%	2.757	100%	142.0	74%	2.386	86%	165.0	86%	2.575	93%
K	181.0	2.203	110.0	61%	2.201	100%	142.0	78%	2.186	99%	158.0	87%	2.197	100%
L	116.0	1.895	92.0	79%	1.894	100%	95.0	82%	1.895	100%	108.0	93%	1.895	100%
M	298.0	5.407	242.0	81%	5.405	100%	206.0	69%	5.377	99%	280.0	94%	5.364	99%
N	633.0	7.582	492.0	78%	7.573	100%	433.0	68%	6.739	89%	543.0	86%	7.139	94%
O	482.0	4.930	271.0	56%	3.898	79%	336.0	70%	4.195	85%	404.0	84%	4.557	92%
P	647.0	6.011	406.0	63%	5.163	86%	388.0	60%	5.403	90%	576.0	89%	5.886	98%
Totals	4,195	56.0	2,971	71%	53.2	95%	2,920	70%	52.0	93%	**3,689**	**88%**	**54.4**	**97%**

Table 1.0 - Sacramento Study Survey Coverage

LEAK INDICATIONS. Following standard operating procedures for leak survey and using traditional handheld instruments, PG&E's Division Leak Surveyors found and graded 117 (92 traditional + 25 found by both) natural gas leaks within the 16 Plat Maps. Picarro's survey results, minus the confirmed 75 false positives ("ghosts LISAs"), were investigated by Division and Quality Control Surveyors who located and graded 138 leaks.



	Both	Picarro	Traditional
Grade 1	0	2	0
Grade 2+	1	3	1
Grade 2	11	21	12
Grade 3	6	24	17
Meter Set	7	83	65
Customer Equip.	0	5	0
Total Found	25	138	92

Figure 3.0 – Sacramento Both, Picarro, Traditional leaks found by Grade

Figure 3.0 shows leaks found by both the traditional and Picarro methods, by Picarro only, and by traditional only. These results illustrate that both methods of survey identified the same leaks, however Picarro was able to locate 138 leaks which traditional survey did not. Conversely, the traditional leak survey method was able to identify 92 leaks (65 were meter set leaks) which Picarro did not identify.

A total of 230 (138 + 92) leaks were found during the side-by-side study. By including leaks found by "both" methods of leak survey the results show that the Picarro method found 2 hazardous leaks and 2 more *Grade 2+* leaks than the traditional survey. It is impossible to determine whether these leaks existed when the traditional survey was conducted because of the time delay between each survey

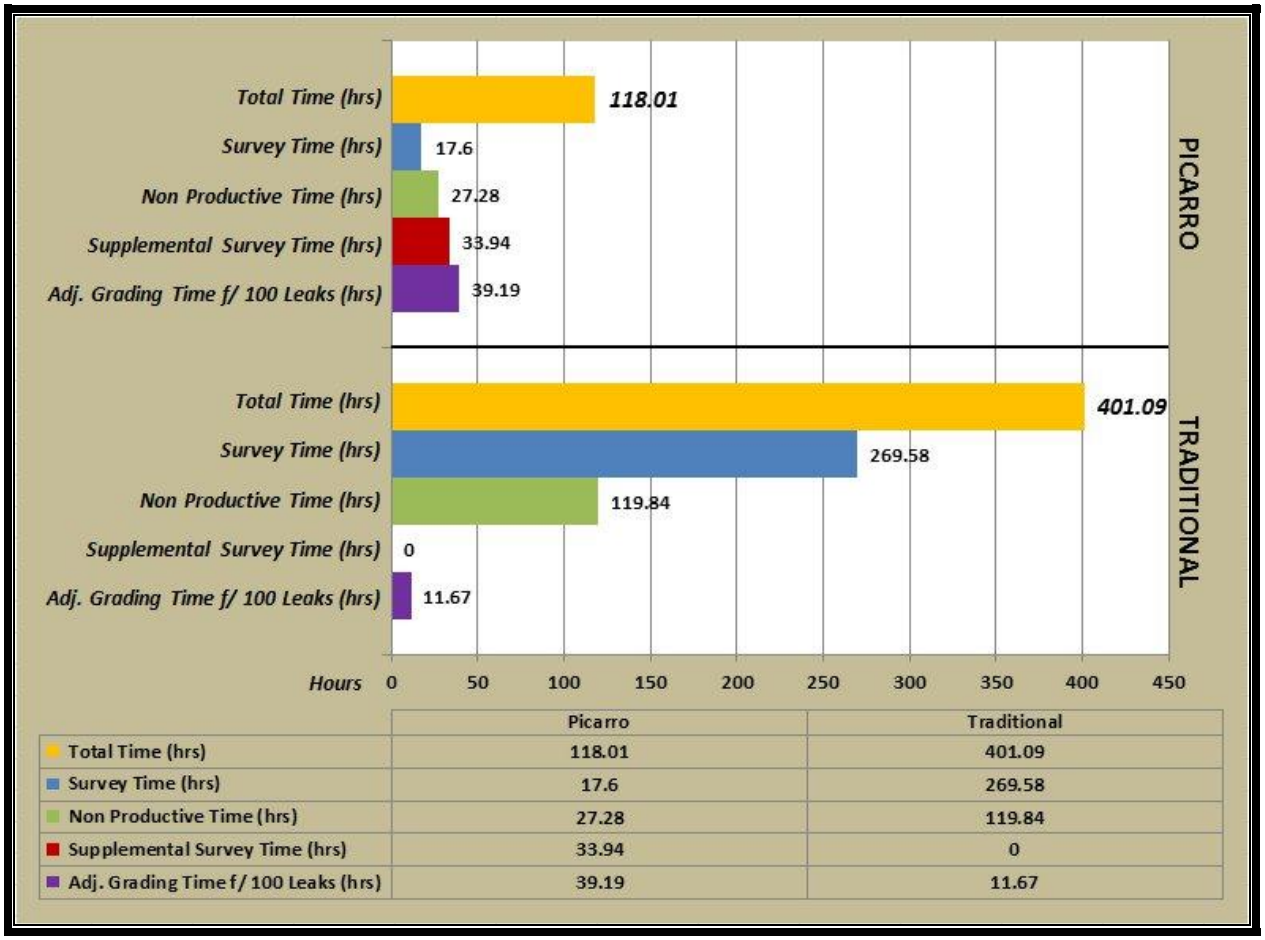
(2.5 months), and it is impossible to determine if Picarro would have detected the hazardous and *Grade 2+* leaks that had already been repaired as a result of the traditional leak survey. Although Picarro identified more, it missed a substantial amount of leaks found by the traditional. In addition, Picarro provided 75 false leak indications where a leak surveyor was dispatched to investigate and determined that no leak was present.

Referring to *Figure 3.0*, a leak "missed" by Picarro is defined as a leak originally found by traditional methods (92) which was within the Survey Coverage (FOV) of Picarro, but was not detected by Picarro. For leaks outside of the FOV of Picarro that were found by traditional, these were not counted as misses and the leak was not counted towards traditional either, as that would not be a true side by side comparison. In addition, leaks found by Picarro include all leaks found within a LISA, and any leak found 150 feet outside of a LISA.

TIME COMPARISON. Total time was segmented into four categories:

1. Survey Time.
2. Non-Productive Survey Time (includes administrative time associated with leak survey).
3. Supplemental Survey Time.
4. Adjusted Grading Time (normalized for 100 leaks or LISAs).

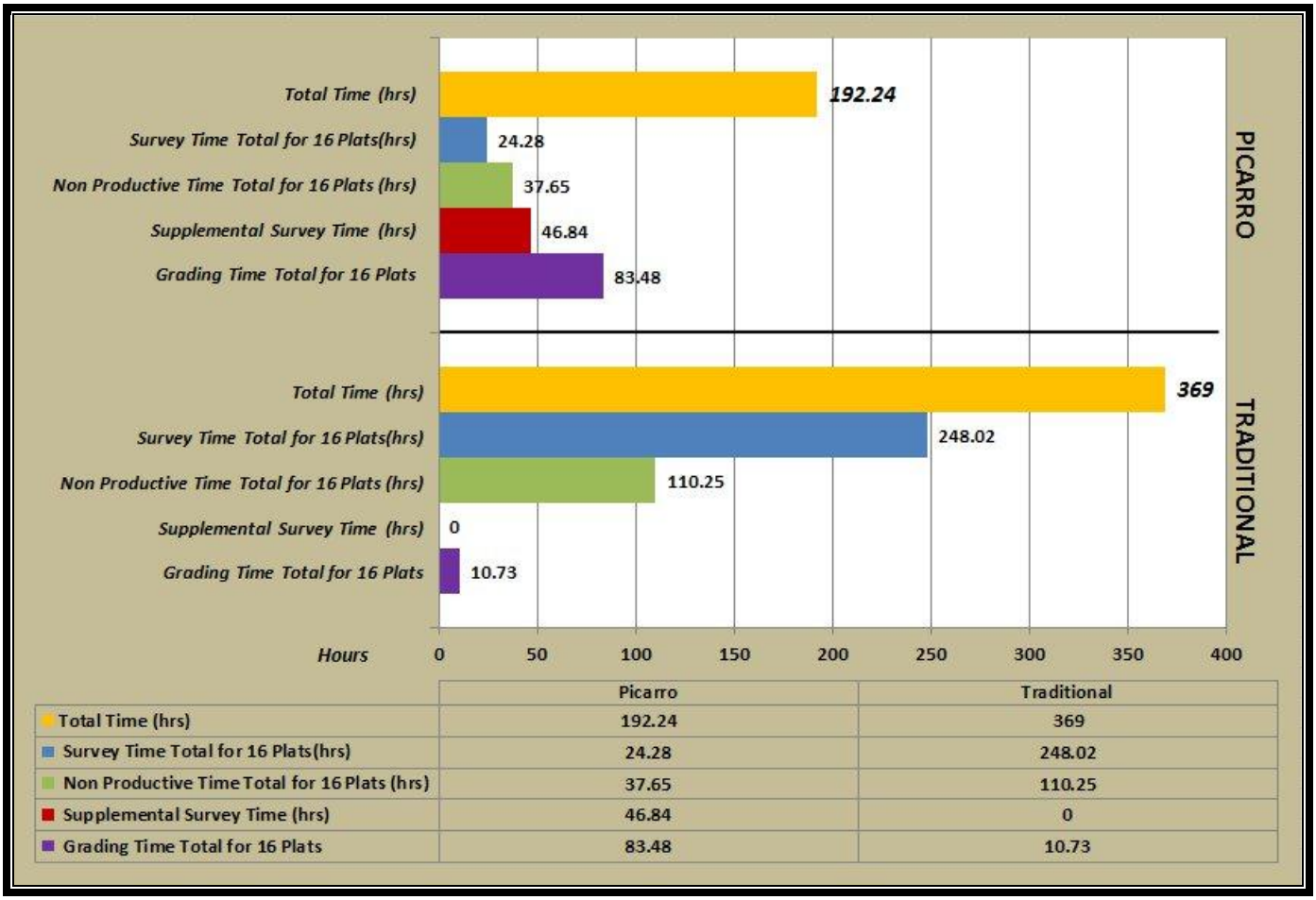
These categories were used to evaluate the production survey rate for each survey method. As shown in *Figure 4.0 (normalized)*, the Picarro method completed the survey for all 16 maps in 118 hours vs. 401 hours for the Traditional ground and Picarro methods, respectively.



	Picarro	Traditional
Total Time (hrs)	118.01	401.09
Survey Time (hrs)	17.6	269.58
Non Productive Time (hrs)	27.28	119.84
Supplemental Survey Time (hrs)	33.94	0
Adj. Grading Time f/ 100 Leaks (hrs)	39.19	11.67

Figure 4.0 – Normalized Sacramento Time Comparison, Picarro vs. Traditional

Traditional leak grading time was fixed at 7 minutes per leak while Picarro's leak indication investigation time was fixed at 23.5 minutes per LISA. Additionally, standard production rates (9.9 services/hr & 2.5 mi/hr) were utilized to calculate the Supplemental Survey Time needed to foot survey the remaining "missed" services and distribution main that were not captured during the Picarro survey coverage. Non-productive survey time was taken directly from time sheets or as recorded by test administrators.



	Picarro	Traditional
Total Time (hrs)	192.24	369
Survey Time Total for 16 Plats(hrs)	24.28	248.02
Non Productive Time Total for 16 Plats (hrs)	37.65	110.25
Supplemental Survey Time (hrs)	46.84	0
Grading Time Total for 16 Plats	83.48	10.73

Figure 5.0 – Non-normalized Sacramento Time Comparison, Picarro vs. Traditional

In addition, *Figure 5.0* shows a non-normalized comparison of the time needed to Leak Survey 16 plats between Picarro and Traditional. Because Picarro found more leaks over the 16 plats than traditional, and because the pinpointing and grading of leaks takes an average of 23.5 minutes per LISA for Picarro, the time to complete a geographic region by the two methods are 192 hours for Picarro and 369 hours for traditional.

4. DISCUSSION. The Sacramento side-by-side testing was the second field study conducted to compare traditional leak survey to Picarro leak survey and was also preceded by two controlled studies in Livermore and Nevada. Throughout the testing of the Picarro Surveyor™ improvements to testing protocols were made, amplitude controls used, Leak Indication Search Algorithms (LISA) were adjusted, and the FOV methodology improved. These improvements were applied to the Sacramento side-by-side study and have contributed to enhancing three fundamental characteristics of this side-by-side study of leak survey:

1. Proficiency to Detect & Locate Leaks.
2. Rate of Survey.
3. Ability to Document Area Surveyed.

*PROTOCOL ENHANCED***.** Several protocol enhancements were incorporated into the Sacramento study from the Diablo side-by-side study which were utilized to increase Picarro survey coverage. These improvements included a shift in the time of day for the Picarro Survey and a requirement to drive both sides of the street.

The Sacramento study took lessons learned from the Diablo study and instructed drivers to drive in two shifts, an evening and early morning time period. This change in protocol was due to atmospheric conditions are best for leak survey when the atmosphere is stable. Atmospheric stability is largely a function of surface wind speeds, solar radiation, and cloud cover which tend to be more favorable during the periods selected to run the Picarro Surveyor. Additionally, since the Picarro Survey requires wind speeds greater than 0 mph to conduct a leak survey, the second early morning drive was required. The early morning drive time was selected as coverage area is directly related to wind direction, which tends to shift direction during different times of the day and night. By adding a second Picarro leak survey shift it was found that coverage was greatly increased over a single daytime leak survey.

To further increase coverage and to ensure a thorough leak survey, drivers were instructed to drive both sides of all streets within the plat maps, an example is shown in *Figure 6.0*. By driving both sides of all streets, incremental increases in coverage were gained while allowing for further coverage by the Picarro Surveyor™ along services and up to meter sets. It was noticed that coverage did not appear to increase dramatically, however this method does increase coverage of mains buried in roadways as surveying both sides of the streets allowed for a wider field of view.

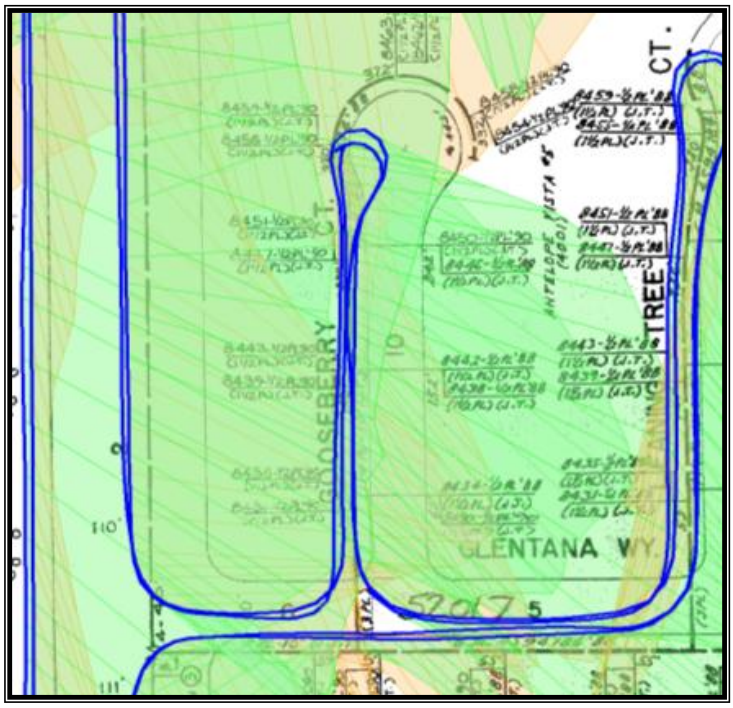


Figure 6.0 – Driving Path (blue) & Coverage (green & tan).

*AMPLITUDE SETTINGS & LEAK INDICATIONS***.** Testing of the Picarro Surveyor™ during the Nevada Control Test at a leak training facility (where leaks could be simulated and controlled) was conducted in part to determine if Picarro's Amplitude settings could be used to adjust the volume of Leak Indication Search Areas. To explain this further it should be noted that a peak in the methane concentration is reported as a leak indication provided that it may be approximated by a Gaussian whose width lies within a certain range. The *Figure 7.0* illustrates how the measured methane concentration (blue line) is approximated by a Gaussian. The amplitude of the peak (at position 20m along the path) is defined as the height of the green curve (approximately 2.47) above the background concentration (shown as 2.0). The concentration reported in the leak indication is defined as the measured concentration at the position of the peak of the Gaussian. In the figure, the reported concentration in the leak indication would be 2.53. Note that because of the way the quantities are defined, it is possible that the concentration in the leak indication may not in fact be the maximum measured concentration in the peak.

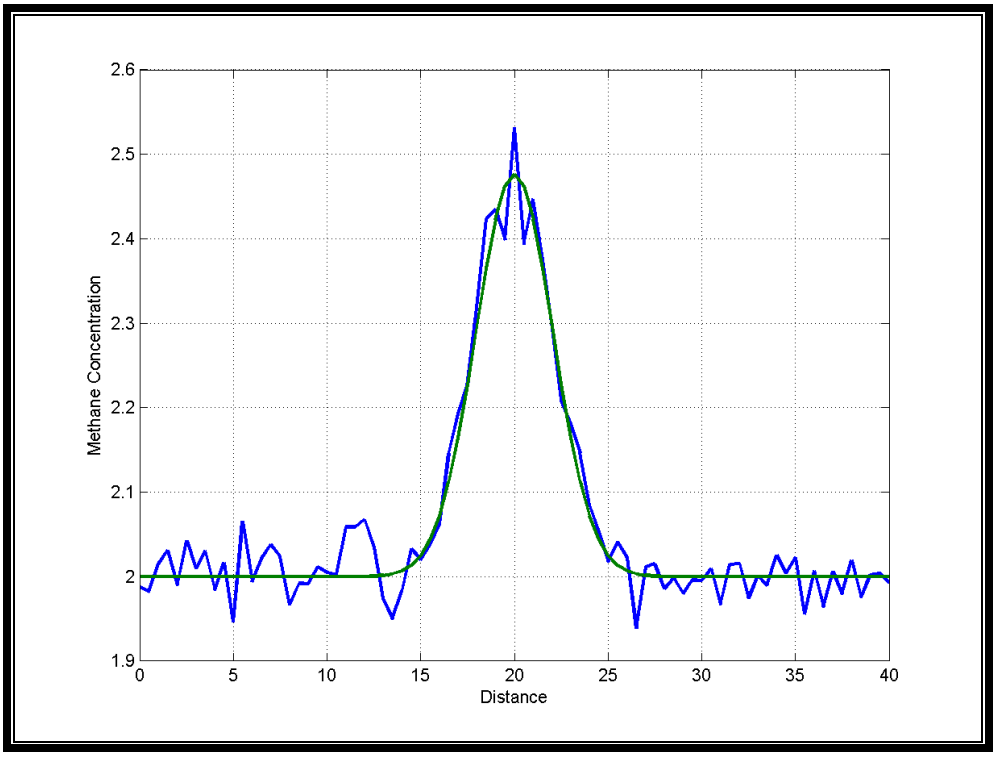


Figure 7.0 – Example of Leak Indication and Gaussian Approximation

A peak in methane is recorded through the P-Cubed software, and by adjusting the amplitude settings the user can effectively adjust the level at which they desire to see leak indications above background methane concentrations as was shown during the Nevada Control Study. It was determined that although the user could use amplitude adjustments to reduce excess leak indications, this was undesired during the Sacramento side-by-side testing as all indications needed investigation to fully compare traditional leak survey to Picarro leak survey. Additionally should Picarro's Surveyor be utilized for gas distribution leak survey it is recommend to leave amplitude at it's lowest setting to allow for the identification and investigation of all leak indications.

FIELD OF VIEW (FOV) DEFINITION. The FOVis a representation of the area surveyed as a result of driving the vehicle past the location of potential leaks. More precisely, points within the FOV are such that the probability of missing a leak located within the FOV is less than a certain value, given the motion of the vehicle, the measured wind velocity and the prevailing atmospheric conditions. If the wind direction is constant and blowing directly from the source to the analyzer (and not blocked by high walls, dense trees, etc), the distance at which a source can be detected is greatest. Under favorable (stable) atmospheric conditions, this distance can approach 100 meters. When the wind is variable, or cannot be determined precisely due to the motion of the vehicle, the distance is correspondingly reduced.

FOV plots generated by Picarro for the Sacramento side-by-side study were generated on electronic plat maps which were in the form of Tiff files. These Tiff files have geographic coordinates associated with the corners of the plat map and are geographically referenced to their physical locations in a GIS or Google Earth mapping system. The P-Cubed software integrates the GPS path which the Picarro vehicle drove and generates a path in blue and a FOV swath in green or tan.

5. CONCLUSION. Studies of the Picarro Surveyor™ and its P-Cubed software have proved the initial concept that Picarro's expertise in gas sampling can be specifically engineered for use by natural gas distribution company's leak survey programs to indicate, target, and confirm the presence of gas pipeline leaks. Furthermore, the P-Cubed software working in unison with Picarro's highly sensitive cavity ring down spectrometer technology allows for data acquisition of leaking distribution systems and an auditable leak survey record unparalleled by current leak survey methods and instruments.

In addition, use of Picarro was shown to reduce the time needed to complete a leak survey by 175 hours, or a savings of 4.4 weeks for a full time employee in the study area. The time savings is a result of being able to drive most or all sections of the distribution system as opposed to walking, however a portion of this time savings can be attributed to the P-cubed software which generated auditable FOV plots showing coverage of the leak survey.

This study confirmed information collected during the Diablo study that Picarro is able to cover 97% of the miles of main and 88% of services by driving the Picarro System early in the morning and late evening on both sides of the street.

The results of this study confirm there is not one leak survey method can identify 100% of all leaks in a single survey. However, this study also indicates that, when used in concert and coordination, traditional and Picarro leak survey methods can be combined to maximize the effectiveness and efficiency of leak detection programs. The delay between the Traditional and the Picarro survey during this test may have led to additional leaks which were not detected during the first survey. Ideally, designating the Picarro Surveyor™ as the leading leak survey method while the traditional leak survey provides support in small targeted areas where Picarro cannot survey as readily.

Additional enhancements to the Picarro Surveyor™, software, and protocols continue to be refined at this time and will further improve its ability to detect and target leaks in natural gas piping systems.

[1] *PG&E Picarro Surveyor™ Leak Detection Study: Livermore Leak Test Facility, 1/13/2012 – 1/19/2012*

[2] *PG&E Picarro Surveyor™ Leak Detection Study: Southern Nevada Leak Test Facility, 3/19/2012 – 3/22/2012*

[3] *PG&E Picarro Surveyor™ Leak Detection Study: Diablo Leak Detection Study 2/21/2012 – 4/11/2012*

Exhibit __ (JF-6):

Clark, Timothy, Picarro Surveyor Leak Detection Study – Diablo Side-by-Side Study (2012)

Picarro Surveyor™ Leak Detection Study

Diablo Side-By-Side Study

2/3/2012 - 4/18/2012

Principal Author: *Timothy Clark - The Spear Group, Pacific Gas & Electric's Project Lead*

Contributing Authors: *Stephen Conley - University of California, Project Scientist*
Eric Crosson - Picarro, Chief Technical Officer
Mike Kerans - Pacific Gas & Electric, Sr. Gas Engineer
Mark Piazza - Pipeline Research Council International, Program Manager

ABSTRACT. The Diablo Side-by-Side Study was conducted from February 3, 2012 through April 18, 2012 and designed to assess the capabilities and accuracy of the vehicle mounted Picarro's Surveyor™ methane detection system and P- Cubed Software which is designed to locate natural gas pipeline leaks emanating from distribution systems. The Study was conducted in Pacific Gas & Electric's Diablo Gas Division and its primary purpose is to compare the performance of the Picarro's Leak Detection System against traditional ground based leak survey methods. This report highlights the leak survey results of sixteen gas facility maps containing 3,545 gas services and 45 miles of distribution main. The report compares the survey rate, ability to locate gradable and non-hazardous gas leaks while assessing the effective survey coverage of both methods.

1. INTRODUCTION. Prior to the conclusion and independent of the Diablo Side-by-Side Study, Pacific Gas & Electric conducted two controlled tests at its Livermore Training Academy[1] and at a similar facility located in Southern Nevada[2]. Although the controlled tests results were reported separately, it's important to note that through these tests Picarro established 'Proof of Concept' and demonstrated the instrument's ability to sense and provide guidance when locating single and multiple gas releases of varying sizes within a controlled setting designed to simulate real world gas leak conditions. Picarro refined the mathematical models and software algorithms using these results and findings to detect and provide guidance to locate the leaks.

In addition, Picarro technology's ability to differentiate between naturally occurring biogenic methane and natural gas and the stability of its calibration ability were independently verified by two third party laboratories: the University of California Davis[3] and Applied Technical Sciences[4] located in San Ramon, CA. These results have also been reported separately.

Picarro's monitoring equipment is a vehicle mounted system including a sonic anemometer, a temperature recorder, and a GPS device to simultaneously record methane concentration, geographic location and atmospheric conditions. All data is wirelessly uploaded to Picarro's Processing Platform or P-Cubed™ for data processing, access and storage. Access to data is real-time via a standard internet connection. A field surveyor is dispatched to investigate any leak detections using GPS coordinates of the corresponding search area and Picarro's Leak Indication Search Area (LISA). Upon confirming a natural gas leak, the field surveyor grades, records and reports the leak.

This study utilizes real world conditions and compares current leak survey methods against the Picarro's Leak Detection System to evaluate its comparable performance. Traditional ground surveys include various mobile platforms (i.e. OMD, Short-Path RMLD) and they're effectiveness is limited by their base technology. Therefore, handheld survey instruments (i.e. FI Units, CGI, RMLD) have become the dominant tools for surveying natural gas distribution and transmission system leaks. In order to assess whether Picarro's technology can conceivably replace current tools and methods, PG&E challenged three aspects of Picarro's leak detection system:

1. Ability to sense natural gas leaks, including very small leaks at extended distances.
2. Pinpoint the leak locations.
3. Provide positive auditable records of areas surveyed.

Although the study's primary focus was to assess each method's ability to sense and locate leaks, the survey rates and each technology's survey coverage was also examined.

2. METHOD. As one of Pacific Gas & Electric's most challenging topographies, the Diablo Gas Division is located in the Diablo Mountain Range situated in San Francisco's east bay area and services approximately 48,000 natural gas customers. Sixteen representative gas facility plat maps were chosen from the division with each plat containing approximately 222 services.

Summit elevations of the Diablo Mountain Range vary between 2,300 and 3,300 feet and are surrounded by rolling hills, punctuated by sudden peaks of hills and plateaus rising out of the valleys to elevations of approximately 1,000 to 2,000 feet. The surrounding foothills range from 400 to 1,000 feet in elevation and include canyons approximately 300 to 400 feet deep with deeper but gentler valleys. A panoramic view of the Divisions operating area is shown in *Figure 1.0*.

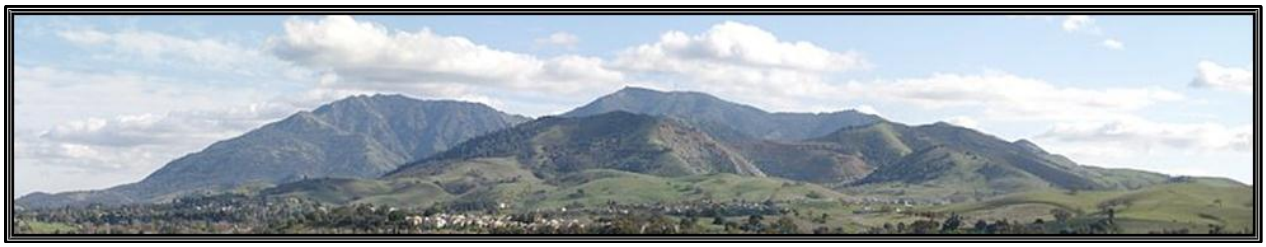


Figure 1.0 - Diablo Mountain Range

Diablo Divisional employees first surveyed and recorded all findings for all 16 plat maps in accordance with PG&E Standard Operating Procedures from 1/3/2012 - 1/31/2012. The Picarro study took place from 2/3/2012 - 4/18/2012. For the purpose of this study, open leaks found prior to the 2012 traditional leak survey were not incorporated into this side-by-side testing or results. Open leaks were not included as the comparison was designed to evaluate the effectiveness of both survey tools and methods to find and identify new leaks in the field.

3. RESULTS. *SURVEY COVERAGE*. The traditional ground survey was conducted first and subsequently followed by a survey (up to several weeks after the ground survey) using Picarro's vehicle mounted Surveyor™. Assessed for gas leaks, survey coverage is defined as the actual services and miles of main. Documentation associated with Survey Coverage must meet the regulatory burden of providing an auditable record with positive confirmation that all gas facilities were inspected. Traditional ground survey derived 100% coverage by manually tallying daily totals of services and miles of main surveyed. As shown in *Figure 2.0*, division employees highlighted areas surveyed in a given day, manually tallied services and miles of main and recorded the information on the back of the example plat map.

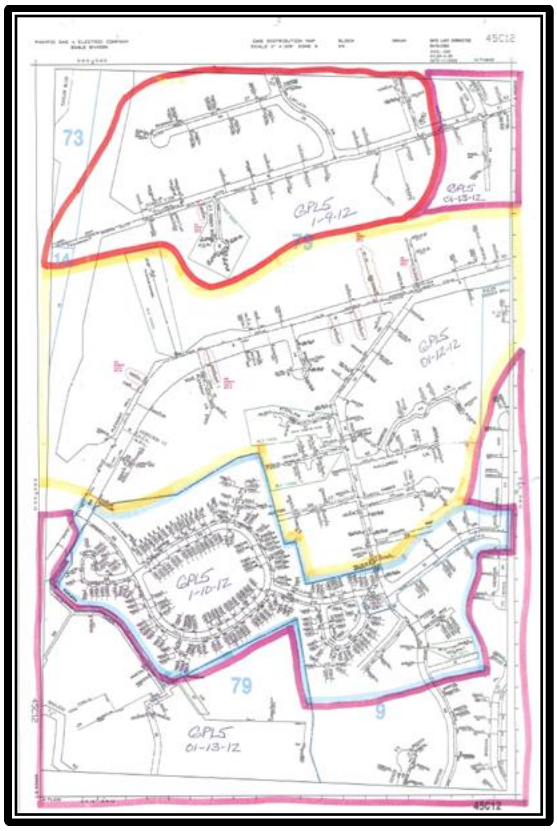


Figure 2.0 - Diablo Study: Example Traditional Survey Coverage for Plat Map

The Picarro survey was performed during regular daytime business hours. Although 77 leak indications and their associated search areas were reported, upon receipt and analysis of Picarro's leak survey data, it became immediately apparent that the field-of-view (FOV) or coverage was unexpectedly low. PG&E consulted with Picarro's Scientists suggesting to re-run the survey during the early morning hours where atmospheric conditions would be more favorable for the survey and found the FOV coverage had significantly increased.

Atmospheric turbulence is greatly enhanced by convective motions due to heating of the earth's Surface and on clear nights is suppressed by cooling of the ground due to outgoing radiation. In order to relate the state of atmospheric convection to simply observable parameters, scientists have developed a simple quantitative rating consisting of six stability classes ranging from highly convective to highly stable flow conditions. Over the course of a day, over land, under clear skies and at sunrise, heating from below sets a convective boundary layer (CBL). When the sun sets the heat loss to space terminates convection creating a stable nocturnal boundary layer (NBL)[5].

Understanding the transport mechanism of natural gas from its point of leakage into the atmosphere to its area of detection by the vehicle is critical to determining natural gas source locations. The primary transport mechanism for moving natural gas from the point source to the vehicle is wind. Natural gas readily disperses into the atmosphere and is carried downwind gradually mixing vertically and horizontally by turbulent eddies generated by wind shear and solar heating of the surface. Although this transport process quickly dilutes the natural gas as it mixes with ambient air, the Picarro Surveyor™ is more sensitive than traditional leak surveying instruments and is able to detect trace (ppb) quantities of gas. As a result of this observation, an overnight, second test run was planned and driven.

Shown in *Figure 3.0* are the Picarro Survey results from the first and second run from example Diablo Plat G, represented by light brown & green swaths. As reported in *Table 1.0*, the coverage for this Plat Map was recorded at 46% and 82% for services while miles-of-main coverage was equal to 96% and 99% respectively for the first and second run.

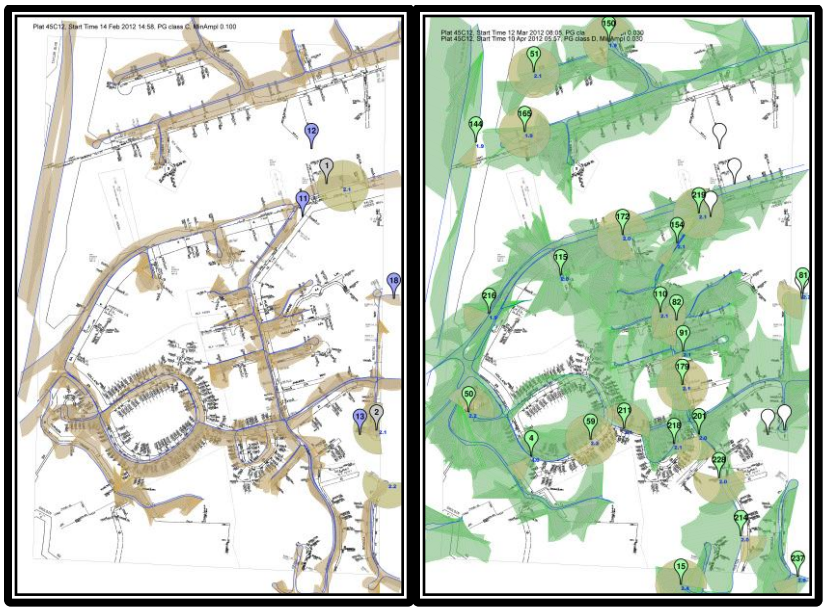


Figure 3.0 - Diablo Survey Coverage: Picarro First & Second Run for Plat G

As represented in *Figure 4.0*, both runs were merged, resulting in an overall increase in survey coverage, aggregate service and miles-of- main, increasing to 89% and 99% respectively.

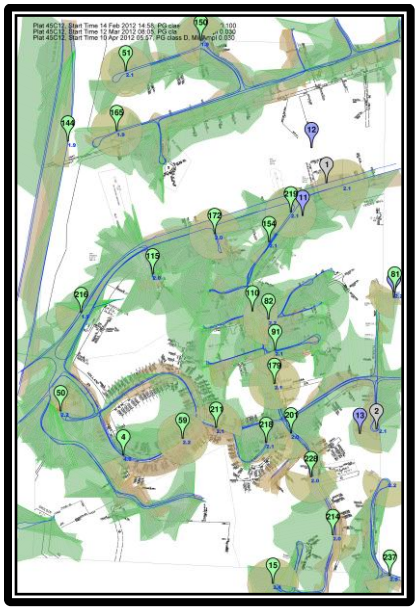


Figure 4.0 - Diablo Plat Map G First & Second Picarro Run Combined

A summary of survey coverage for all 16 Diablo plat maps is shown in *Table 1.0*. The percentage of mains and services not covered by Picarro's FOV plot would need to be leak surveyed by traditional leak survey technicians in order to obtain 100% coverage of all gas facilities and fulfill regulatory requirements.

PG&E Plat Map Data			Picarro - 1st Run				Picarro - 2nd Run				Picarro Combined Runs			
Plat	Services	Miles of Main	Services	% Service	Miles of Main	% Miles of Main	Services	% Service	Miles of Main	Miles of Main	Services	% Service	Miles of Main	% Miles of Main
A	16.0	0.26	7.0	44%	0.198	76%	8.0	50%	0.230	88%	9.0	56%	0.260	100%
B	103.0	2.58	68.0	66%	2.487	96%	79.0	77%	2.548	99%	90.0	87%	2.550	99%
C	153.0	1.95	77.0	50%	1.863	96%	129.0	84%	1.950	100%	140.0	92%	1.935	99%
D	44.0	1.26	25.0	57%	1.247	99%	38.0	86%	1.245	99%	36.0	82%	1.243	99%
E	134.0	2.46	75.0	56%	2.221	90%	102.0	76%	2.244	91%	114.0	85%	2.384	97%
F	45.0	0.7	27.0	60%	0.670	96%	34.0	76%	0.564	81%	40.0	89%	0.609	87%
H	299.0	2.89	137.0	46%	2.786	96%	244.0	82%	2.860	99%	265.0	89%	2.888	100%
I	410.0	5.25	173.0	42%	5.136	98%	192.0	47%	4.080	78%	367.0	90%	5.248	100%
J	371.0	3.98	186.0	50%	3.741	94%	318.0	86%	3.919	98%	339.0	91%	3.980	100%
K	492.0	5.96	243.0	49%	5.915	99%	445.0	90%	5.960	100%	464.0	94%	5.960	100%
L	259.0	3.91	132.0	51%	3.836	98%	227.0	88%	3.800	97%	227.0	88%	3.874	99%
M	85.0	1.06	38.0	45%	1.032	97%	70.0	82%	1.005	95%	72.0	85%	1.060	100%
N	248.0	3.97	63.0	25%	3.656	92%	207.0	83%	3.925	99%	215.0	87%	3.909	98%
O	296.0	4.21	25.0	8%	3.862	92%	254.0	86%	4.149	99%	266.0	90%	4.176	99%
P	340.0	4.54	87.0	26%	4.063	89%	289.0	85%	4.423	97%	297.0	87%	4.442	98%
G	250.0	3.02	159.0	64%	2.887	96%	218.0	87%	3.012	100%	230.0	92%	3.020	100%
Totals	3,545	48.0	1,522	43%	45.6	95%	2,854	81%	45.9	96%	**3,171**	**89%**	**47.5**	**99%**

Table 1.0 - Diablo Study Survey Coverage

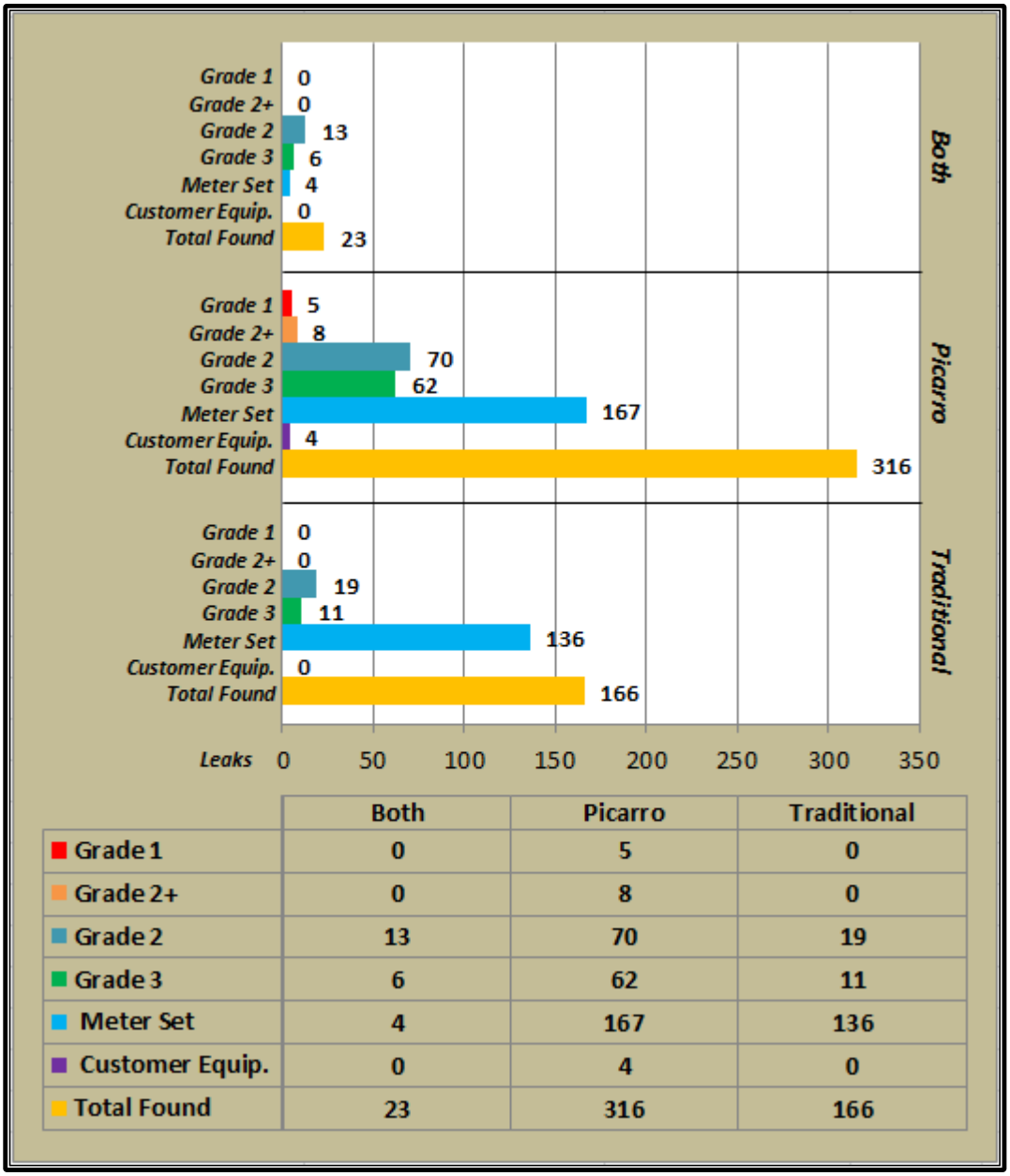
LEAK INDICATIONS. PG&E's Divisional surveyors found and graded 189 (166 found by traditional + 23 found by both) natural gas leaks within the 16 Plat Maps while following Standard Operating Procedures for leak survey using traditional handheld instruments of their choice. Leaks found with the Picarro Surveyor™, minus the confirmed 80 false positives ("ghosts LISAs"), were investigated by divisional and quality control technicians who located and graded 316 leaks.

To compare and further evaluate the effectiveness of the two survey methods refer to *Figure 5* which shows leaks found by both the traditional and Picarro methods, by Picarro only, and by traditional only by grade, meter set and customer equipment.

Referring to *Figure 5.0*, a leak "missed" by Picarro is defined as a leak originally found by traditional methods (166) which was within the Survey Coverage (FOV) of Picarro, but was not detected by Picarro. For leaks outside of the Picarro FOV that were found by traditional, these were not counted as misses and the leak was not counted towards traditional either, as that would not be a true side by side comparison. In addition, leaks found by Picarro include all leaks found within a LISA, and any leak found 150 feet outside of a LISA.

Referring to *Figure 5.0,* Picarro successfully identified 1.9 times more leaks when compared to the traditional survey methods (316 vs. 166 leaks). 5 *Grade 1* and 8 *Grade 2+* leaks were detected by Picarro's Surveyor™ within the 16 surveyed plats area that were previously undetected by traditional survey methods, therefore unknown to PG&E. *Grade 1* and *2+* leaks were found by traditional methods but because the leaks were repaired prior to the Picarro survey, they were not counted in this report.

The Surveyor™ failed to detect 1 *Grade 2* and 2 *Grade 3* leaks, the *Grade 2* was repaired by field construction crews prior to the investigation. 1 of the *Grade 3* leaks was obscured by heavy evergreens while the second *Grade 3* leak was easily found by the Picarro Surveyor™ during subsequent investigations but still categorized as a miss. As represented by *Figure 4.0,* the Picarro Leak detection system is able to provide documented auditable survey coverage (89% for services and 99% for miles-of-main).



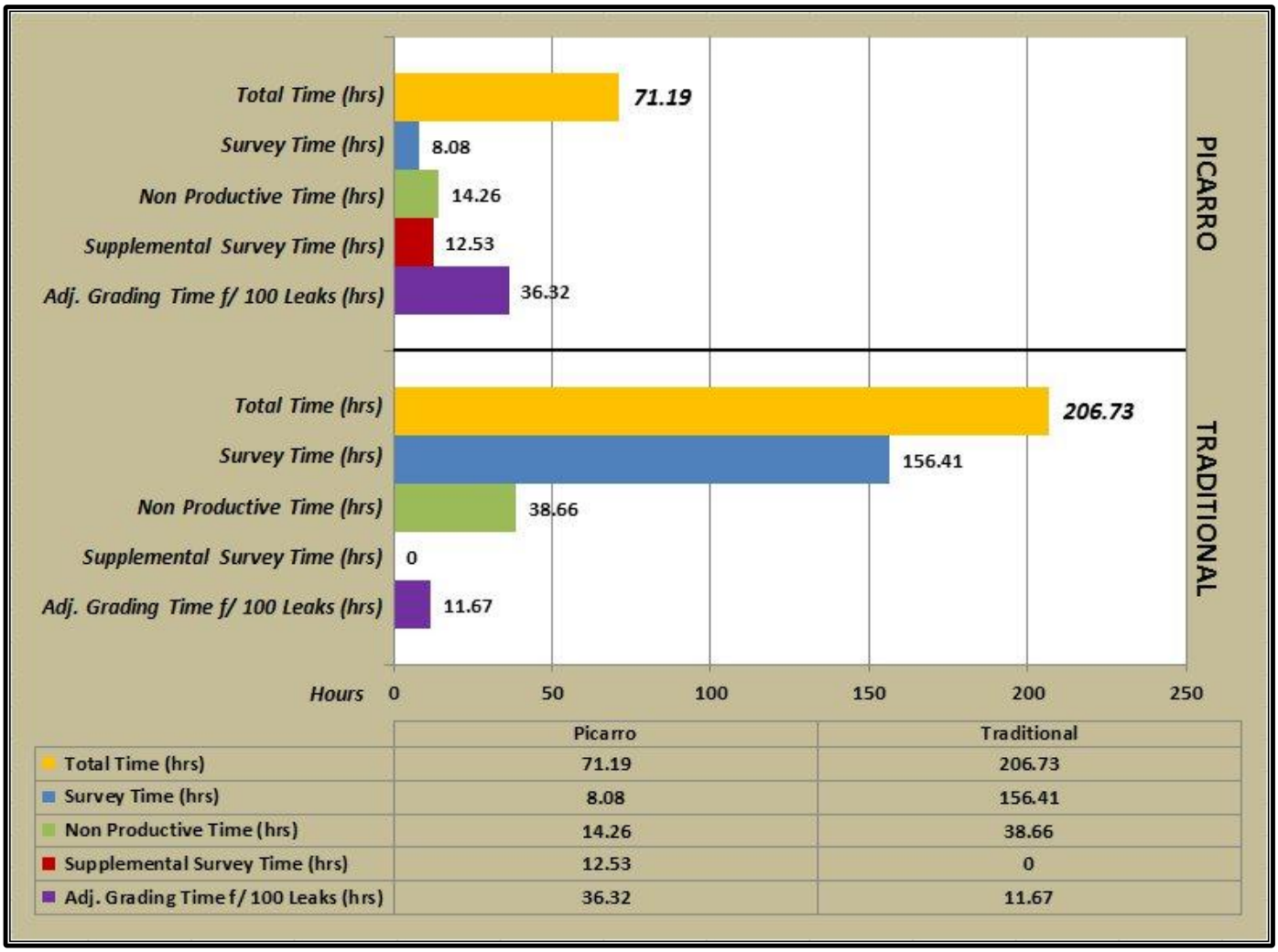
	Both	Picarro	Traditional
Grade 1	0	5	0
Grade 2+	0	8	0
Grade 2	13	70	19
Grade 3	6	62	11
Meter Set	4	167	136
Customer Equip.	0	4	0
Total Found	23	316	166

Figure 5.0 - Diablo Study: Leak Classification (Detected by Both, Traditional, Picarro)

TIME COMPARISON. For the purpose of this study the Total Survey Time is defined by the sum of five segments:

1. Survey Time.
2. Non-Productive Survey Time (includes administrative time associated with leak survey).
3. Supplemental Survey Time.
4. Adjusted Grading Time (normalized for 100 leaks or LISAs).

As shown in *Figure 6.0 (normalized)*, using the Picarro Surveyor™ all 16 maps were comprehensively surveyed and graded in 71 hours vs. 207 hours Total Survey Time for the Traditional ground and Picarro methods, respectively.

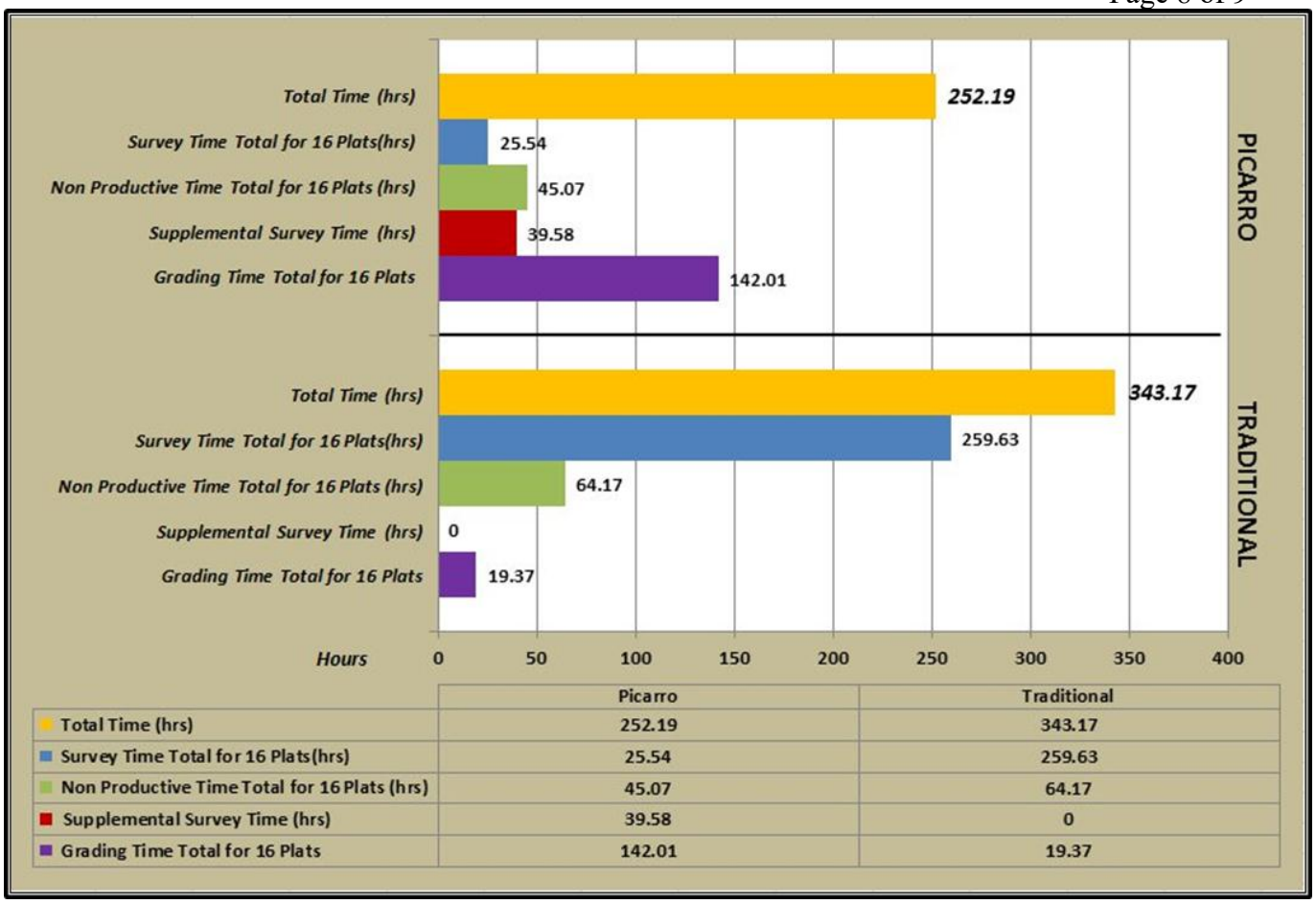


	Picarro	Traditional
Total Time (hrs)	71.19	206.73
Survey Time (hrs)	8.08	156.41
Non Productive Time (hrs)	14.26	38.66
Supplemental Survey Time (hrs)	12.53	0
Adj. Grading Time f/ 100 Leaks (hrs)	36.32	11.67

Figure 6.0 - Diablo Study: Normalized Comparison of Productive Survey Rates, Picarro vs. Traditional

For the purpose of conducting side-by-side productivity analysis the number of leaks were normalized to 100 because the Picarro Surveyor™ located more leaks than Traditional Survey methods. Therefore, all computations relating to Grading Time are based on 100 leaks and leak indication search areas. Traditional leak grading time was fixed at 7 minutes per leak. Grading time following the Picarro Surveyor™ protocol time was fixed at 22 minutes per LISA since leak survey and grading processes are decoupled with the Picarro approach. In order to obtain the 22 minutes per LISA time, an average was taken of multiple LISA investigations and grading. Additionally, standard production rates (9.9 services/hr & 2.5 mi/hr) were utilized to calculate the Supplemental Survey Time needed to foot survey the 11% of services and 1% of distribution main not covered by the Picarro vehicle. Non-productive survey time was taken directly from time sheets or as recorded by test administrators.

The rate of survey provided by Picarro was about 2.9 times faster than traditional survey. However, the grading time is much longer than traditional survey, as it takes a separate dispatch of a surveyor using a different survey instrument to pinpoint and grade the leak.



	Picarro	Traditional
Total Time (hrs)	252.19	343.17
Survey Time Total for 16 Plats(hrs)	25.54	259.63
Non Productive Time Total for 16 Plats (hrs)	45.07	64.17
Supplemental Survey Time (hrs)	39.58	0
Grading Time Total for 16 Plats	142.01	19.37

Figure 7.0 - Non-normalized Diablo Study Comparison of Productive Survey Rates, Picarro vs. Traditional

In addition, *Figure 7.0* shows a non-normalized comparison of the time needed to Leak Survey 16 plats between Picarro and Traditional. Because Picarro found more leaks over the 16 plats than traditional, and because the pinpointing and grading of leaks takes an average of 22 minutes per LISA for Picarro, the time to complete a geographic region by the two methods are 252 hours for Picarro and 343 hours for traditional.

4. DISCUSSION. As mentioned in the introduction**,** there are three fundamental characteristics of leak survey:

1. Proficiency to Detect and Locate Leaks.
2. Rate of Survey.
3. Ability to Document Areas Surveyed.

As demonstrated in earlier studies and reinforced during this test, atmospheric conditions are key components of surveying with the Picarro Surveyor™. In *Table 1.0,* daytime service coverage averaged 43% while overnight coverage rose to 81% with a combined service coverage approaching 90%. To minimize the role of atmospheric conditions on main coverage, the protocol of surveying both sides of each street was implemented.

While the Picarro Surveyor™ captures and stores all measured methane concentration values, the user can adjust the filtering threshold to identify leak indications and eliminate natural fluctuations in methane concentrations. During the first run, the amplitude over baseline was fixed at 0.10 ppm, yielding

53 leak indications, ultimately resulting in 143 natural gas leaks found with no false positives.

The amplitude on the second run amplitude was set at 0.030 ppm, generating 215 indications resulting in 197 leaks found with 80 "false" positives (informally termed "ghost LISAs").

5. **CONCLUSION**. Studies of the Picarro Surveyor™ and its P-Cubed software have proved initial concept. Picarro's expertise in gas sampling can be specifically engineered for use in leak survey programs by natural gas distribution companies to indicate, target, and confirm the presence of pipeline leaks. Furthermore, the P-Cubed software, working in unison with Picarro's highly sensitive cavity ring down spectrometer technology allows for data acquisition of leaking distribution systems and provides an auditable leak survey record unparalleled by current leak survey methods and instruments.

It has been shown in this side-by-side study that the Picarro Surveyor™ increases leak survey productivity by reducing time needed to leak survey a plat map. Specifically, use of Picarro to supplement current leak survey methods was shown to reduce the time needed to complete a leak survey by 91 hours, or a reduction of 2.3 weeks for a full-time employee, not including other related field operation costs overheads. The increase in logistical efficiency is a result of vehicle sampling as opposed to foot sampling for most or all sections of the distribution system. However, a portion of this time savings can be attributed to the P-cubed software which generated auditable FOV plots showing coverage of the leak survey. The studies' results support PG&E's suggestion and Picarro's assertion that overnight surveys would be much more productive as the inherent benefit of nighttime atmospheric conditions allow for an increase in survey scope and/or the instrument's FOV.

The results of this study confirm there is not one leak survey method can identify 100% of all leaks in a single survey. However, this study also indicates that, when used in concert and coordination, traditional and Picarro leak survey methods can be combined to maximize the effectiveness and efficiency of leak detection programs. While more testing of the Picarro Surveyor™ is necessary to understand its capabilities and limitations for production leak surveying, the data suggests that this powerful instrument warrants consideration as a primary leak detection tool, with the traditional leak survey method and instruments being retained for completion of unable-to-survey mains and services by the Picarro Surveyor™. Traditional equipment is still required for pinpointing and grading leaks, as the Picarro Surveyor™ does not provide this functionality.

During the course of this study a far better understanding of the Surveyor's™ sensitivity and adjustments was attained, particularly how it influences and relates to our future test development and implementation plan. As Picarro field sampling processes continue, and with close collaboration with Picarro engineering staff, a better understanding of the device's assets and weaknesses will lead to method, technique, performance and training improvements which will ultimately perfect the overall process. In addition, enhancements to the Picarro Surveyor™, software, and protocols continue to be refined at this time and will further improve its ability to detect and target leaks in natural gas piping systems.

[1] *PG&E Picarro Surveyor™ Leak Detection Study: Livermore Leak Test Facility, 1/13/2012 – 1/19/2012*

[2] *PG&E Picarro Surveyor™ Leak Detection Study: Southern Nevada Leak Test Facility, 3/19/2012 – 3/22/2012*

[3] *UC Davis iCH4 Isotopic Report, 2/23/2012*

[4] *ATS Calibration Report FCDS2006, 1/1/2012*

[5] *Wikipedia Planetary Boundary Layer: http://en.wikipedia.org/wiki/Atmospheric_boundary_layer*